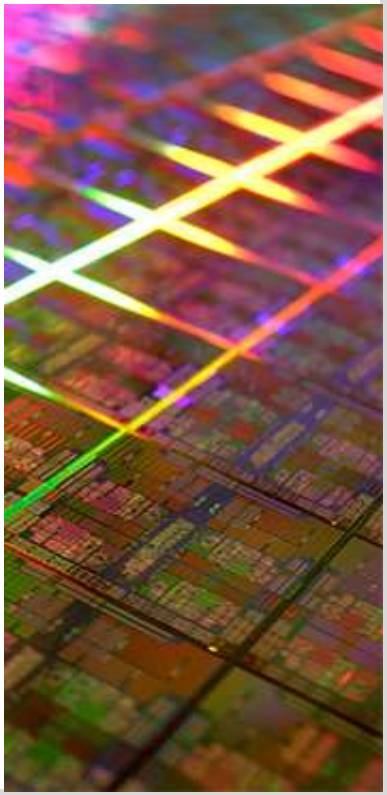
Specialty Chemicals and Gases for Semiconductor Devices Fabrication (Lithography Process)



Dr. Nithi Atthi

Fab Manager

Thai Microelectronics Center (TMEC)
National Science and Technology Development Agency (NSTDA)

Dr. Nithi Atthi

Fab Manager | Thai Microelectronics Center (TMEC)



Ph.D., Tokyo Institute of Technology
Multiple International Awards

23+ Years
Semiconductor Experience

Expertise
CMOS | 3D FinFETs
MEMS | Si Photonics

45+ Publications
Peer-Reviewed Journals

27 Patents
Innovation & IP

nithi.att@nstda.or.th

Outline

1. Introduction to Lithography
- Role of lithography in semiconductor fabrication
- Resolution, overlay, CD/CDU, DOF, LER/LWR and aspect ratio
- Lithography & patterning technologies
- Overall lithography process flow

2. Wafer Preparation & Adhesion Promotion
- Wafer pre-clean and dehydration bake
- HMDS priming
- Surface hydrophilicity / hydrophobicity and PR adhesion

3. Photoresist & Lithography Materials
- Positive / negative photoresist
- PR composition: resin, sensitizer/PAG, solvent and additives
- g-line, i-line, KrF, ArF and EUV resist
- CAR, MOR and e-beam resist
- Thick PR, dry-film resist and lift-off resist
- PI / PBO for advanced packaging
- BARC & TARC

4. Photoresist Coating Process
- Spin, spray, slit and other coating methods
- Coating kinetics and thickness control
- Coater/developer track
- EBR & BSR
- PGMEA and other EBR/BSR solvents
- Pre-bake and coating defects

5. Photomask & Reticle
- Photomask structure and fabrication
- Binary / Cr-on-quartz mask
- EUV reflective mask
- Chemicals used in photomask fabrication

6. Alignment & Exposure
- Mask aligner, stepper and scanner
- Contact / proximity / projection exposure
- Exposure wavelength and numerical aperture
- Alignment & overlay
- Energy–Focus matrix and process window

7. PEB, Development & Hard Bake
- Post-exposure bake and standing-wave control
- Developer types and application methods
- TMAH 2.38 wt%
- Developer kinetics
- Developer for PR / PI / PBO
- Hard bake

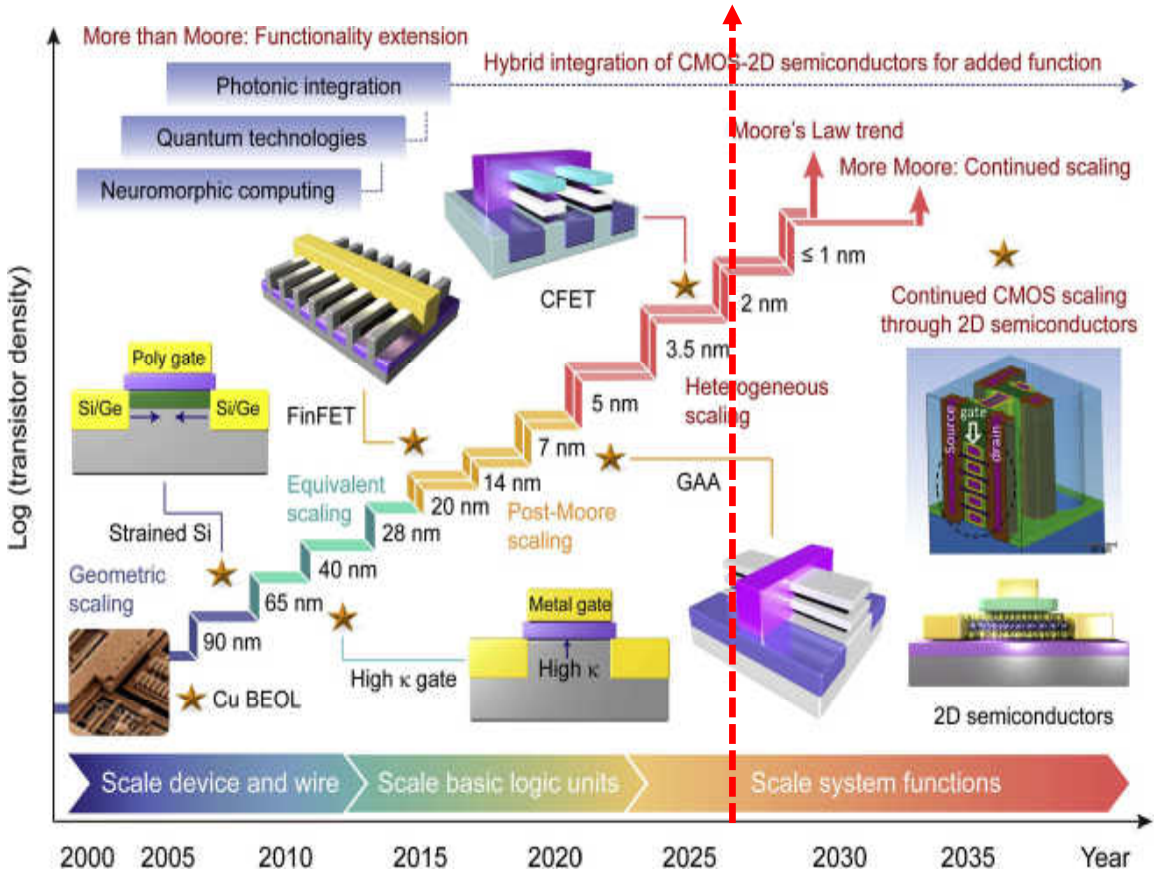
8. Lithography Metrology & Inspection
- CD / CDU
- Overlay
- LER / LWR
- Sidewall profile
- Defect inspection
- CD-SEM, FE-SEM, OCD, AFM and overlay metrology

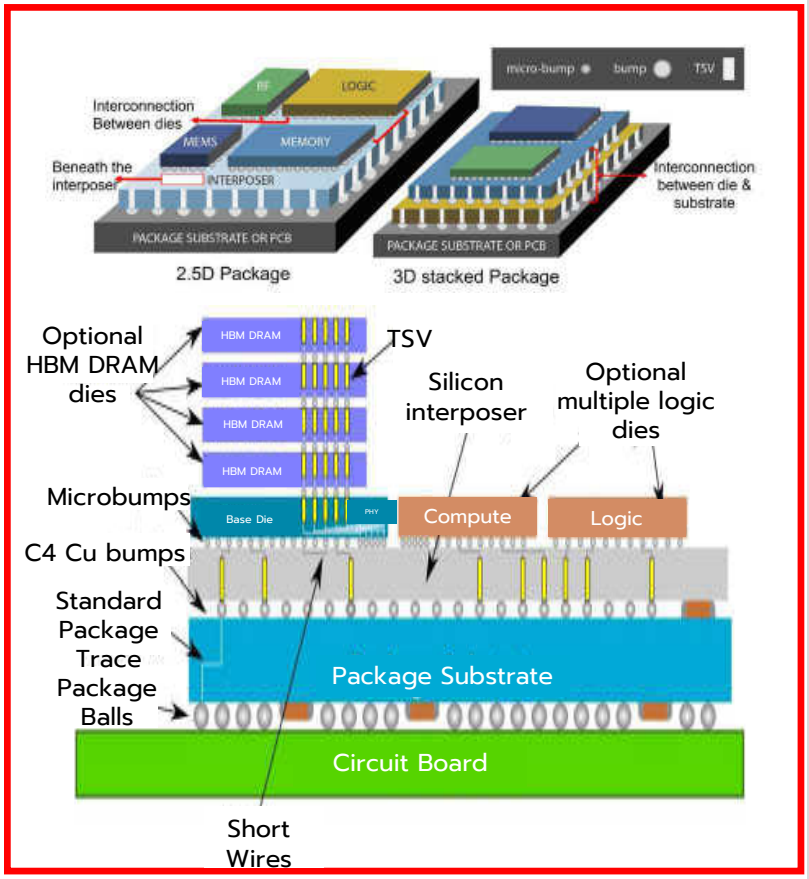
9. PR Stripping & Lithography Rework
- PGMEA vs. NMP vs. DMSO
- Wet PR stripping and residue removal
- ALEG / EKC and formulated strippers
- Rework chemistry by process stage
- Flood exposure for positive-PR rework

Lithography Process

Moore's law scaling to Hyper-scaling era

(Physical scaling x Advanced Packaging) → 3D nano-structures

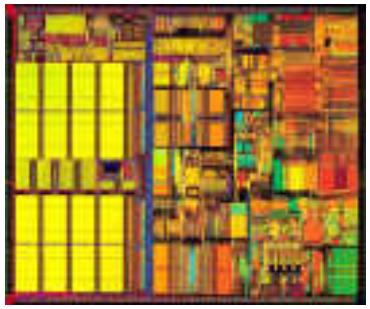




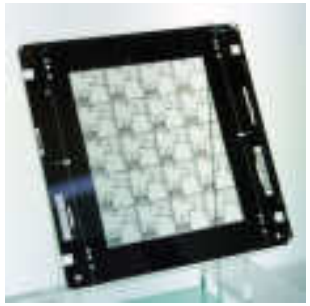
https://blog.delmic.com/more-than-moore-the-next-steps-for-the-semiconductor-industry
https://doi.org/10.1016/j.isci.2022.105160

Electronics devices fabrication

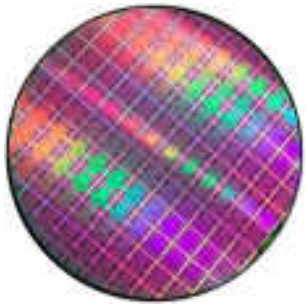
Front-end
(Upstream)



Circuit & layout
Design
(FabLess, Design House)



Photomask fabrication
(Mask shop)







Wafer fabrication
(IDM, Foundries,
Fablite)

Back-end
(Downstream)



Electronics products
(EMS/Box build)



Assembly
(PCB, PCBA)

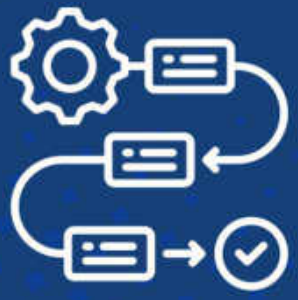


Packaging
(OSAT, ATP)

PCB: Printed Circuit Board
PCBA: Printed Circuit Board Assembly
IDM: Integrated Device Manufacturer (Design, Manufacturing, sell)
Foundries: A semiconductor manufacturer that makes chips for other companies
OSAT: Outsourced Semiconductor Assembly and Test
ATP: Assembly, Test, and Packaging

Wafer fabrication process (CMOS)

MEMS is built on semiconductor technology & extends it into the mechanical world.



IC fabrication Process cycle

(ถ่าย) 3 — PR / Thin Film / Silicon — **Lithography**

Thin Film / Silicon — **Etching** (กัด) 4

Thin Film / Doping Doping / Silicon — **Diffusion & Ion Implantation + Thermal process** (ยิง + อบ) 5

Thin Film / Silicon — **Thin Film Deposition** (ปลูก) 2

Silicon — 1 **Wafer cleaning & wet etching** (ล้าง)

ล้าง → ปลูก → ถ่าย → กัด → ยิง → อบ

FEOL: front-end-of-line
BEOL: back-end-of-line

Not including
- Surface Planarization (CMP)
- Wafer bonding
- Metrology & inspection
- Electrical Characterization

https://semiengineering.com/breaking-the-2nm-barrier/
https://www.pngwing.com/en/free-png-tpesj

Lithography Process

The Heart of Semiconductor Manufacturing

1. Pattern transfer technology

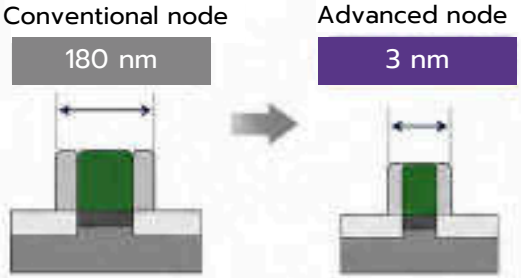
Lithography transfers the circuit design from mask to the wafer through a series of precise steps



Lithography accurately prints billions of pattern on a wafer, defining the size, shape, and placement of every device

2. Enables scaling and device functionality

Advanced lithography enables smaller features, higher density, and more powerful devices



Better lithography = smaller nodes
= More transistors
= Higher performance
= Lower power

3. Impact on key performance metrics

Lithography performance directly drives chip quality, functionality and competitiveness

YIELD
Accurate patterning reduces defects and variability, resulting in higher yield

RESOLUTION
Determines the smallest feature size that can be printed on the wafer

OVERLAY
Ensures every layer is aligned precisely. Better overlay = better device performance

DEVICE PERFORMANCE
Impacts speed, power, leakage, and reliability of the final chip

WHY IT MATTERS

Accurate pattern transfer → Smaller, more reliable devices → Better performance & lower power → Competitive products

Lithography is the foundation of every chip. Better lithography enables scaling, drives performance, and powers the future

Lithography Requirements

To print the right pattern at the right place, with the right cost

TMEC Thai Microelectronics Center

1. Resolution
Print small features clearly

Ability to print the minimum feature size with high image contrast



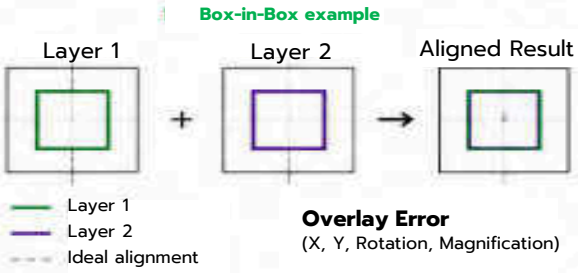
Key Factors

- High NA (Numerical Aperture)
- Short Wavelength (e.g. ArF, EUV)
- Good focus & process window
- OPC/ Phase shift/ Immersion

Better resolution enables smaller technology nodes and higher density

2. Overlay accuracy
Print each layer in the right place

Ability to align each layer precisely to ensure device performance and yield



Key Factors

- Stable equipment & stage accuracy
- Low vibration environment
- Thermal control
- Advanced metrology & process control

Better overlay improves device performance and reduces defects

3. Economic
Manufacture at the right cost

Achieve high productivity and low cost per good die while meeting quality targets

High Throughput
Short cycle time, high productivity

High Yield
Low defectivity, High first-pass yield

Low Cost of Ownership (CoO)
Efficient tools, low energy, low maintenance

Process window
Robust process, Less sensitivity

Better economics ensure competitiveness and sustainable manufacturing

The Goal
Right pattern
Right place
Right cost

Balanced Excellence
Resolution x Overlay x Economics
= High Yield, Performance & Profitability

Interdependent
Improving one aspect often impacts others
Optimization and balance are key success.

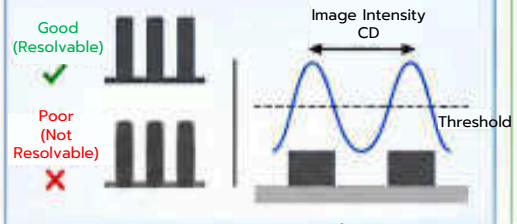
K. Ronse , Optical lithography—a historical perspective ,C. R. Physique 7 (2006) 844–857

Key Lithography Metrics

Critical parameters that define pattern fidelity and device performance

1. Resolution

The ability to print the smallest features as separate lines



Key Metric

Rayleigh Resolution Limit

$$R = k_1 \frac{\lambda}{NA}$$

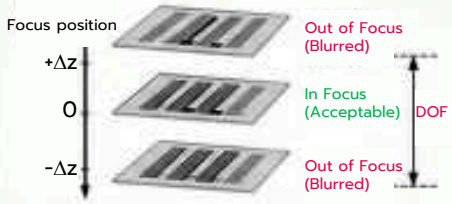
Impact

Determines minimum feature size

λ = Wavelength, NA = Numerical Aperture
K_1 = Process factor (0.25-0.35)

2. Depth of Focus (DOF)

The range of focus over which features can be printed acceptably



Key Metric

$$DOF = k_2 \frac{\lambda}{NA^2}$$

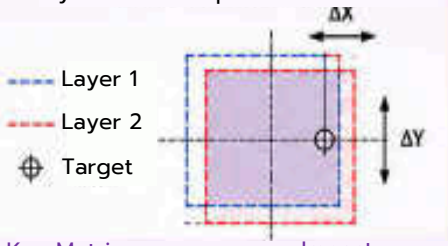
Impact

Determines focus latitude & process robustness

λ = Wavelength, NA = Numerical Aperture
K_2 is a Process factor (0.5-1.0)

3. Overlay Accuracy (OVL)

The ability to align patterns from different layers within a specified tolerance



Key Metric

- Overlay Error (3σ)

$$\sigma_{overlay} = \sqrt{\sigma^2_{wafer} + \sigma^2_{tool} + \sigma^2_{process}}$$

- Reported in nm (e.g. 3σ = 2nm)

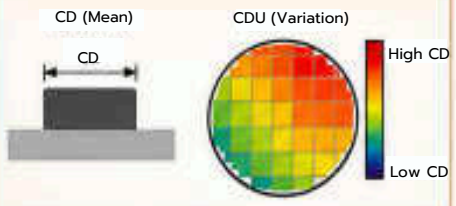
Impact

Critical for multi-patterning tight pitch alignment; affects yield

4. CD/CDU

CD: Mean feature size

CDU: Variation of CD across the wafer (WiW) and from wafer to wafer (W2W)



Key Metric

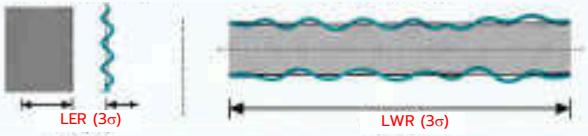
- CD: Mean of measured CD (nm)
- CDU (3σ): Standard deviation of CD (nm)

Impact

- Affects device performance, design rules margin, and yield

5. LER, LWR

LER: High-frequency random deviation of the edge
LWR: Low-frequency variation of line width along the feature.



Key Metric

- LER (3σ): nm (typically 1-10 nm)
- LWR (3σ): nm (typically 0.1 to 100 μm)

Impact

Impacts device variability, leakage and performance at advanced nodes

6. SWA, SWR

SWA: Sidewall angle, The angle of the patterned sidewall from vertical
SWR: Sidewall roughness, The roughness of the sidewall profile



Key Metric

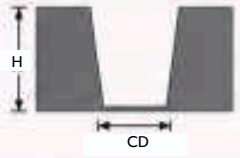
- SWA: Degrees (°)
- SWR (3σ): nm

Impact

Affects etch transfer, line resistance, leakage, and reliability

7. Aspect Ratio (AR)

AR: The ratio of feature height (H) to critical dimension (CD)



Aspect ratio = H/CD

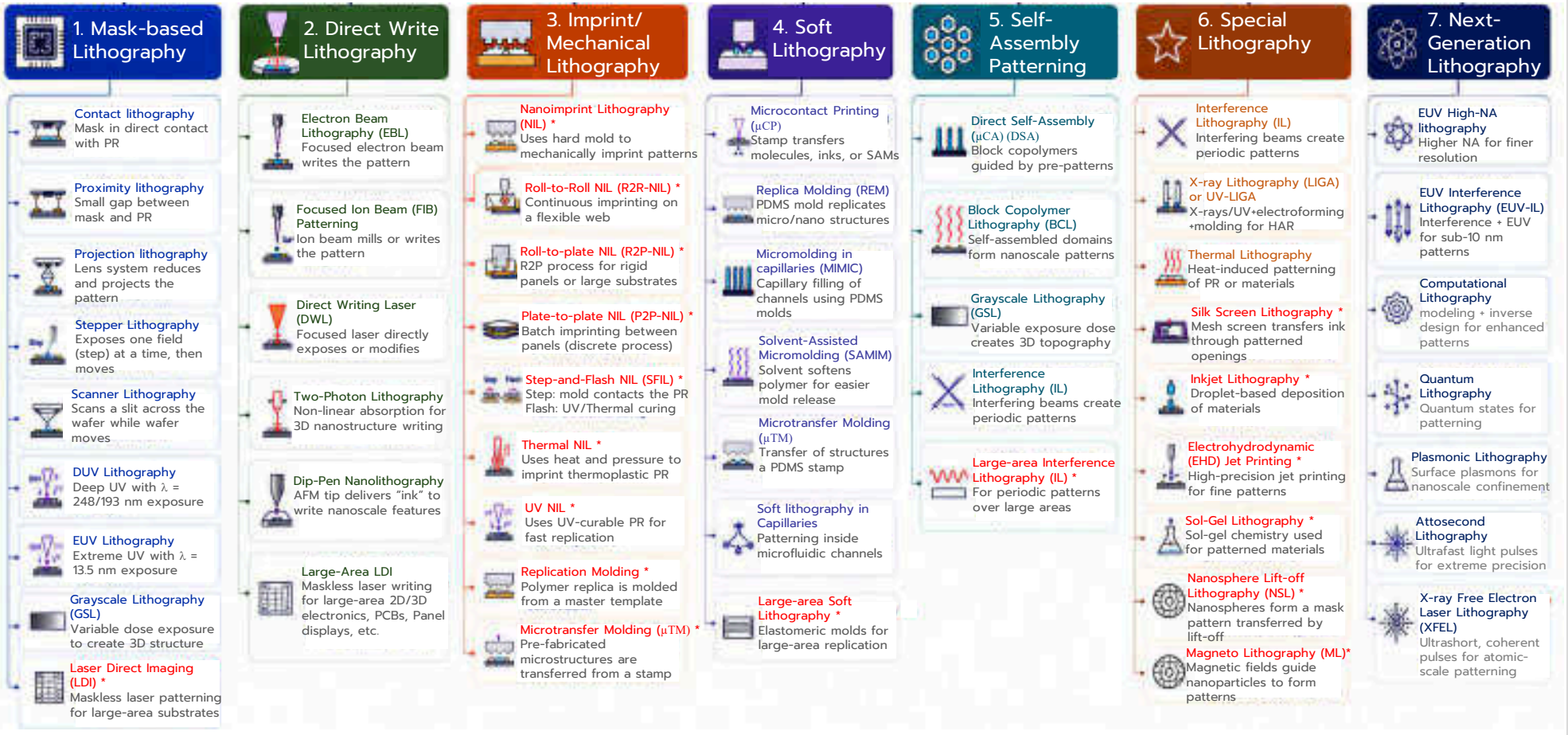
Key Metric

- Aspect ratio (AR) = H/CD (Dimensionless)

Impact

Higher AR increases process complexity, impacts etch, deposition, and planarization

Keys Lithography-based Patterning Methods

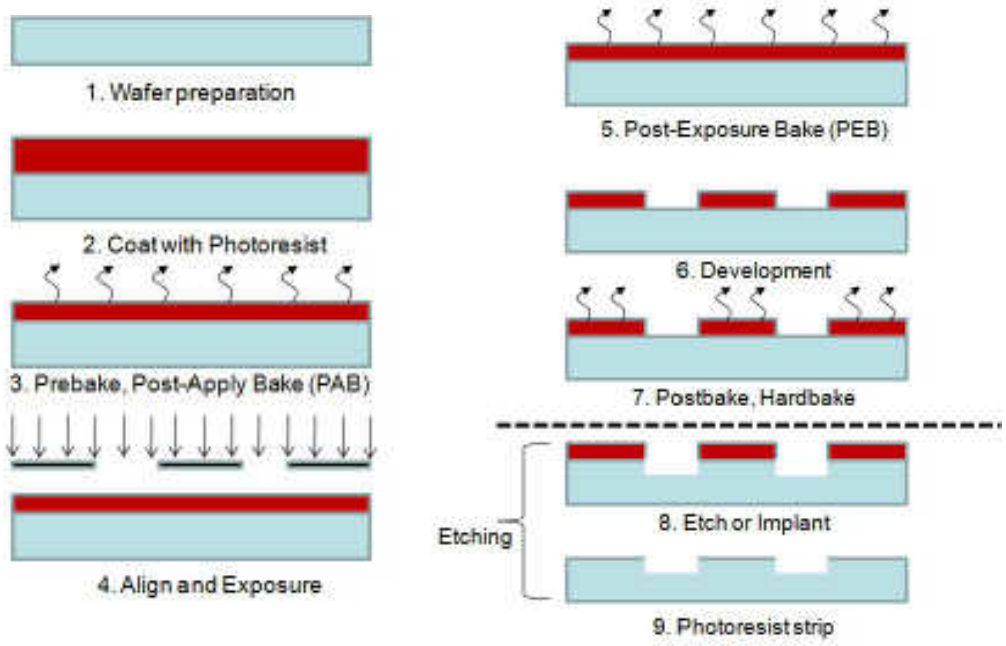


Lithography process

Patterning = Photolithography+ Etching

"Lithos" = Stones, "Graphia" = to write → Lithography = "Writing on stones".

The sub-micron patterns are written by small wavelength "light" into a light-sensitive polymer called "Photoresist" coated on "Si wafer".

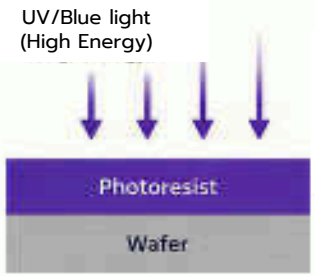




Why Lithography Room Must be Yellow Light

To protect photoresist from premature exposure

(1) Photoresist (PR) is light sensitive

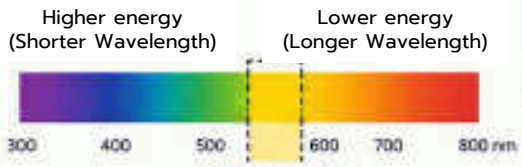


Photoresist (PR) changes chemistry when exposed to shorter wavelength light (UV, blue and some visible light)

Unintended exposure = undesired patterning

(2) Yellow Light is Safe

Yellow light has a longer wavelength (lower energy) that is above absorption range of photoresist



Yellow Light
(λ = 560-590 nm)



(3) Prevents Defects & Yield Loss

Stray exposure before the intended step can cause:



Critical Dimension (CD) shift | Pattern distortion or bridging | Poor etch performance | Reduced yield & reliability

(4) Industry standard

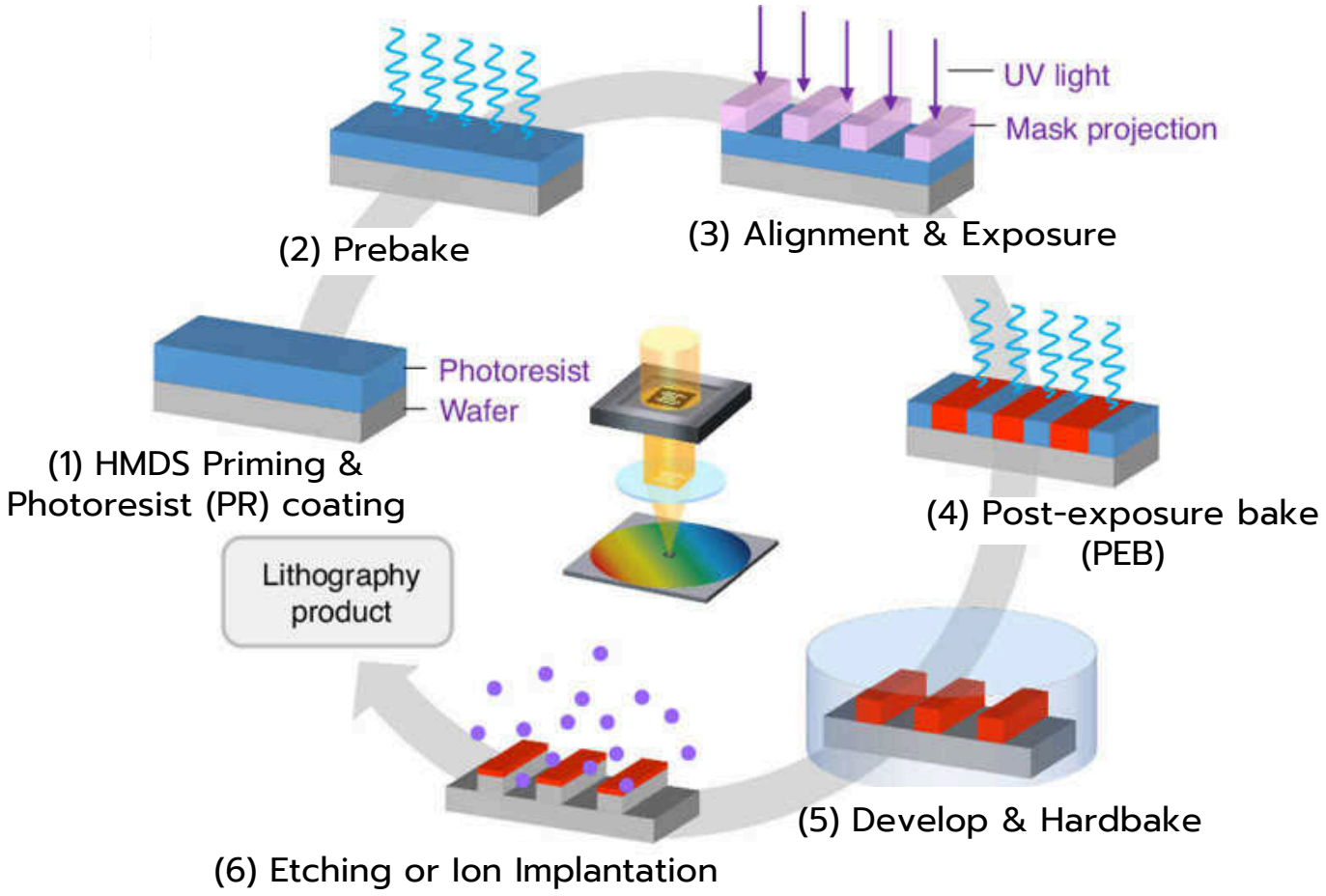
SEMI standards (SEMI C24) recommend yellow or amber lighting (λ > 500 nm) in all lithography areas

Typical specification
Peak wavelength: 560-590 nm
Illuminance: ≤ 200 lux

Key Benefits

- Maintains pattern fidelity
- Prevents premature photoresist exposure
- Improves yield and device performance
- Ensures process consistency and reliability

Lithography process flow



https://doi.org/10.1038/s41377-025-01923-w

Chemicals & Gases in Lithography Process

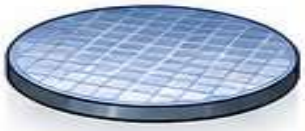
1. Wafer Pre-clean	Solvent	Acetone, IPA	✓	✓	Organic contamination removal
	SC-1	$NH_4OH + H_2O_2 + H_2O$	✓	△	Particle / organic removal
	SC-2	$HCl + H_2O_2 + H_2O$	✓	△	Metallic contamination removal
	Dilute HF	$HF + H_2O$	✓	△	Native oxide removal
2. Adhesion Promotion	HMDS	Hexamethyldisilazane	✓✓✓	✓✓	Improve PR adhesion
3. Bottom ARC	BARC	Bottom Anti-Reflective Coating	✓✓✓	△	Reduce substrate reflection / standing wave
4. Resist Coating	Photoresist	i-line / KrF / ArF PR	✓✓✓	✓	
	Thick PR	Positive / negative thick PR	–	✓✓✓	RDL / bump plating mold
	PI / PBO	Photosensitive PI / PBO	–	✓✓✓	Passivation / dielectric
5. Top ARC	TARC	Top Anti-Reflective Coating	✓✓	△	Reduce top-surface reflection
6. EBR	EBR Solvent	PGMEA / proprietary EBR	✓✓✓	✓✓✓	Edge bead removal
7. Development	TMAH	Typically 2.38 wt%	✓✓✓	✓✓✓	Positive PR development
	Solvent Developer	Vendor-specific	△	✓	Negative / thick PR development
8. Rinse	DI Water	UPW	✓✓✓	✓✓✓	Post-develop rinse
9. Descum	O_2 plasma	O_2	✓✓	✓✓✓	Remove residual resist
	O_2/CF_4 plasma	$O_2 + CF_4$	✓	✓	Descum / surface treatment
10. PR Strip / Residue Removal	ALEG™ stripper	Commercial formulated wet stripper	✓✓✓	✓✓	PR / post-etch residue removal
	EKC™ stripper	Commercial formulated wet stripper	✓✓✓	✓✓	PR / post-etch residue removal
	NMP-based	NMP formulations	✓✓	✓✓✓	PR / thick PR stripping
	DMSO-based	DMSO formulations	✓✓	✓✓✓	PR / thick PR stripping
	Amine-based	Proprietary mixtures	✓✓✓	✓✓✓	PR / polymer residue removal
11. Post-strip Clean	SPM	$H_2SO_4 + H_2O_2$	✓✓	△	Organic residue removal
	IPA / DIW	IPA / UPW	✓	✓✓	Final clean / rinse

Dehydration Bake & HMDS Priming

Dehydration bake

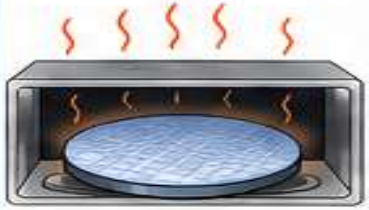
To improve photoresist coating uniformity and pattern fidelity

(1) Wafer Load



Cleaned wafer loaded into dehydration bake chamber

(2) Dehydration bake



Bake at 150-200 °C for 60-120 sec. (Typical)

(3) Moisture Removal



Physically adsorbed water desorbs. Surface –OH density is reduced

(4) Cool & Transfer

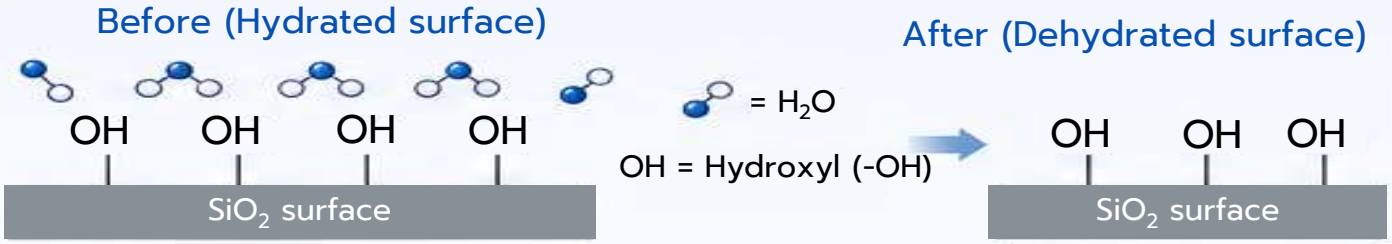


Wafer cooled and transferred to HMDS priming module

Key Benefits

- Removes adsorbed moisture
- Reduces defects (Blisters, scumming, resist lifting)
- Improves resist coating uniformity
- Enhances adhesion control consistency

Surface Chemistry before vs after dehydration bake



Why it Matters

A dry surface is essential for effective HMDS bonding and stable, hydrophobic termination

Dehydration bake does not completely remove water from surface of Silica substrate because Si atoms bond strongly with a monolayer of water forming Silanol groups (SiOH).

Hexamethyldisilazane (HMDS)

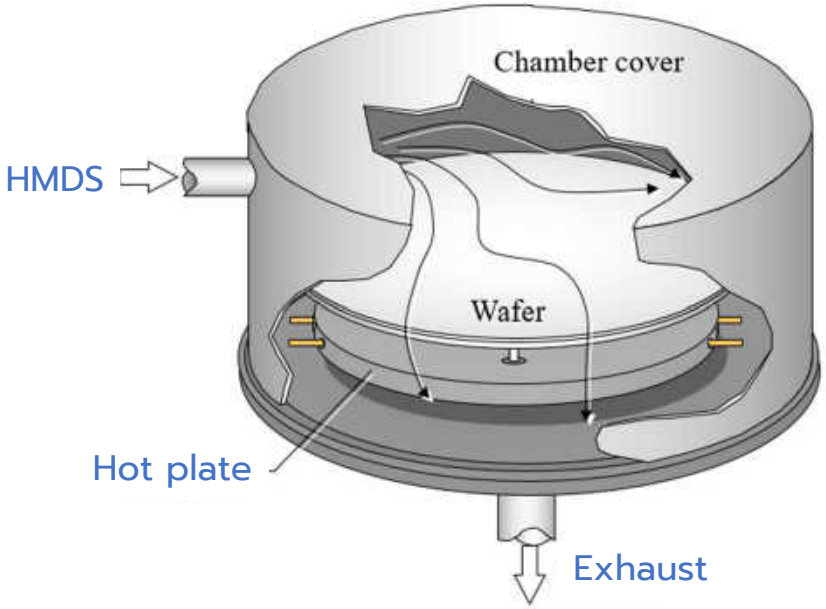
HMDS = Adhesion Promoter → Surface Hydrophobization → Better PR Adhesion → Highly Flammable & Moisture Sensitive

Process flow as: Wafer Clean → Dehydration Bake → HMDS Vapor Prime → BARC (if required) → PR Coat → TARC (if required) → Exposure

Chemical / Formula	Hexamethyldisilazane (HMDS), $C_6H_{19}NSi_2$
Physical Properties	Colorless, volatile liquid; BP ~125°C; density ~0.77 g/cm³
Main Application	Photoresist adhesion promoter; makes wafer surface hydrophobic
Process Step	Dehydration bake → HMDS prime → PR coating
Application Method	Vapor priming (preferred) or spin coating
Typical Substrate	Si, SiO_2, glass / hydroxylated surfaces
Container / Storage	Sealed compatible container; cool, dry, well-ventilated area; protect from moisture
Safety Concerns	🔥 Highly flammable; harmful by exposure; vapor/skin-contact risk; moisture sensitive
Handling / PPE	Local exhaust; chemical-resistant gloves; goggles; keep away from heat, sparks & ignition sources

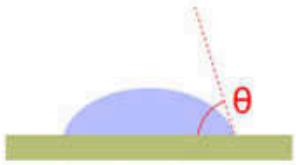
Hexamethyldisilazane (HMDS) Priming

- HMDS priming used to promote the adhesion for photoresist (200-250°C for 60 sec)
- Vapor priming with HMDS leaves a mono-layer of TMS (trimethylsilyl) on the Si or SiO_2 surface
- Dehydrates the substrate surface and lowers the surface energy to promote better wetting
- The molecular formula for Hexamethyldisilazane, or bis(trimethylsilyl)amine, is $C_6H_{19}NSi_2$

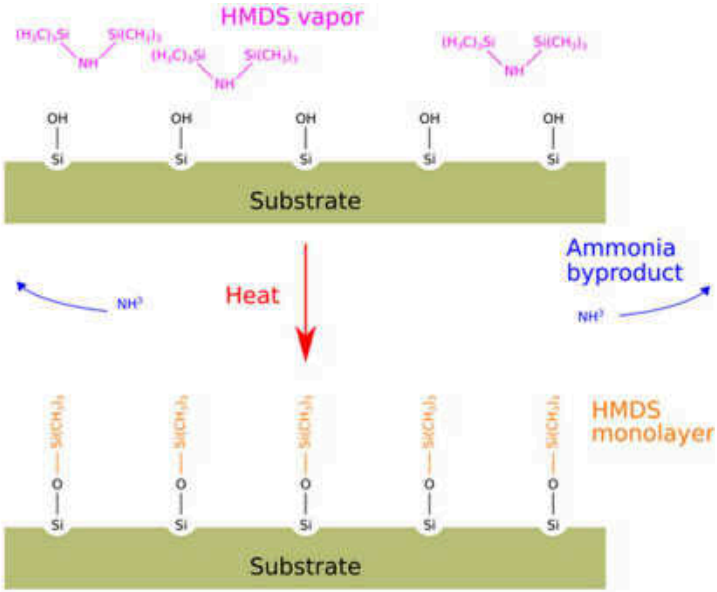




Hydrophilic surface
- Low contact angle
- High surface energy



Hydrophobic surface
- High contact angle
- Low surface energy



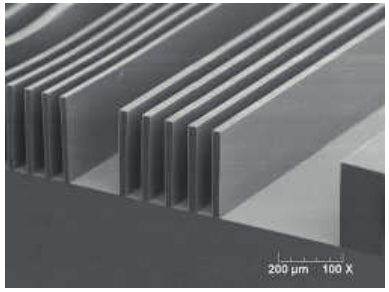
Photoresists

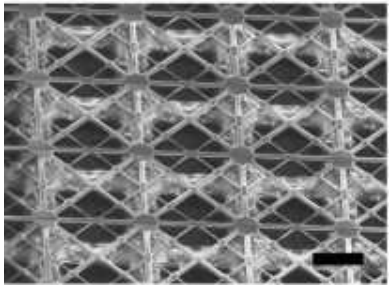
What is Photoresist?

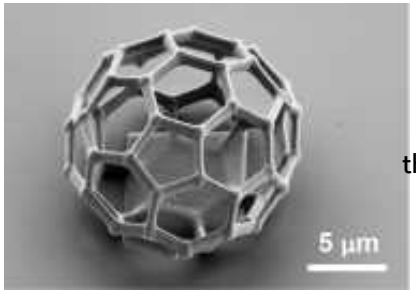
Photoresist (PR) is a light-sensitive organic compound that have been used in the printing industry since the 1920s.

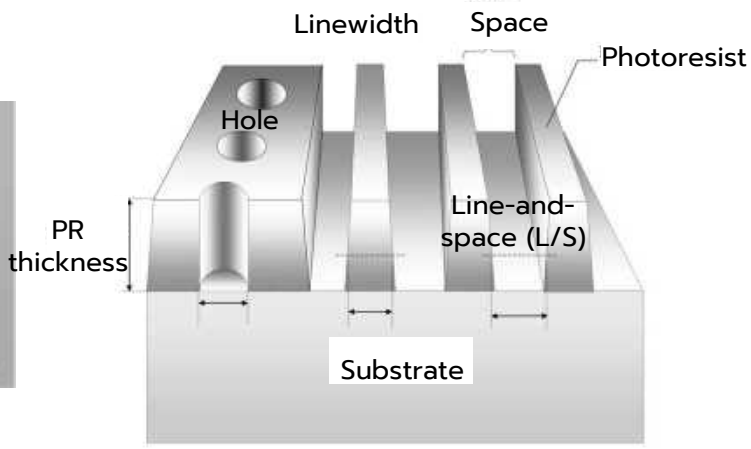
Use to transfer the mask pattern to the resist on the top layer of the substrate surface.

Protect the underlying material during subsequent processing such as etch and ion implantation.



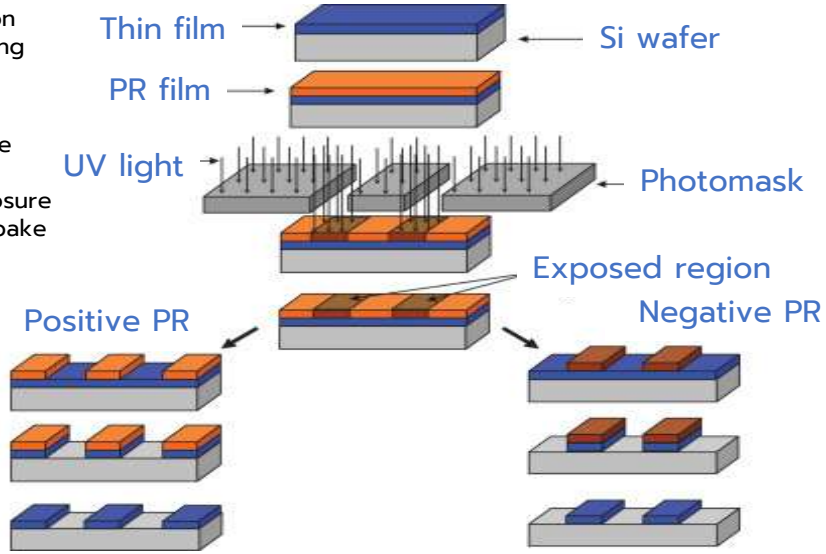






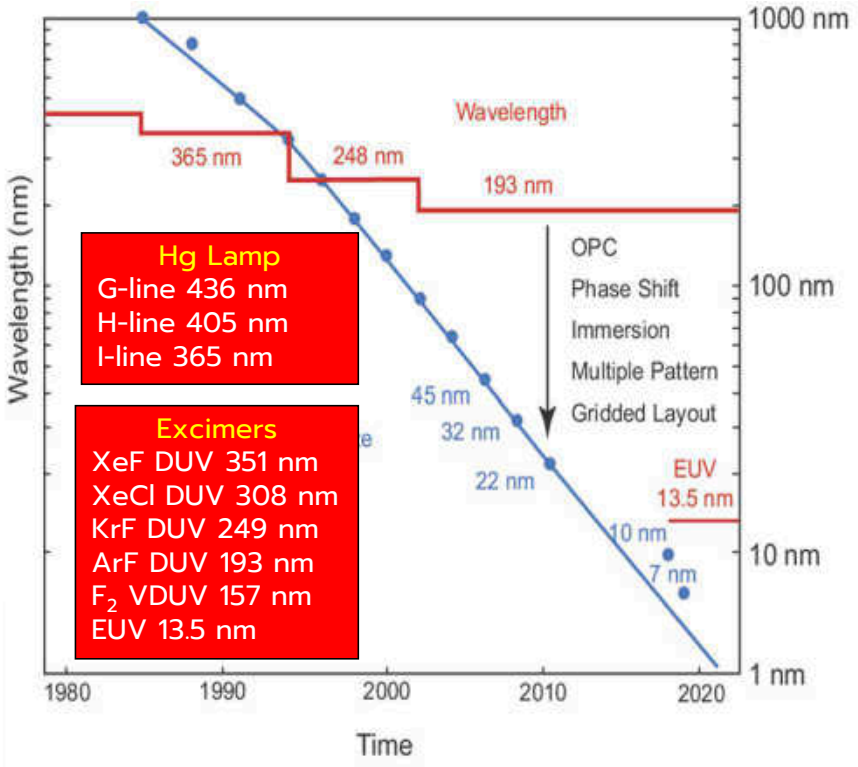
Type of Photoresist (PR) film

1. Thin-film deposition & substrate cleaning & HMDS priming
2. Photoresist (PR) coating & Prebake
3. Alignment & Exposure & Post-exposure bake (PEB)
4. Develop & Postbake
5. Thin-film etching
6. PR film removal



Four components of PR

- Resin: organic molecules matrix that binds the materials in PR (+PR: Novolak resin), (-PR: Polyisoprene rubber)
- Sensitizer: photosensitive or photoactive compound (PAC) that responds to UV light (+PR: Diazonaphthoquinone (DNQ), (-PR: Photoactive sensitizer)
- Solvent: keep PR in a liquid state for spin coating application.
- Additives: modified specific performance of PR.



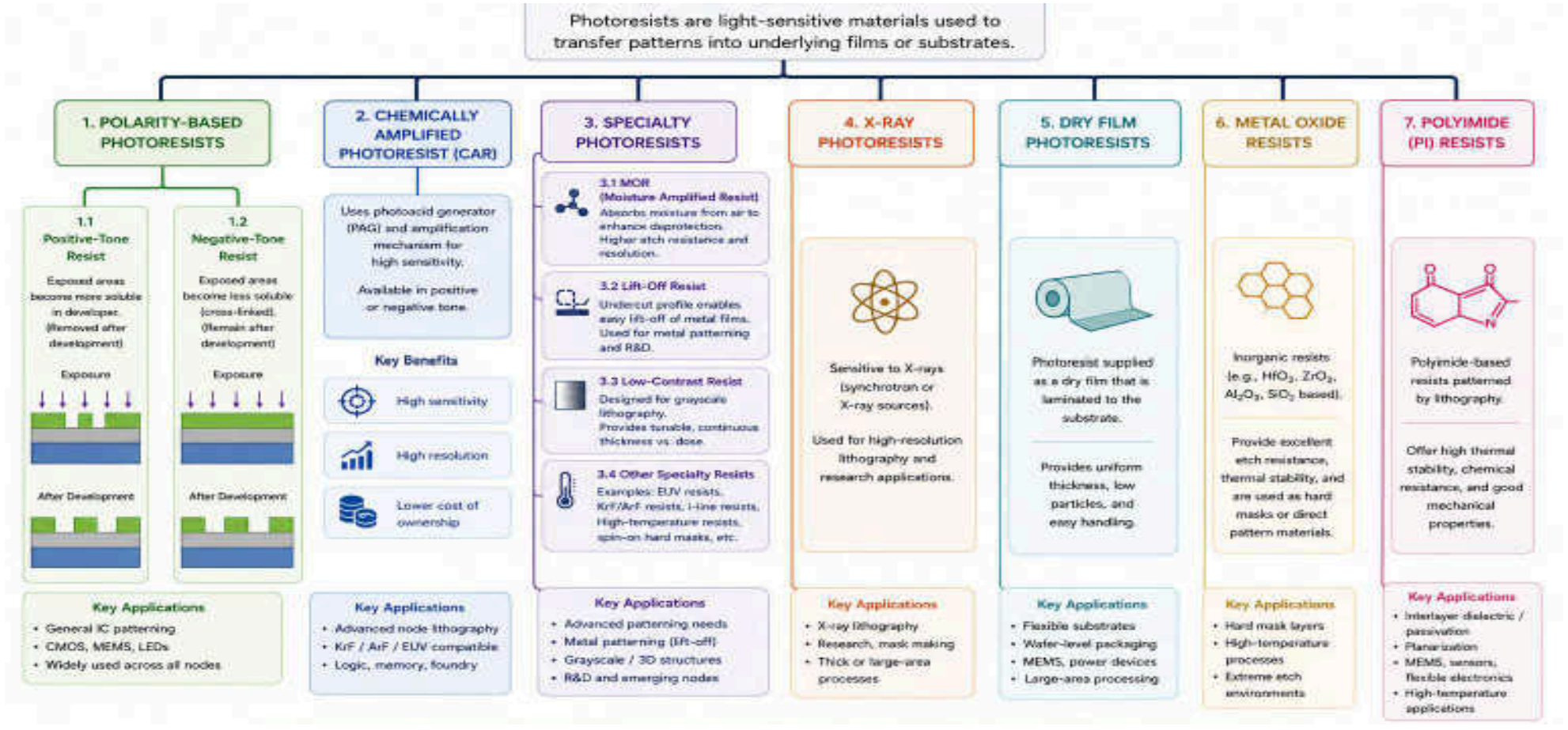
(+) PR has a higher resolution than (-) PR but more expensive

Quirk, M. and J. Serda. 2001, Semiconductor Manufacturing Technology, Prentice hall, New Jersey.

Photoresist Comparison by Lithography Wavelength

Feature	g-line 436 nm	i-line 365 nm	DUV 248 nm (KrF), 193 nm (ArF)	EUV 13.5 nm
Wavelength (source)	436 nm (Mercury arc lamp)	365 nm (Mercury arc lamp)	248 nm (KrF), 193 nm (ArF) (Excimer Laser)	13.5 nm (EUV Source)
Exposure tool	Stepper/Mask aligner	Stepper	Stepper/scanner (Dry/immersion)	EUV scanner (Reflective optics)
Photoresist type	Positive (Novolak/DNQ)	Conventional positive or CAR	CAR (Advanced polymer + PAG)	CAR (High-purity polymer + PAG)
PR film thickness	0.5-2.0 μm	0.5-2.0 μm	0.1-1.5 μm	20-60 nm
Resolution	1.0-1.5 μm (mature nodes)	0.35-0.8 μm (500 nm to 90 nm nodes)	20-100 nm (90 nm to 7 nm nodes)	10-20 nm (7 nm and below nodes)
Sensitivity (mJ/cm^2)	High (>10)	Medium-High (5-20)	Medium (1-10)	Low (0.1-3.0)
LER	High (poor)	Medium	Low	Very low
Key advantages	• Simple & Low cost • High sensitivity • Thick film capability • For mature process	• Better resolution that g-line • Used in MEMS, power device, analog IC • Good process window	• High resolution • Good CD control • Suitable for advanced CMOS, DRAM, Logic	• Highest resolution • Sub-10 nm nodes • Excellent pattern fidelity • High process uniformity
Key challenges	• Low resolution • High LER • Not for advanced nodes	• Limited resolution for advanced nodes • Standing wave effect	• Acid diffusion (Blur) • LER/LWR • Stochastic defects	• Stochastic defects • Outgassing/contamination • High-cost materials
Typical applications	MEMS, sensor, PCB, microfluidics	Power Devices, Analog IC, MEMS, LEDs	Advanced CMOS, DRAM, Logic, CIS	7 nm and below nodes, Logic, DRAM

Photoresist (PR) film categories



Chemical Amplified Resist (CARs)

CARs use a catalytic cycle to achieve high sensitivity in semiconductor lithography. During exposure, light activates a Photoacid Generator (PAG) to release an acid. During a PEB, this acid acts as a catalyst to cleave protecting groups on thousands of polymer chains, causing a dramatic change in solubility.

Exposure (Acid Generation)

DUV directly activates PAG, while EUV generates secondary electrons that trigger photoacid formation.

Post-Exposure Bake (PEB) (Catalytic Deprotection)

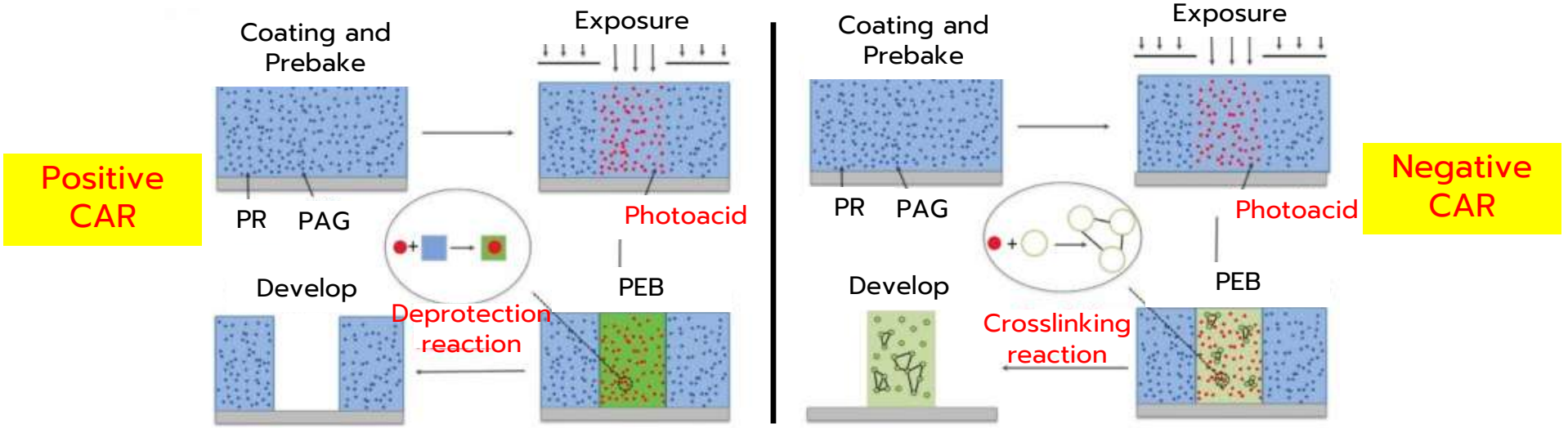
PEB drives acid diffusion and deprotection, making exposed resist soluble in TMAH.

Quenching (Precision Control)

Base quenchers limit acid diffusion, reducing acid blur and improving resolution.

Development

Development removes exposed positive-tone CAR or retains cross-linked negative-tone CAR



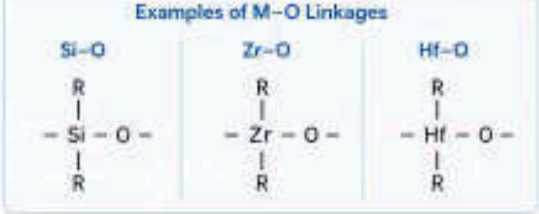
Metal Oxide Resist (MOR)

Inorganic-organic hybrid PR for etch resistance and excellent thermal stability

(1) What is MOR?

Contains metal-oxygen (M-O) linkage (e.g. Si-O, Zr-O, Hf-O) in an organic matrix, providing superior resistance to plasma etching and high-temperature processing

Examples of M–O Linkages

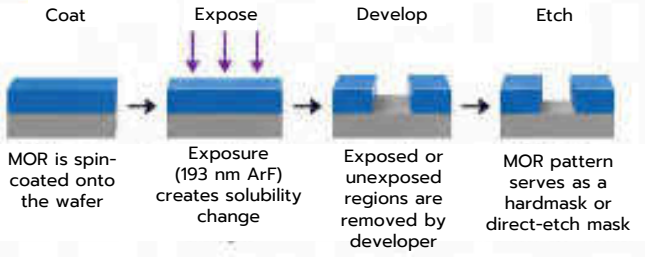


R: Organic group

(2) Key Advantages

- **High Etch Resistance**
  Dense, inorganic-like structure & low etch rates
- **Excellent Thermal Stability**
  Maintains structure and performance at elevated temperatures
- **Low Dielectric Constant (Low-κ)**
  Beneficial for advanced interconnects
- **Excellent Pattern Fidelity**
  High resolution, low line-edge roughness (LER) and dimension control
- **Hardmask Capability**
  Can be used as an etch resistant hardmask for aggressive plasma etching

(3) Working Principle



Why MOR is Etch Resistant?

During plasma etching, the M-O bonds form a stable, inorganic-rich surface (e.g. metal oxides) that protects the underlying material

(4) Typical properties

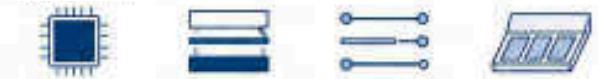
Property	Typical range	Benefit
Metal content (Si, Zr, Hf, etc.)	10-50 wt%	Improves etch resistance and thermal stability
Etch rate	5-20 nm/min (SiO2 etch)	High resistance to aggressive plasma
Dielectric constant (k)	2.2-3.5	Low-k for interconnect applications
Glass transition temp (T_g)	120-200 ℃	Thermal stability in processing
Decomposition temp. (T_d)	> 300 ℃	High temperature robustness
Resolution (CD)	< 20 nm (1:1 line/space)	Enables advanced lithography nodes

(5) Chemistry & Applications

By Metal type

Si-based MOR	Zr-based MOR	Hf-based MOR	Others
(Polysilessquioxane, Siloxane)	(Zirconium compounds)	(Hafnium compounds)	(Al, Ti, Ta, etc.)

By application



Advanced logic (< 7 nm node)	DRAM/3D NAND patterning	Interconnect hardmask/Low-k patterning	MEMS/Sensors/ optoelectronics

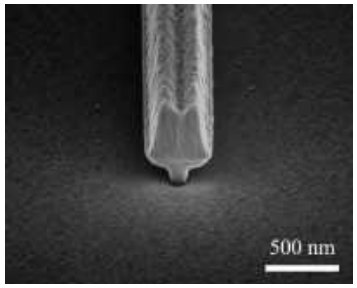
(6) Key Takeaway

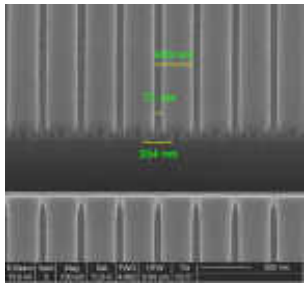
MORs combine the best of inorganic strength and organic processability, delivering unmatched etch resistance, thermal stability and performance for next-generation semiconductor patterning

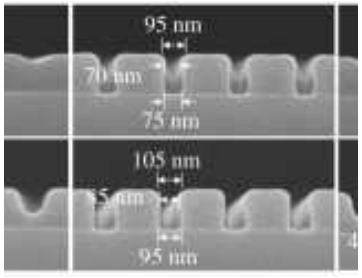
Electron Beam Photoresist (For EBL)

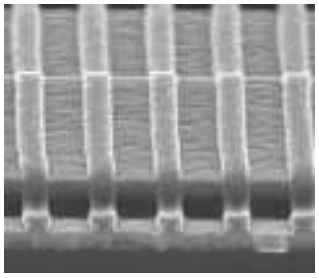
Electron-beam resist is an electron-sensitive material used for direct-write, high-resolution nanoscale patterning and photomask fabrication.

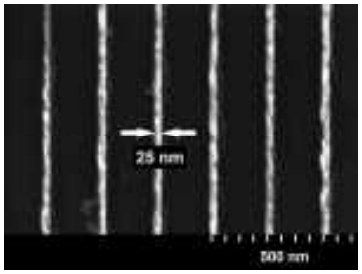
Resist Type	Tone	Mechanism	Typical Example	Application
PMMA	Positive	Electron-induced chain scission	PMMA	R&D, nanolithography
ZEP	Positive	Polymer chain scission	ZEP520A	High-resolution patterning
HSQ	Negative	Electron-induced cross-linking	HSQ	Sub-10 nm patterning
CSAR	Positive	Chemically amplified / high sensitivity	CSAR 62	High-resolution EBL
CAR E-beam resist	Positive/negative	Acid-catalyzed reaction	Various	High-throughput EBL











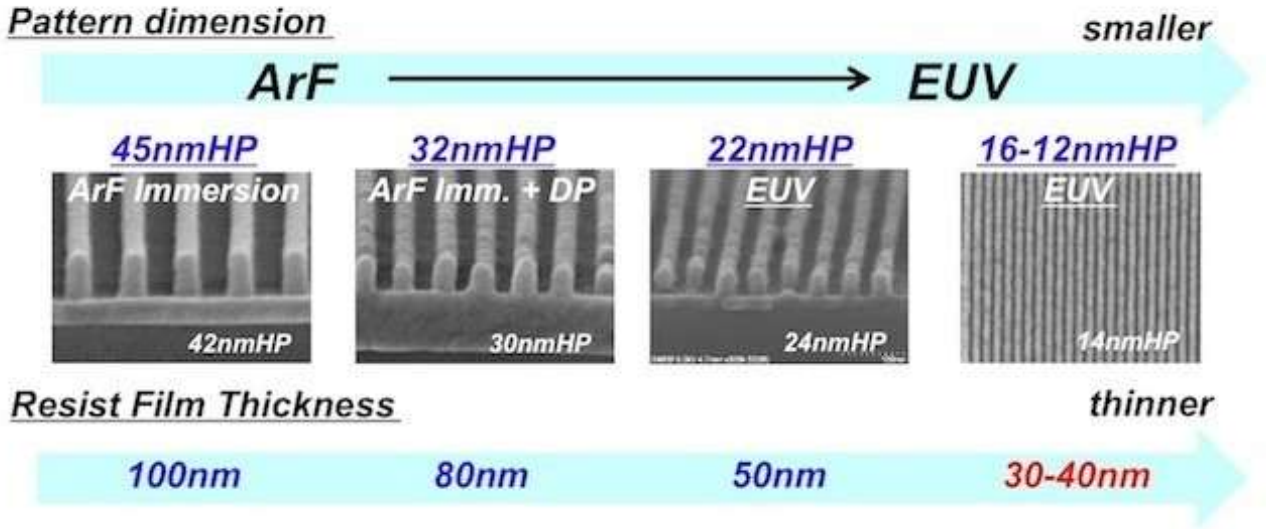
Extreme Ultraviolet (EUV) Photoresist

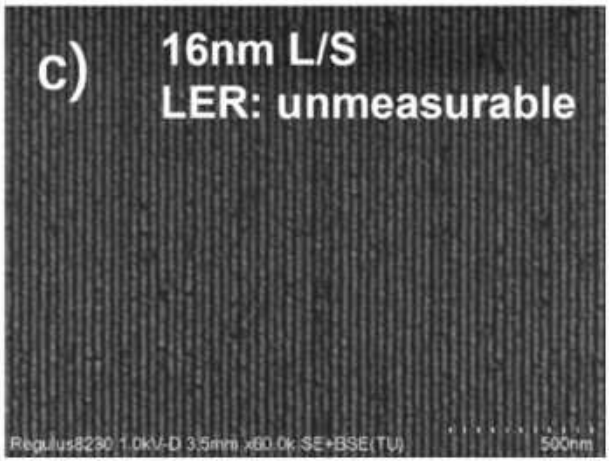
- EUV photoresists: high-energy EUV photons generate photoacid (PAG).
- Acid-catalyzed deprotection increases developer solubility of exposed region.
- Chemical amplification enhances sensitivity and high-resolution patterning.
- Scaling challenges: resolution, sensitivity, and pattern fidelity

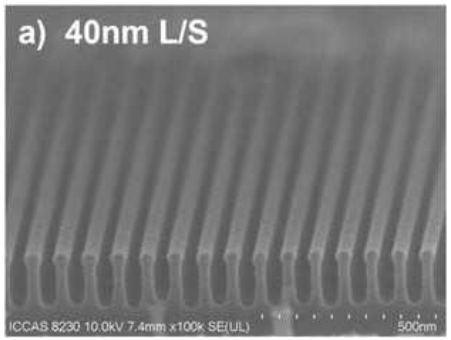
Type of EUV resist	Principle	Resolution potential	Design and optimization	Processing challenges
Chemical Amplified Resist (CAR)	Photoacid generator (PAG) absorbs the energy and undergoes a photolysis reaction	Sub-10 nm resolution	PAG release acid upon exposure to light, initiating the acid-catalyzed reactions	Pattern collapse Line-edge roughness Stochastic effects
Inorganic EUV Photoresists (Metal Oxide Resist, MOR)	Organic polymers that undergo a chemical change when exposed to light	Sub-5nm; High-NA EUV at 2nm and below	It covert the incoming EUV photons into the necessary energy for the desired photochemical or physical reactions	Etch resistance Sensitivity and dose requirements Thermal stability
Non-chemical Amplified resists (NCAR)	NCAR uses mechanisms like cross-linking or phase change to alter its solubility properties	Sub-20 nm resolution	It undergo controlled physical changes upon exposure, leading to the desired pattern formation	Resolution Sensitivity Contrast and latent image formation
Hybrid EUV photoresist	Combination of both organic and inorganic materials to enhance lithographic performance	Sub-10 nm resolution	Optimizing the ratio of components, compatibility between the both materials and interactions	Adhesion Compatibility Pattern transfer

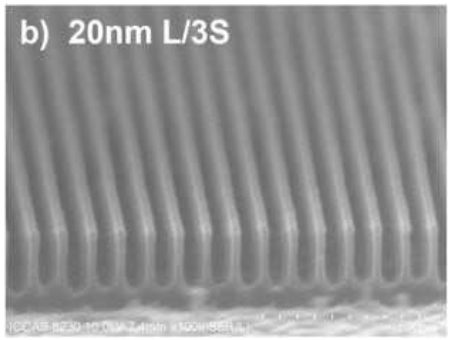
https://iebrain.com/advancing-semiconductor-processing-with-novel-extreme-uv-photoresist-materials/

Extreme Ultraviolet (EUV) PR Patterns





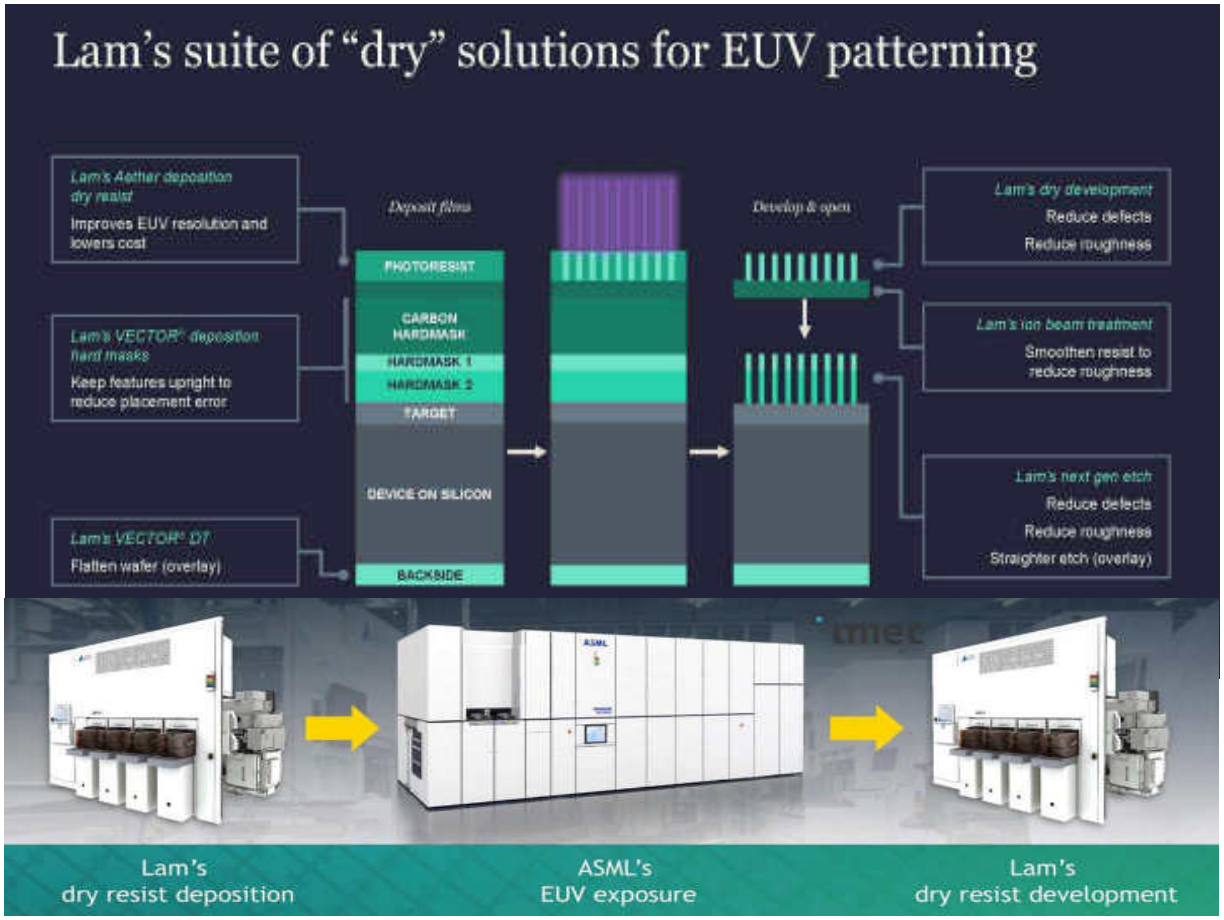


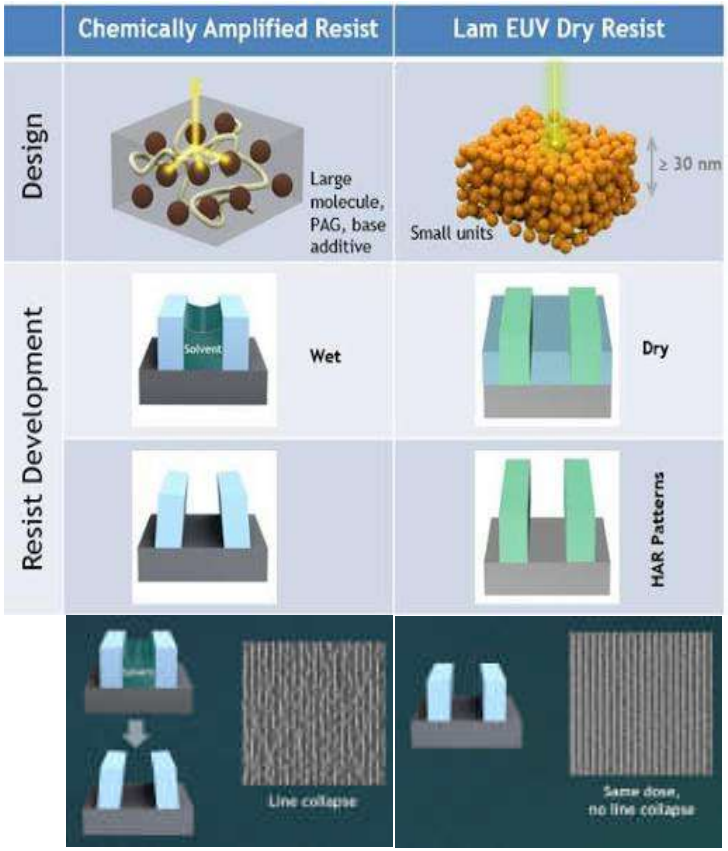


DOI: 10.1039/d4lp00136b

https://www.jsrmicro.com/electronic-materials/lithography/euv

Dry Resist & Dry Development for EUV Pattern

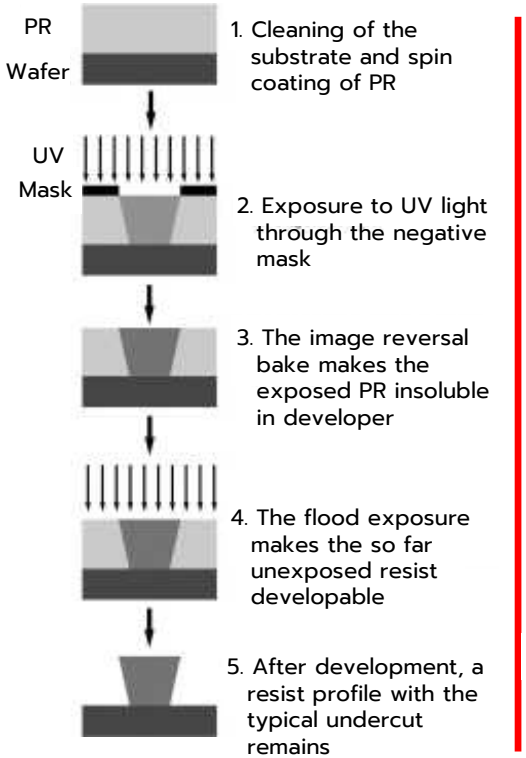




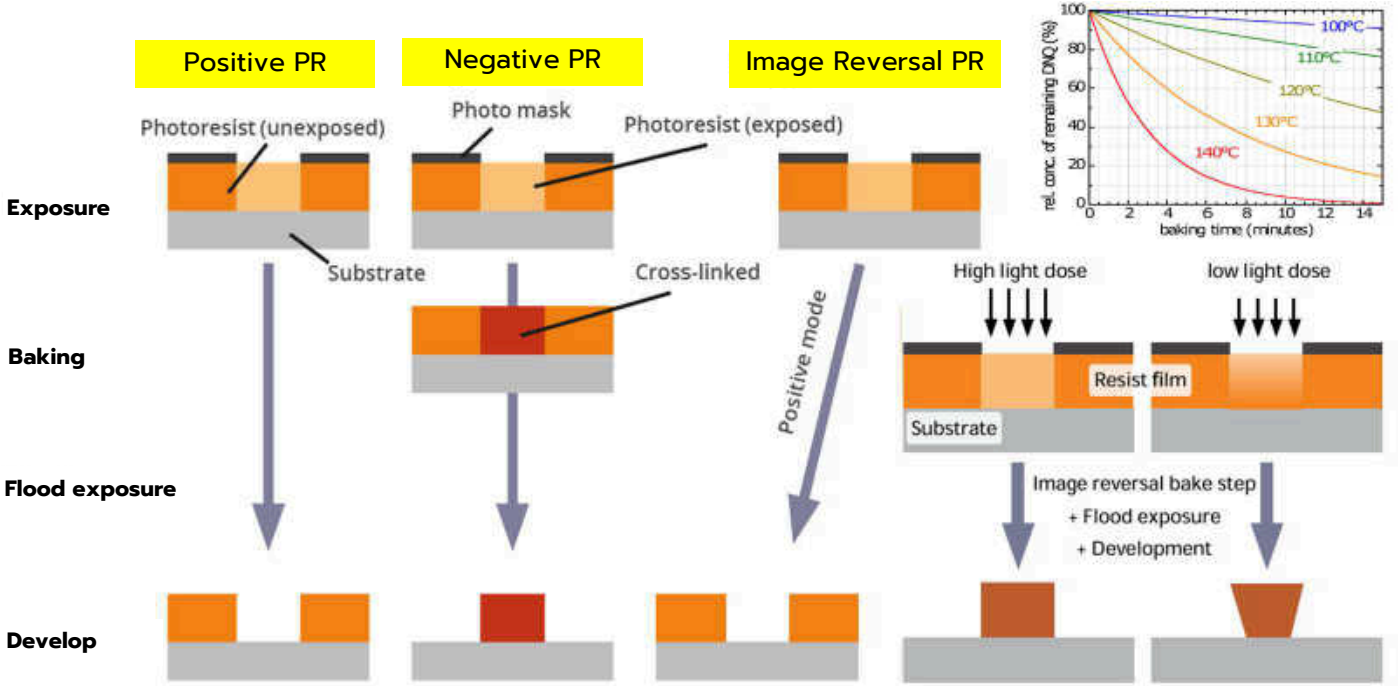
https://newsroom.lamresearch.com/everything-you-need-to-know-about-aether?blog=true
https://newsletter.semianalysis.com/p/lam-research-lrcx-dry-deposit-and

Image Reversal Photoresist (For Metal Lift-off)

Image reversal photoresist (IR PR) such as AZ® 5214E is a photoresist that can reverse the pattern polarity from positive-tone to negative-tone using a **special bake and flood-exposure process**. a negative-tone image and **undercut profile** using a positive-resist-type process, making it highly suitable for metal lift-off.



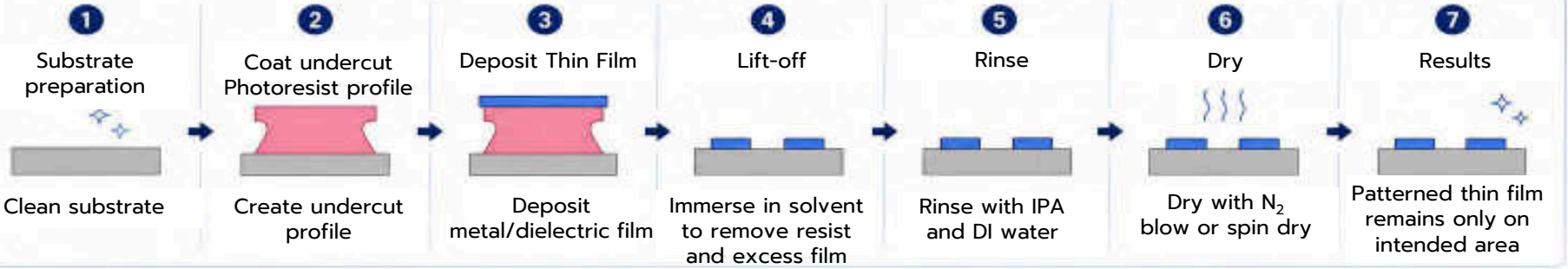
1. Cleaning of the substrate and spin coating of PR
2. Exposure to UV light through the negative mask
3. The image reversal bake makes the exposed PR insoluble in developer
4. The flood exposure makes the so far unexposed resist developable
5. After development, a resist profile with the typical undercut remains



https://www.microchemicals.com/dokumente/application_notes/image_reversal_resists.pdf

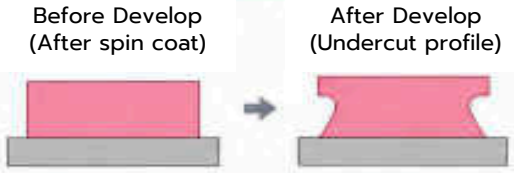
Photoresist Lift-off Process

PROCESS FLOW (COMMON TO ALL)



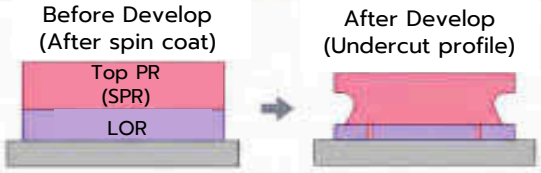
RESIST STRATEGIES FOR UNDERCUT PROFILE

SINGLE-LAYER PHOTORESIST



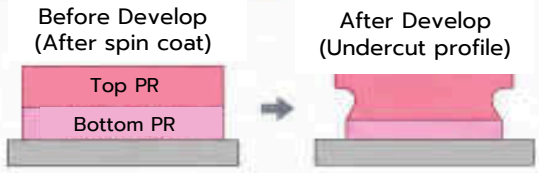
- Use thick positive PR and optimize develop time to form undercut
- Example: AZ nLOF, S1813 (thick)

LIFT-OFF RESIST (LOR)



- LOR (Lower is more soluble in developer, creating undercut under the top PR
- Example: LOR 3A + S1813

DOUBLE-LAYER PHOTORESIST



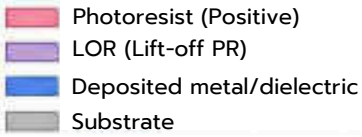
- Bottom resist is developed faster to form undercut beneath the top resist
- Example: S1813 (bottom) + S1813 (top) with controlled develop time

PROS

- Simple and widely used for MEMS fabrication
- Enables clean, residue-free metal/dielectric patterns
- Compatible with a wide range of deposition methods (e-beam, sputter, evaporation)
- Cost-effective and scalable

CONS

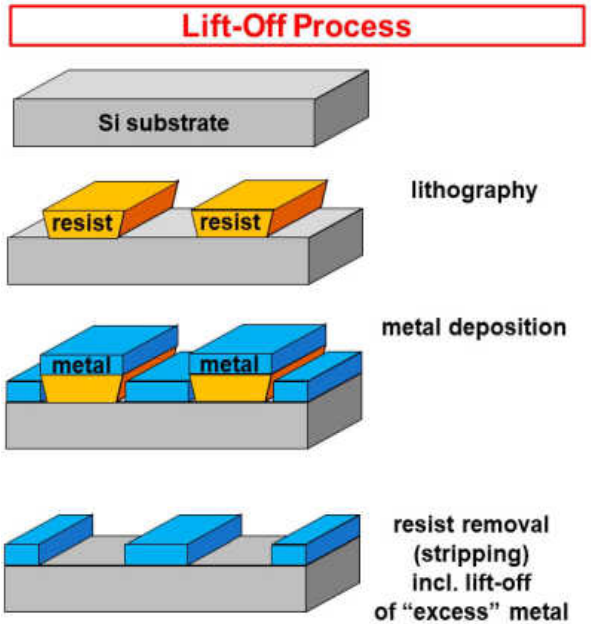
- Requires good control of undercut and film thickness
- Poor lift-off if undercut is insufficient or film is too conformal/thick
- Not suitable for extremely high step coverage (very conformal) films
- Residual resist or scumming if solvent agitation/develop is inadequate



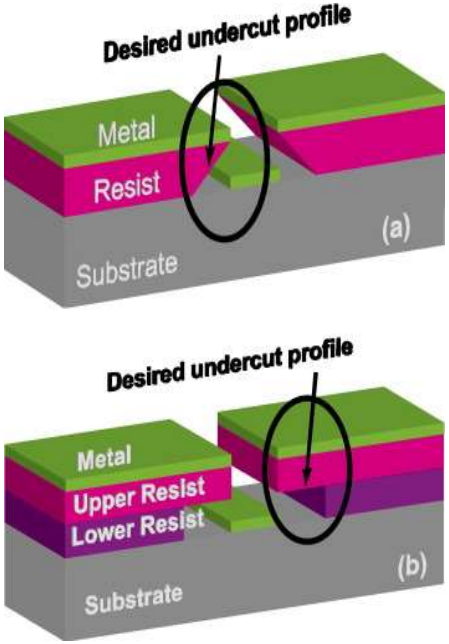
Lift-off Photoresist (LOR) & Bi-layer PR

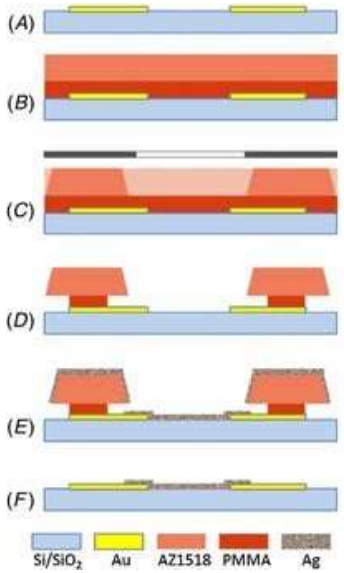
Lift-off photoresist (LOR / lift-off resist) is a resist system specifically designed to create an undercut or re-entrant sidewall profile for metal lift-off.

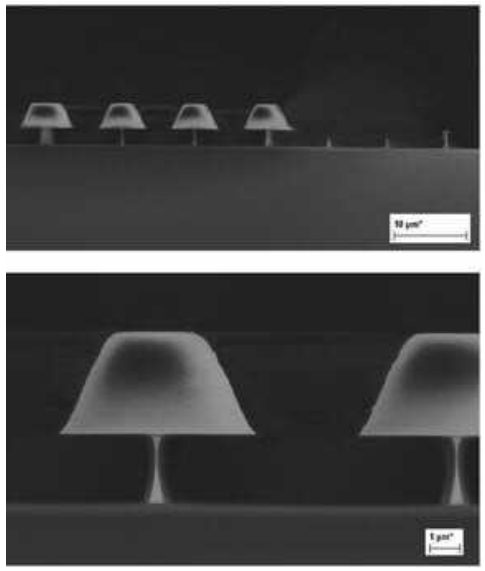
Single Layer LOR



Bi-Layer LOR







https://www.sciencedirect.com/science/article/pii/S0167931709006753 DOI: 10.1088/0960-1317/24/1/015001

Image Reversal PR vs Lift-Off Resist

Image reversal PR creates undercut by changing resist polarity; LOR creates undercut through a dedicated lift-off layer and differential development.

Feature	Image Reversal PR	Lift-Off Resist (LOR)
Main purpose	Reverse image polarity	Metal lift-off
Resist system	Usually single layer	Often bi-layer
Tone	Positive → negative image	Underlayer + imaging PR
Undercut	Created by reversal process	Controlled undercut by differential development
Typical flow	Exposure → reversal bake → flood expose	LOR → PR → expose → develop
Typical products	AZ 5214E	LOR / PMGI
Best application	Lift-off, pattern reversal	Clean, high-yield metal lift-off

Thick Photoresist (5- 200 μmT)

For HAR patterns with excellent pattern fidelity, mechanical strength and chemical resistance

(1) What is thick photoresist?



Thick photoresist is the same material as conventional photoresist but applied in much greater thickness to form robust high-aspect-ratio (HAR) structures

It provides excellent pattern fidelity, mechanical strength and chemical resistance during subsequent process

(2) Key Benefits

High-aspect-ratio (HAR) patterning
Enables deep and vertical structures

Excellent Etch Resistance
Withstands aggressive etching during process steps

Planarization Capability
Helps fill topography and achieve uniform surfaces

Process Robustness
Offers good mechanical strength and handling stability

(3) Applications

Deep Silicon Etching (DRIE)



Used as an etch mask for deep trenches, TSVs, and MEMS structures

Wafer Level Packaging (WLP)



Defines thick mold cavities and redistribution layer openings

MEMS



Fabricates HAR structures such as beams, comb drives, and micro-channels

Power Devices



Enables isolation trenches and vertical structures in power semiconductors (IGBT, MOSFET)

Photonic Devices



Used for defining waveguides, grating, and other photonic structures

(4) Typical Thickness Range



- Thickness depends on application requirements
- Higher thickness → higher aspect ratio capability

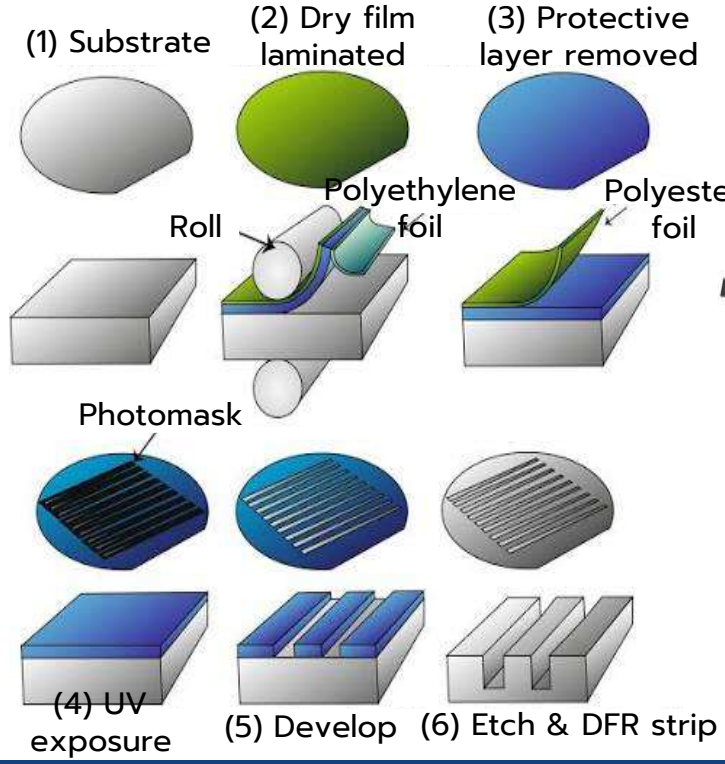
(5) Common suppliers

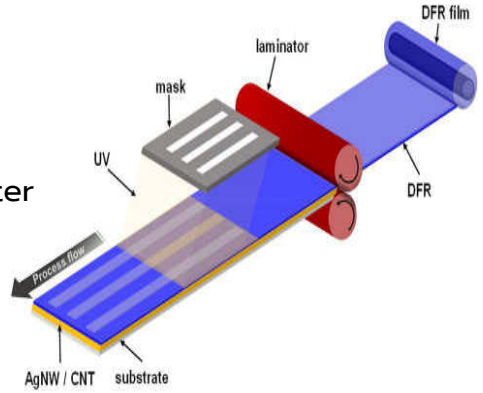


- Positive-tone thick PR (AZ-P4620, SPR220, Microresist Technology, etc.)
- Negative-tone thick PR (SU-8, SLAR, etc.)
- Dry film thick PR

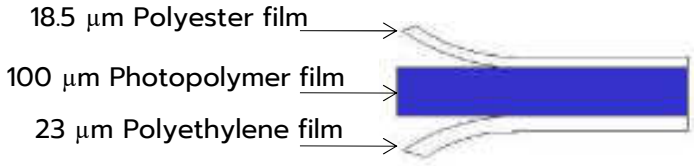
Dry Film Photoresist (DFR)

Dry film photoresist is one kind of negative photoresist comprises with three layers which are Polyester, Photopolymer and Polyethylene. The total film thickness is 30 um to 100 um. The method is use to apply dry film PR to the substrate is roller coating or Laminate dry film.











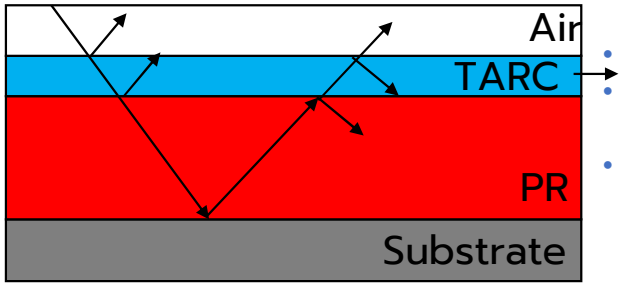
https://kayakuam.com/product/structsure/su-8-dry-film

TARC & BARC Materials

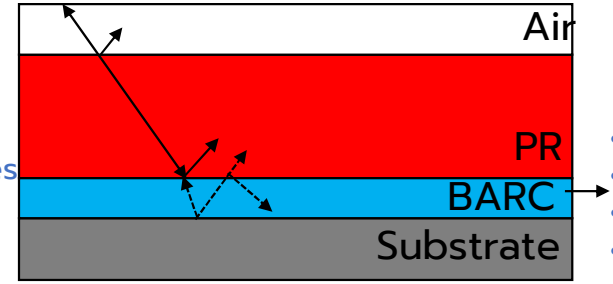
TARC and BARC in lithography process

Light reflected by the substrate can cause

- Decreased **exposure latitude** (EL) and **depth of focus** (DOF)
- Sharp edges patterns can scatter reflected light, leading to **notching** and **standing wave**
- Wafer topography → localized thickness variation → **CD uniformity due to swing curve effects**
- **Iso/dense bias**



- DIW-based solution
- Surfactants + UV additives
- alcohol-based mixtures



- SiN
- Poly(hydroxystyrene)
- SiON
- α-C

TARC (Top antireflective coating) → 30-100 nmT

use the refractive index and thickness of two films (resist and TARC) to control reflectivity through destructive interference only.

Benefit

- Simplified process

Drawback

- Environmental hazard (PFAS)
- No reflective notching and topography control
- Intermixing can affect photoresist
- Not optimum n value to reduce thin film interference

BARC (Bottom antireflective coating) → 20-200 nmT

use the refractive index, thickness and absorption of light to control substrate non-reflective.

Benefit

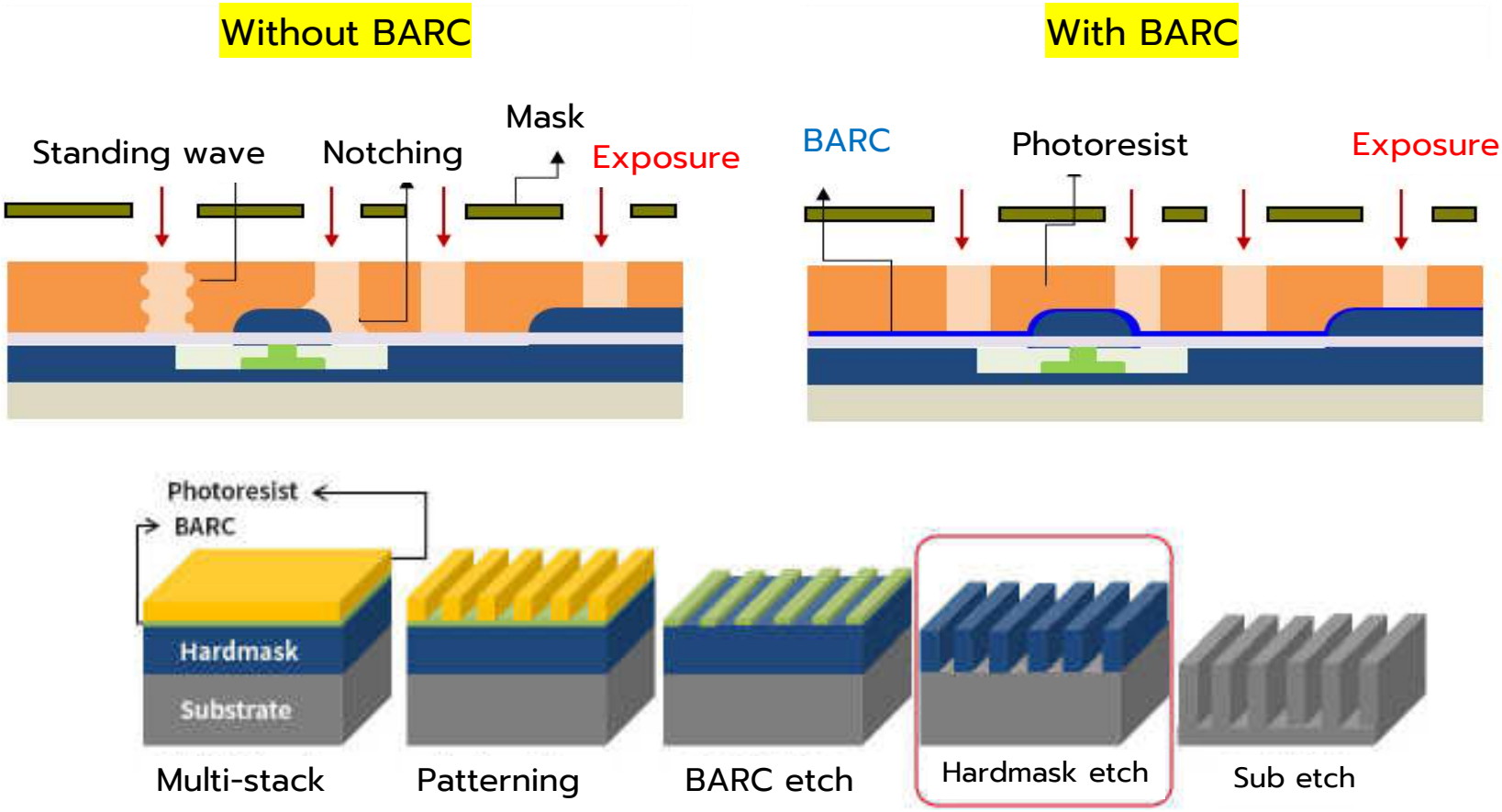
- Better reflection control and process stability
- Larger process windows (good CD control) + PFAS-free
- Compatible with PR waste drains
- Reflective notching and topography control
- No etch open for wet developable BARCs

Drawback

- Requires etch open step for dry BARCs

https://www.brewerscience.com/products/tarc-vs-barc/

Lithography with BARC layer



https://chemknock.com/view.do?no=194&pgMode=view&dvsn=pdt&idx=36871

BARC (Bottom Anti-Reflective Coating) materials

Organic BARC → spin-coated polymer/chromophore formulation → most relevant to lithography chemicals
Inorganic BARC → SiON / SiN_x deposited by CVD/PVD → more appropriately considered a thin-film process material, rather than a wet chemical.

Organic polymer + chromophore	Organic BARC	i-line / KrF / ArF	Most common spin-on BARC; absorbs reflected UV light
Polyimide-based / aromatic polymer	Organic BARC	i-line / KrF	Aromatic structure provides UV absorption
Polyester / acrylic-based polymer	Organic BARC	KrF / ArF	Tunable optical properties (n, k)
Si-containing polymer (Si-BARC)	Organic/inorganic hybrid	ArF / multilayer lithography	Good etch selectivity; useful for advanced pattern transfer
SiON (Silicon Oxynitride)	Inorganic BARC	KrF / ArF	Deposited ARC; optical properties can be tuned
SiN_x (Silicon Nitride)	Inorganic ARC	Selected processes	High durability; optical constants controlled by deposition
TiN	Inorganic ARC / hard mask	Metal/interconnect lithography	Can combine anti-reflective and hard-mask functions

TARC (Top Anti-Reflective Coating) materials

TARC (Top Anti-Reflective Coating) is also a **formulated coating**, rather than one specific chemical. Unlike BARC, TARC is applied **on top of the photoresist** and therefore must be compatible with the PR and subsequent development/rinse.

Fluorinated polymer	Organic polymer	KrF / ArF	Low refractive index; reduces reflection at PR surface
Fluorinated acrylic polymer	Organic polymer	ArF / advanced lithography	Good optical transparency and low surface energy
Poly(acrylic acid)-based polymer	Water-soluble polymer	i-line / KrF	Can be removed during aqueous development/rinse
Water-soluble polymer blends	Organic polymer	KrF / ArF	PR-compatible; easy removal
Surfactant-containing polymer	Formulated TARC	Various lithography	Controls surface/interface properties
Fluoropolymer topcoat	Specialty TARC/topcoat	ArF immersion	Low surface energy; helps control water–resist interaction

BARC vs. TARC Materials — Lithography

Item	BARC	TARC
Full Name	Bottom Anti-Reflective Coating	Top Anti-Reflective Coating
Position	Below PR	Above PR
Material Type	Organic polymer-based or inorganic ARC	Typically polymer-based coating
Composition	Polymer + absorbing chromophore + solvent/additives	Polymer + optical-control additives + solvent
Chemical Formula	No single formula; proprietary formulation	No single formula; proprietary formulation
Main Function	Absorbs light reflected from substrate	Reduces reflection from PR top surface
Key Benefits	Reduces standing waves, reflective notching & CD variation	Improves CD/profile control and reduces top-surface reflection
Typical Solvent	PGMEA, PGME or proprietary solvent*	Water- or solvent-based formulation*
Application	Spin coat → bake → PR coat	PR coat → TARC coat
Removal	Usually plasma etch / BARC open	Typically removed during/around development or dedicated rinse, depending on formulation
Typical Lithography	i-line, KrF (248 nm), ArF (193 nm)	i-line/KrF/ArF depending on resist stack; topcoat concepts also used in immersion lithography
Front-End Use	★★★ Very common	★★ Process-dependent
Advanced Packaging	★–★★ Process-dependent	★ Limited/process-dependent
Safety Concern	Depends on formulation; solvent-based products may be flammable and irritating	Depends on formulation; check product-specific SDS
Storage / Handling	Controlled temperature; sealed container; avoid contamination	Controlled temperature; sealed container; follow supplier specification

Polyimide (PI) & Polybenzoxazole (PBO)

Polyimide (PI) for Patterning

Excellent thermal stability, mechanical strength, chemical resistance

(1) General Process Flow



1. Coat
PI solution is spin-coated onto the wafer to form a uniform film

2. Expose
UV light passes through a mask to expose the PI film

3. Bake (Imidization)
Thermal bake converts the PI film to imide form, improving stability and properties

4. Develop/Etch
Exposed or unexposed region are removed (Depending on PI type) to form the pattern

5. Result
PI pattern remains as an etch mask or functional layer for subsequent process

PI types for patterning
- Positive-tone PI: exposed area is removed
- Negative-tone PI: exposed area remains

(2) Key Properties

High Thermal Stability
Stable at high temp (> 350 °C)
For demanding semiconductor process

Excellent Chemical Resistance
Resistant to acids, bases, solvents, and plasma environments

Mechanical Strength & Adhesion
High film strength with excellent adhesion to various substrates

Good Dielectric Properties
Low dielectric constant and low moisture absorption

Pattern Fidelity
High resolution and good dimensional stability for fine patterning

(3) Applications in Semiconductor



Interlayer Dielectric (ILD)
Used as low-k dielectric material between metal layers for interconnect insulation



Passivation Layer
Protects devices from moisture, contaminants, and mechanical stress



Planarization Layer
Provides smooth and flat surfaces for multi-layer fabrication



Cu Pillar Growth
Serves as a passivation and diffusion barrier layer with openings for Cu electroplating (ECP) to form Cu pillars for 3D IC and advanced packaging



Etch stop/Barrier Layer
Acts as an etch stop or diffusion barrier in advanced architectures



Wafer-level packaging (WLP)
Used as an insulation and protection layer in Fan-out and 3D packaging



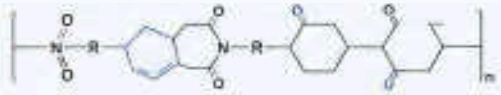
RDL Insulation Layer
Insulates redistribution layers in advanced packaging technologies

(4) Advantages

- Enables high-resolution patterning with excellent dimensional control
- Withstands harsh process conditions
- Supports advanced device scaling and 3D integration
- Versatile for multiple roles in front-end and back-end processes

(5) Typical PI material systems

- Photosensitive PI for patterning (e.g. positive/negative-tone)
- Non-photosensitive PI for insulation and protection
- Semi-aromatic and aromatic PI for high-temperature stability
- Low-k PI for advanced interconnect applications



Polybenzoxazole (PBO)

PBO = Photosensitive Low-k Dielectric + Passivation Material for Advanced Packaging

Material	PBO – Polybenzoxazole, high-performance aromatic polymer
Material Type	Polymer dielectric / passivation material; available as photosensitive PBO (PS-PBO)
Key Properties	Low-k, high thermal stability, good mechanical strength, chemical resistance, low moisture absorption
Main Application	Passivation and interlayer dielectric (ILD) in advanced packaging
Typical Uses	RDL, Cu pillar, bumping, WLP/FOWLP, 2.5D/3D packaging
Lithography Role	Photosensitive PBO can be directly patterned by photolithography, reducing the need for a separate PR layer
Typical Process	Coat → Soft Bake → Exposure → Develop → Cure
Developer	Often TMAH-based for compatible photosensitive formulations; chemistry is formulation-dependent
Curing	High-temperature curing converts the precursor into the final robust PBO film; exact temperature depends on formulation
Key Advantage	Low-k + high thermal/mechanical reliability + photopatternability
vs. Polyimide (PI)	Generally attractive for lower dielectric constant and lower moisture uptake; actual properties depend on formulation
Main Concern	High-temperature cure, adhesion/stress control, chemical compatibility and complete development/residue removal

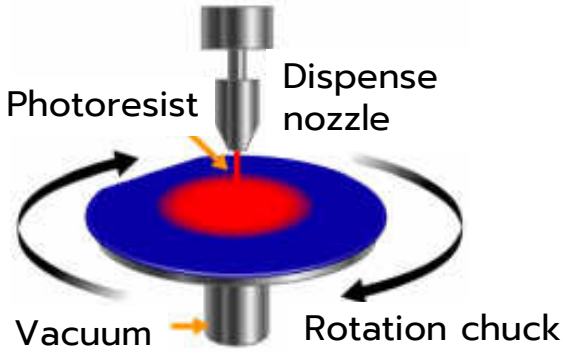
Comparison: PI vs. PBO vs. PR vs. SiO_2

PR = temporary photosensitive patterning material,
PI/PBO = permanent polymer dielectric/passivation materials that can also be made photosensitive.
SiO_2 = permanent inorganic dielectric/passivation (requires PR lithography + etching to create a pattern).

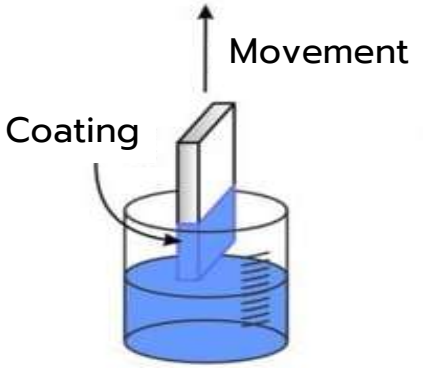
Parameter	PI (Polyimide)	PBO (Polybenzoxazole)	PR (Photoresist)	SiO_2
Material Type	Organic polymer	Organic polymer	Photosensitive polymer	Inorganic dielectric
Dielectric Layer	✅ Excellent	✅ Excellent	❌ Not normally used	✅ Excellent
Passivation Layer	✅ Yes	✅ Yes	❌ Temporary mask	✅ Yes
Photosensitive / Direct Patterning	✅ PSPI available	✅ PS-PBO available	✅ Primary function	❌ No
Permanent Layer	✅ Yes	✅ Yes	❌ Normally stripped	✅ Yes
Dielectric Constant (k)	~3–3.5	~2.7–3.2*	~3–4*	~3.9
Moisture Resistance	Moderate	Good / better than conventional PI	Low–moderate	Excellent
Thermal Stability	High	High	Low–moderate	Very high
Mechanical Flexibility	Excellent	Good	Moderate	❌ Brittle
Typical Thickness	Few–tens of μm	Few–tens of μm	<1 to >100 μm	nm to few μm
Patterning Method	Direct if PSPI; otherwise PR + etch	Direct if PS-PBO; otherwise PR + etch	Exposure + development	PR lithography + etching
Developer	Formulation-dependent	Formulation-dependent; often aqueous alkaline for PS grades	TMAH or solvent, depending on PR	N/A
High-Temperature Cure	✅ Required	✅ Required	Usually ❌	Deposition / thermal growth
RDL / Bumping	✅ Common	✅ Very common	✅ Process mask	△ Used depending on stack
Front-End IC	△ Specialized	△ Limited	✅ Essential lithography material	✅ Essential dielectric
Main Role	Permanent dielectric + passivation	Permanent dielectric + passivation	Temporary patterning mask	Permanent dielectric + passivation

PR/PI/PBO Coating Methods

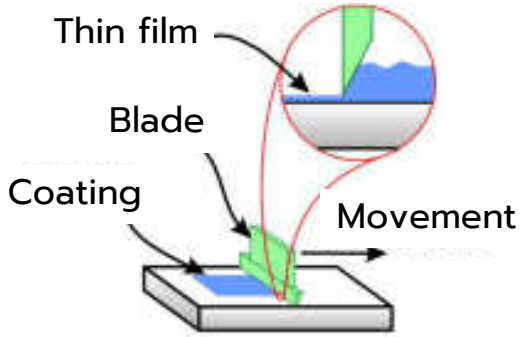
Photoresist (PR) film coating methods



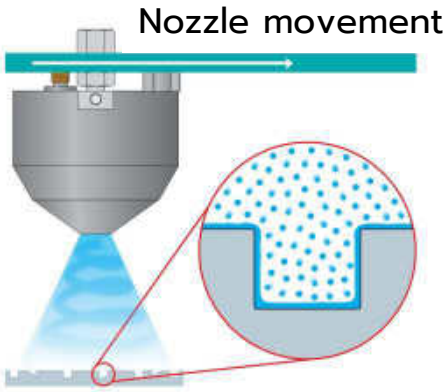
Spin coating



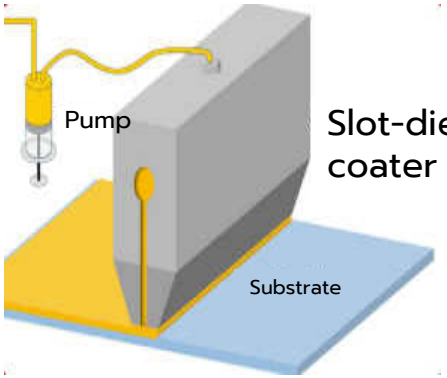
Dip coating



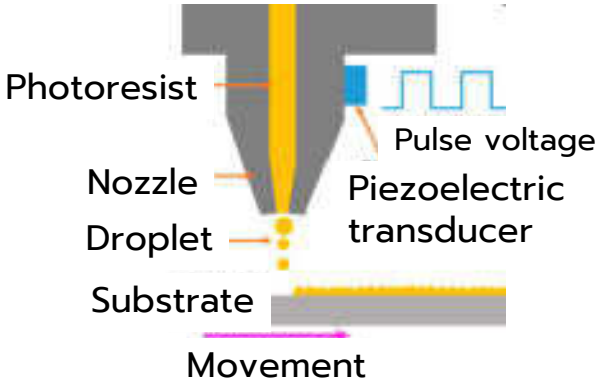
Doctor blade coating



Spray coating

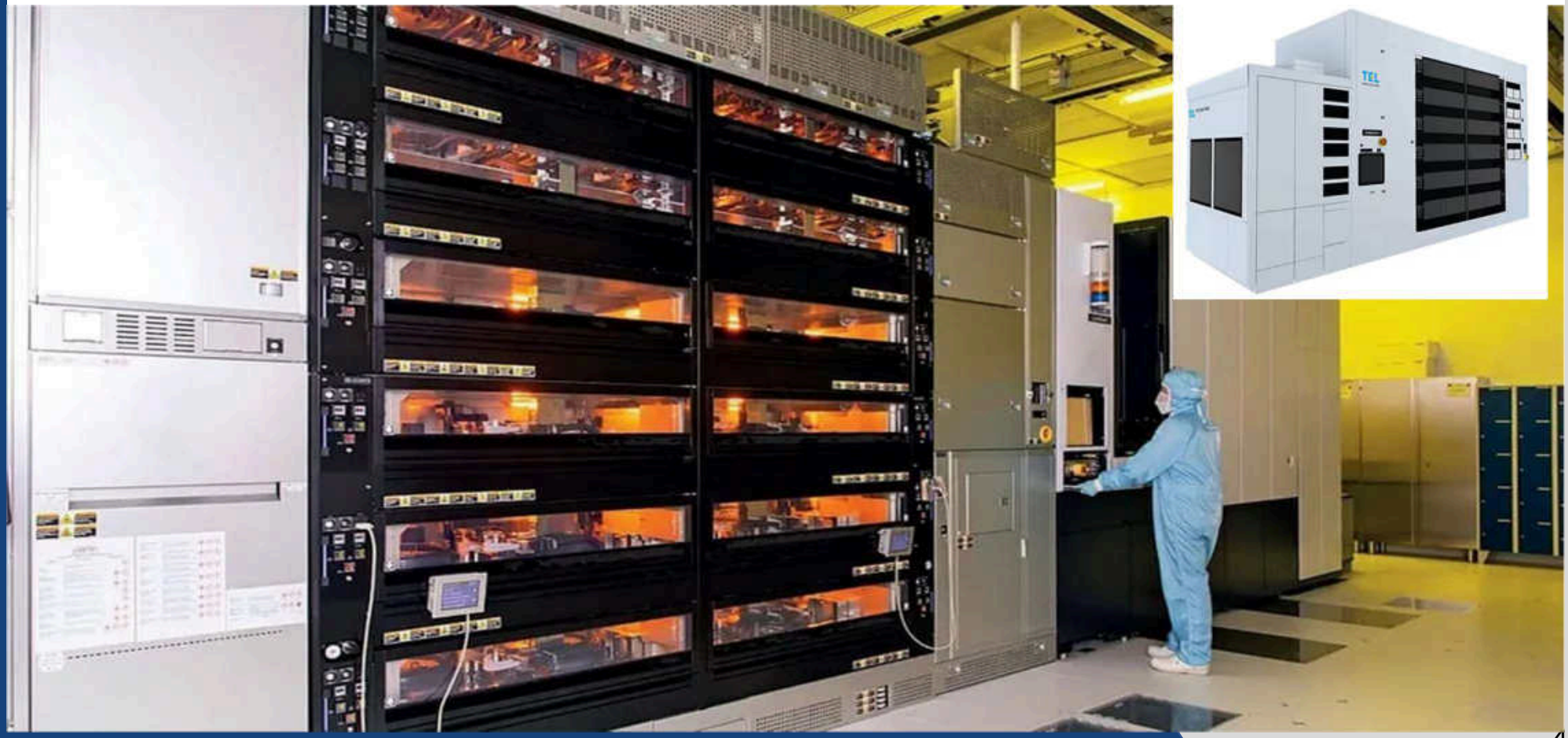


Slot-die coating



Inkjet coating

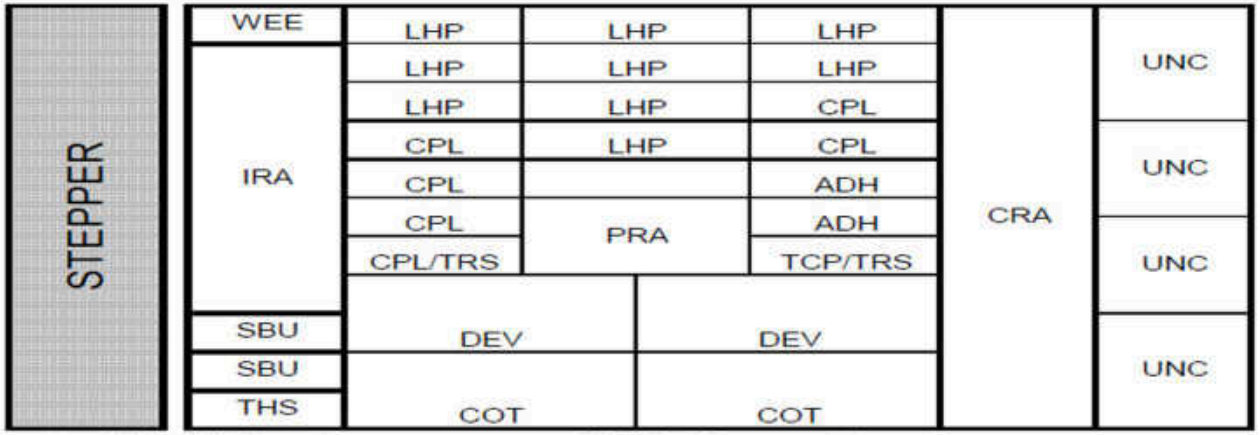
Photoresist coater/developer (C/D) track system



https://www.tel.com/product/lithius.html

Configuration of C/D track system

For HMDS, Dehydration bake, PR coating + BSR + EBE, Pre-bake, PEB, Develop, Hard Bake



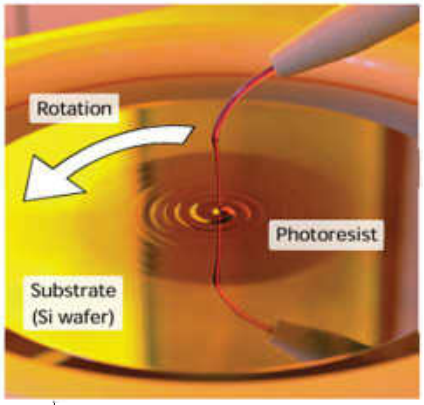
UNC: UNI Cassette
CRA: Cassette Block Robotic Arm
PRA: Process Block Robotic Arm
IRA: Interface Block Robotic Arm
TRS: Transition
TCP: Transition Chill Plate
SBU: Station Buffer Unit

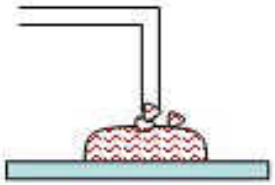
COT: Coat Process station
DEV: Develop Process Station
LHP: Low Temperature Hot Plate Process Station
ADH: Adhesion Hot Plate Process Station
CPL: Chill Plate Process Station
WEE : Wafer Edge Exposure
THS: Temporary Holding Stage

	CSB	TRS	ADH	CPL	COT	LHP	CPL	WEE	BUF	EXP	TRS	LHP	CPL	DEV	LHP	TCP	CSB
Process Temp			110	23		90	23					100	23		120		
Process Time			60	60	75	100	60	30				120	60	80	100	30	
OHT			9	6	8	6	8	10				6	6	8	6	8	
Module			2	2	2	3	2	1				3	2	2	3		
T.Time/unit			34.5	33	41.5	35	34	40				42	33	44	35		
Arm cycle			3.1	4.4	3.1	3.1	4.1					3.1	4.4	3.5	3.1		
Throughput	79																

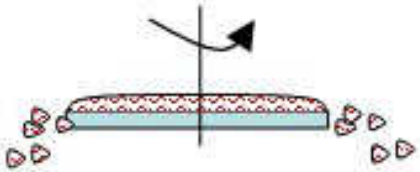
https://www.tel.com/product/lithius.html

Photoresist spin coating

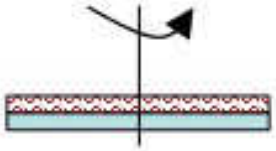




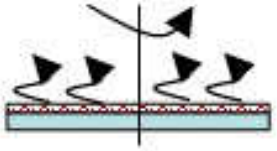
(1) Dispensing



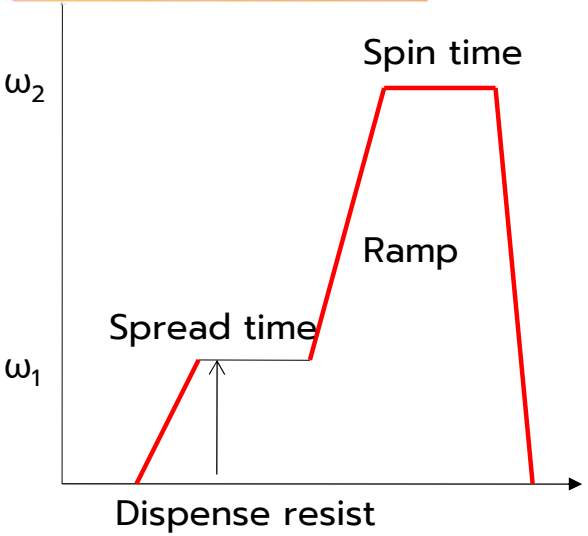
(2) Spin-up

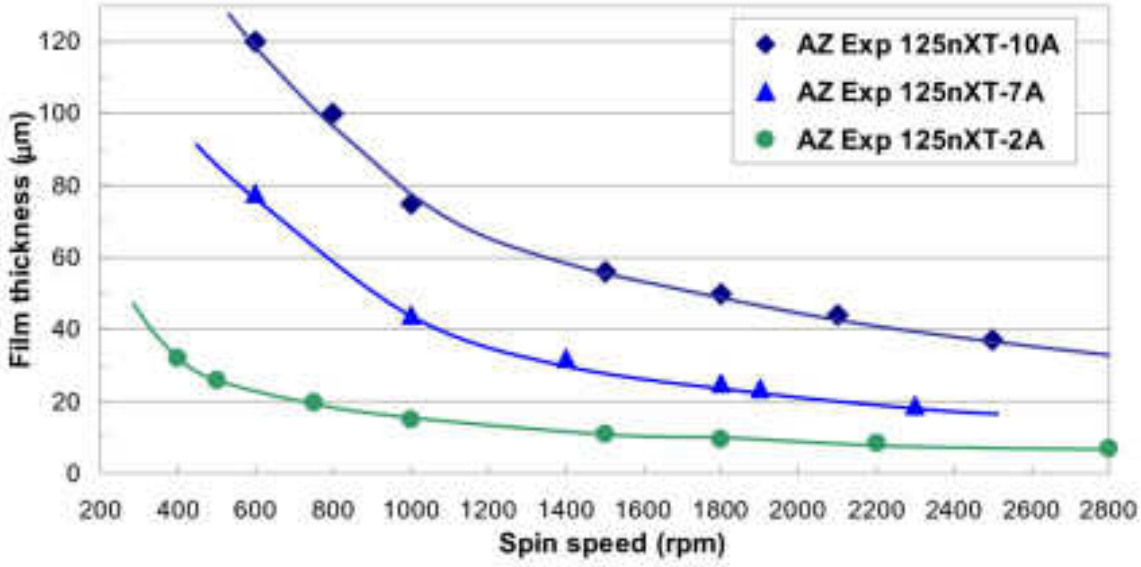


(3) Spin-off



(4) Evaporate





DOI: 10.1117/12.773986

Photoresist Coating Kinetics — Spin Coating

PR Coating Kinetics = Competition between centrifugal thinning + viscous flow + solvent evaporation → determines final PR thickness and uniformity.

↑ Spin Speed → ↓ PR Thickness
↑ PR Viscosity / Solid Content → ↑ PR Thickness
↑ Solvent Evaporation Rate → faster film solidification

1. Dispense	PR deposited on wafer	Dispense volume, viscosity, temperature	Initial PR coverage
2. Spreading	Centrifugal flow spreads PR outward	Spin acceleration, RPM, viscosity	Wafer coverage
3. Spin-off	Excess PR flows off wafer edge	Spin speed, viscosity	Rapid film thinning
4. Flow / Thinning	Radial flow continues	RPM, viscosity, surface tension	Controls film thickness & uniformity
5. Solvent Evaporation	PR solvent evaporates	Solvent volatility, airflow, temperature, humidity	Film becomes concentrated / solidifies
6. Film Stabilization	Flow becomes very small	Polymer concentration, solvent remaining	Final coated PR thickness

Main factors controlling coating kinetics

Spin Speed (RPM) ↑	PR thickness ↓
Spin Time ↑	Thickness ↓ initially, then approaches stable value
PR Viscosity ↑	Thickness ↑
Solid Content ↑	Thickness ↑
Solvent Volatility ↑	Faster drying / earlier film stabilization
Wafer Temperature ↑	Evaporation ↑
Airflow ↑	Solvent evaporation ↑; may affect uniformity
Humidity	Can affect coating behavior and some resist chemistries
Surface Energy / HMDS	Changes wetting and PR adhesion
Acceleration	Influences initial spreading and uniformity

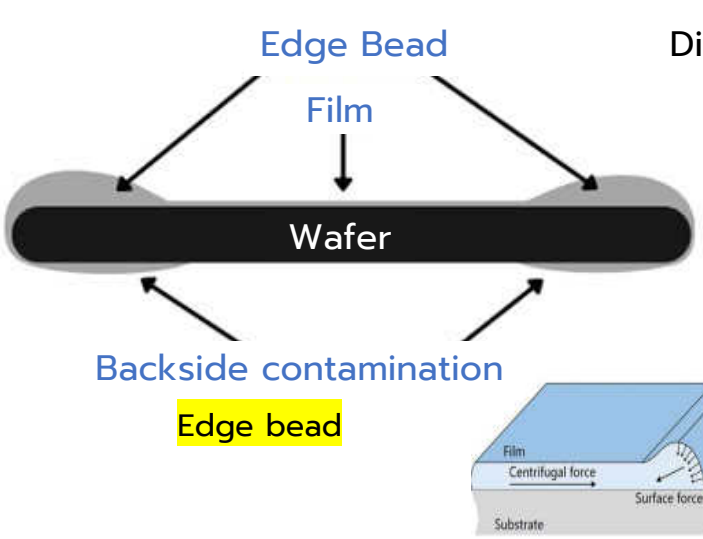
Comparison Table of (PR) film coating methods

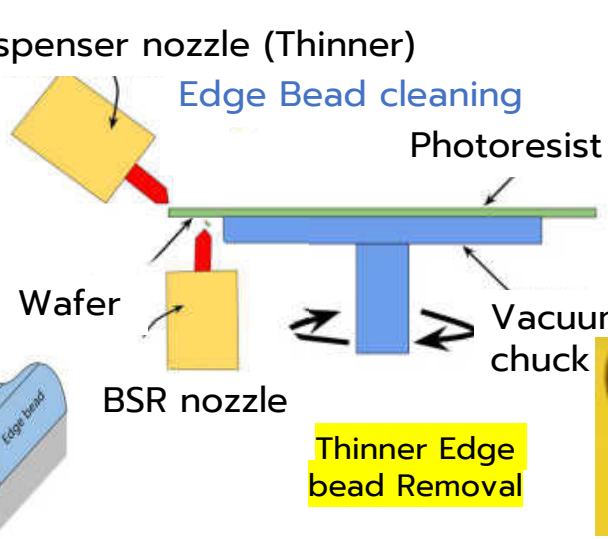
Spin coating (wafer)	50 nm–10 μm	★★★★★	★	★★★★★	Low	CMOS, photonics	Poor step coverage, edge bead
Spin coating (photomask)	50 nm–1 μm	★★★★★★	★	★★★★	Medium	Reticle fabrication (standard)	Square-edge effects, resist waste, chuck design critical
Spray coating	1–50 μm	★★★	★★★★★	★★★	Medium	MEMS, TSV	Lower CDU, droplet defects (Orange peel), roughness
Dip coating	100 nm–10 μm	★★★	★★★	★★	Low	R&D, optics	Poor thickness control, low throughput
Slot-die coating	100 nm–10 μm	★★★★	★★	★★★★	High	Panels, packaging	Edge uniformity, complex setup, sensitive to gap control
Doctor blade	1–100 μm	★★	★★	★★★	Low	Thick resist (MEMS)	Non-uniformity, streaking, edge effects
Inkjet printing	Variable	★★	★★	★★	High	Maskless lithography	Resolution limits, droplet defects
Electrophoretic deposition (EPD)	1–20 μm	★★★★	★★★★★	★★	High	TSV, 3D	Limited materials, complex chemistry

Edge bead removal (EBR) and Backside rinse (BSR)

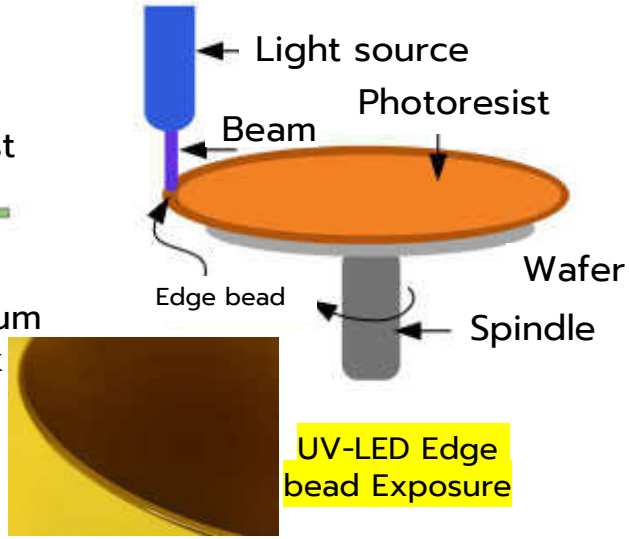
The spin coating of PR, ARC, and PI coating on wafer is often formed the unwanted accumulation of the coated material at the outer edges of the substrate in form of a bump called "**Edge Bead**". It may wrap around the "**back-side edges**" of the wafer causing particle contamination from the flak off a dry PR particle.

The unwanted residues on the outer reaches of the substrate should be removed by using a stream of high purity solvents to clean and to level the edges to the same thickness as the rest of the substrate surface area.





Thinner Edge bead Removal



UV-LED Edge bead Exposure

Propylene Glycol Methyl Ether Acetate (PGMEA)

The main purpose of EBR is to remove the thick photoresist bead formed at the wafer edge during spin coating, improving wafer handling, reducing particle/contamination risk, and avoiding problems in subsequent lithography and process steps.

Chemical Name	PGMEA – Propylene Glycol Methyl Ether Acetate
Chemical Formula	$C_6H_{12}O_3$
CAS No.	108-65-6 (commonly supplied isomer)
Physical Properties	Clear, colorless, volatile liquid; BP ~146°C, flash point ~42°C
Main Lithography Application	EBR (Edge Bead Removal) and backside resist cleaning
Other Uses	Major solvent/carrier for photoresist; equipment/nozzle cleaning depending on process
EBR Function	Dissolves and removes excess PR at wafer edge, preventing edge contamination and handling problems
Process Step	PR Spin Coat → PGMEA EBR / Backside Rinse → Soft Bake
Application Method	Dispensed through an EBR nozzle near wafer edge while wafer rotates
PR Compatibility	Commonly compatible with many positive PR systems; actual compatibility is resist-specific
Container / Storage	Sealed compatible chemical container; store in cool, ventilated flammable-liquid area
Safety Concerns	🔥 Flammable liquid/vapor; vapor inhalation and eye/skin exposure concerns
Handling / PPE	Ventilation/local exhaust, chemical-resistant gloves + goggles; keep away from heat, sparks and ignition sources

EBR & BSR Chemical Summary- Lithography

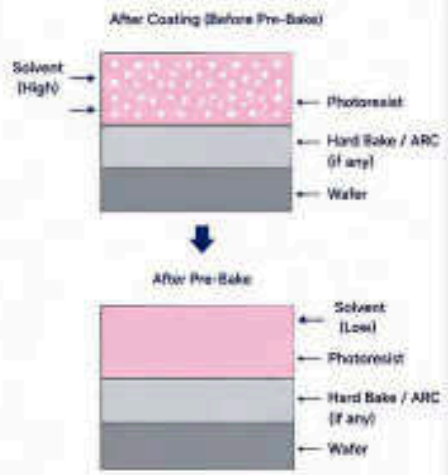
PGMEA → PGME → Ethyl Lactate → n-Butyl Acetate, with **PGMEA** being the key representative chemical for EBR/BSR.

PGMEA	Propylene Glycol Monomethyl Ether Acetate	$C_6H_{12}O_3$	★★★	★★★	Most common EBR/BSR solvent; dissolves fresh PR effectively
PGME	Propylene Glycol Monomethyl Ether	$C_4H_{10}O_2$	★★	★★	Often blended with PGMEA; controls dissolution and evaporation
EL	Ethyl Lactate	$C_5H_{10}O_3$	★★	★★	PR-compatible solvent for selected resist systems
nBA	n-Butyl Acetate	$C_6H_{12}O_2$	★★	★★	Used with certain PR formulations
EBR/BSR Solvent	Proprietary Edge-Bead-Removal / Backside-Rinse Solvent	Proprietary	★★★	★★★	Supplier-optimized blend matched to specific PR chemistry
Acetone	Propan-2-one / Acetone	C_3H_6O	△	△	Strong, fast-evaporating solvent; mainly lab/R&D
IPA	Isopropyl Alcohol / 2-Propanol	C_3H_8O	△	△	Mainly cleaning/rinsing; generally not primary PR-removal solvent

Pre-bake (after photoresist coating)

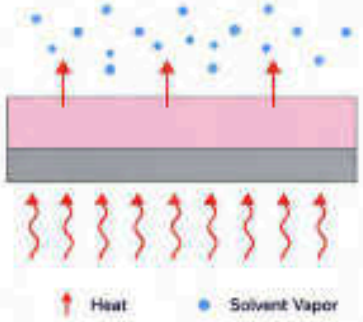
1. PURPOSE OF PRE-BAKE

- Remove remaining solvent from the PR film
- Improve PR film adhesion to the substrate
- Stabilize film to prevent swelling or boiling during exposure
- Improve dimensional control and pattern fidelity
- Reduce defects such as blistering, footing, and pattern collapse



2. HOW PRE-BAKE WORKS

Heat drives off solvent from the photoresist film.



Key Point
Controlled heating removes solvent without causing resist flow, cracking, or poor adhesion.

3. TYPICAL PRE-BAKE CONDITIONS (EXAMPLE)

Parameter	Typical Range
Temperature	90 – 120 °C (depending on resist type and thickness)
Time	60 – 120 seconds
Hotplate	Stable and level hotplate with uniform temperature
Ambient	Clean, low humidity environment

* Always follow resist vendor recommendation.

4. WHAT HAPPENS IF PRE-BAKE IS NOT PROPER?

Under-Bake (Insufficient)



- Solvent remains in film
- Swelling / boiling during exposure (T-top effect)
- Poor CD control
- Pattern defects

Over-Bake (Too Hot or Too Long)



- Resist hardens too much
- Poor adhesion
- Cracking / lifting
- Difficult development

Non-Uniform Bake (Hotplate Variation)



- Thickness non-uniformity
- CD variation across wafer
- Poor overlay

Particle / Contamination During Bake



- Particle defects
- Pinholes
- Pattern printing failure

5. BEST PRACTICES



Use correct temperature and time



Ensure hotplate uniformity and calibration



Use clean environment and proper filtration



Do not stack wafers (single wafer bake)



Follow resist vendor's recommendation

Photoresist Film Defects & Quality Concerns

1. COMMON PHOTORESIST FILM DEFECTS

DEFECT	DESCRIPTION	TOP VIEW (EXAMPLE)	CROSS-SECTION (EXAMPLE)	POTENTIAL IMPACT
1 Particle	Foreign particles on wafer or in PR film			•Print defects (missing or extra features •Yield loss
2 Pinhole	Small holes in PR film reaching to underlayer			•CD variation •Leakage/pattern defect
3 Thickness Non-uniformity	Variation of PR film thickness across wafer or within die			•CD variation •Focus/DOF sensitivity
4 Topping/ Scumming	Thin polymer layer formed on PR surface			•Poor adhesion •Pattern collapse or bridging
5 Cracking/ Crazing	Crack or craze lines due to stress or mechanical damage			•Pattern break •Reliability concern
6 PR residue/ Streaks	Organic residue or streaks after coating or bake			•Pattern defects •Poor etch profile

2. QUALITY CONCERNS (KEY PARAMETERS)

		Target
Total thickness Uniformity (TTV)	Variation of PR thickness across wafer	± 2-5%
Surface roughness (Ra)	Surface smoothness affects pattern transfer	< 1.0 nm
Particle/Defects	Particles and defects degrade yield and reliability	As low as possible
Adhesion	Good adhesion prevents peel-off and pattern failure	Strong (No peel-off)
Stress	Film stress (tensile/ compressive) affects CD and reliability	Low & well controlled
Chemistry stability	Resist chemistry stability ensure consistent performance	Stable

3. ROOT CAUSES (EXAMPLES)

Contamination
- Particles, metallic ions
- Dirty tools/wafer

Coating Issues
- Incorrect dispense
- Poor chuck vacuum
- High acceleration

Bake Issues
- Incorrect temperature
- Unevent hotplate
- Under/oven bake

Environment
- High humidity
- Airflow/vibration
- Temperature fluctuation

Resist/process
- Aged resist
- Wrong solvent
- Incompatible materials

4. HOW TO MINIMIZE DEFECTS & IMPROVE FILM QUALITY

- Maintain clean environment & proper filtration
- Optimize coating parameters (spin, dispense, acceleration)
- Control bake conditions (Temp, time, uniformity)
- Use proper resist chemistry & handling
- Monitor & Inspect regularly (Thickness, defects, adhesion)

Photoresist coating defects

Category	Defect	Definition	Root Cause	Impact on Pattern	Measurement / Detection
Adhesion-related	Peeling / delamination	Resist detaches from substrate	Poor HMDS, moisture	Pattern loss	Optical inspection, SEM
	Lifting during develop	Pattern detaches locally	Weak adhesion, stress	CD failure	SEM
Bake-related	Residual solvent	Incomplete solvent removal	Underbake	CD variation, footing	FTIR, CD-SEM
	Reticulation (cracking)	Crack network in resist	Overbake	Pattern distortion	Optical inspection
	Skin formation	Hardened top layer	Improper bake profile	Development non-uniformity	SEM
Film thickness/ uniformity	CDU (Thickness non-uniformity)	Variation of resist thickness across wafer	Spin instability, viscosity drift, temp variation	CD shift, EPE	Ellipsometry, reflectometry
	Striation	Radial thickness oscillation	Spin dynamics, solvent evaporation instability	CDU degradation	Optical inspection, thickness mapping
Edge-related	Edge bead	Thick resist at wafer edge	Surface tension during spin	Focus error, exposure non-uniformity	Optical inspection
Coating defects	Streaks / swirl marks	Radial flow marks	Poor dispense, airflow turbulence	CD variation	Optical inspection
	Bubbles / voids	Air trapped in resist film	Poor degassing, dispense issue	Missing pattern	Optical inspection, SEM
	Pinholes	Microscopic holes in resist	Contamination, solvent issues	Etch penetration → killer defect	Bright-field inspection, SEM
	Resist mottle (micro non-uniformity)	Small-scale thickness variation	Phase separation, solvent evaporation	LER, CD variation	Scatterometry, SEM
Particles / contamination	Embedded particles	Particles inside resist film	Dirty chemicals, environment	Pattern defect (bridge/open)	Bright-field / dark-field inspection
	Surface particles	Particles on resist surface	Airborne contamination	Lithography defect	Inspection tools (BF/DF)
Optical interaction	Standing wave	Vertical interference pattern	Reflective substrate	CD variation	CD-SEM
	Swing effect	CD variation vs thickness	Optical reflectivity	CDU issue	Scatterometry

Defects of the Photoresist Film

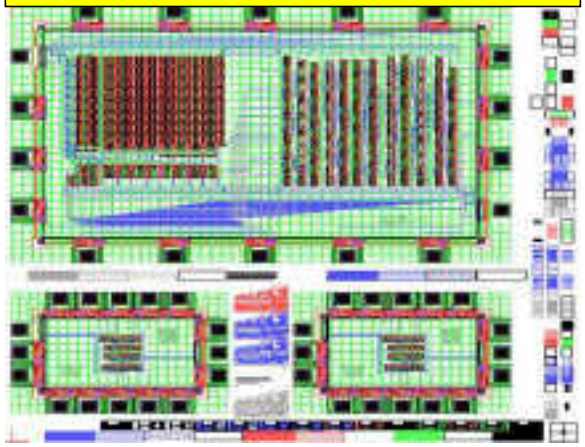
Defect	Description	Typical Cause
Thickness Non-uniformity	PR thickness varies across wafer	Spin speed, viscosity, airflow
Edge Bead	Thick PR at wafer edge	Surface tension during spin coat
Comet / Streak	Tail-shaped coating defect	Particle or resist agglomerate
Striation	Radial / spiral thickness lines	Solvent evaporation, airflow
Pinholes	Small openings in PR film	Particle, poor wetting, bubbles
Bubble / Void	Trapped air in resist	Dispense or degassing issue
Dewetting	PR retracts from wafer surface	Surface contamination, poor adhesion
Center / Dispense Mark	Defect at wafer center	Dispense condition
Backside / Edge Contamination	PR on wafer edge or backside	EBR failure, splash
Particles / Gel	Local bumps or defects	PR contamination, poor filtration
Crack / Peeling	PR film cracks or delaminates	Stress, thick PR, poor adhesion
Mottle / Cloudiness	Non-uniform film appearance	Solvent, humidity, bake issue



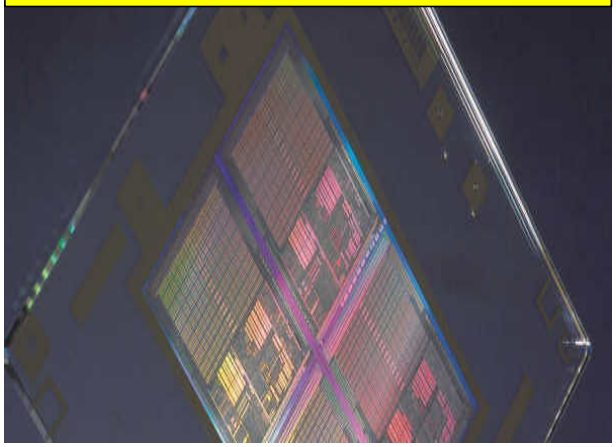
Photomask Making Process

Photomask and Reticle

CAD layout



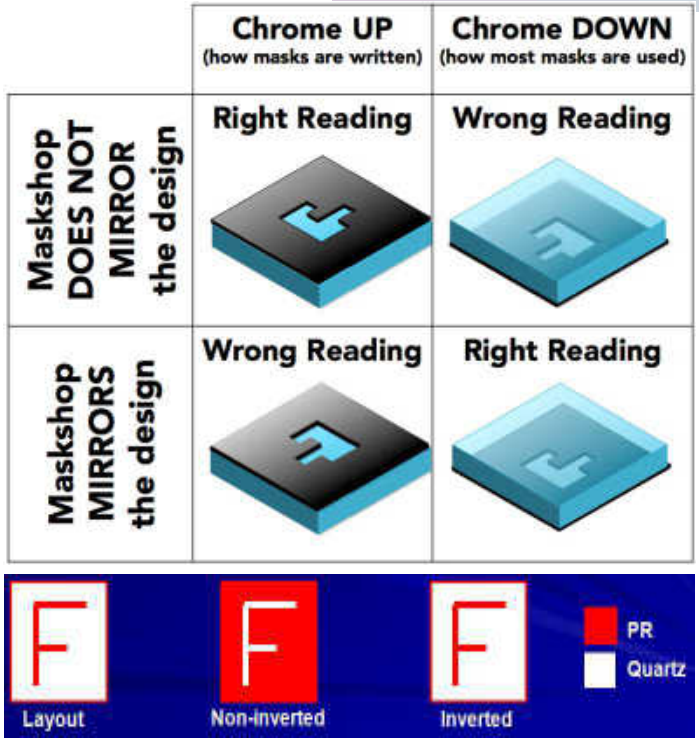
Photomask, Mask, Reticle



- Blank mask: chrome-on-quartz (COG).
- Typical size: 3, 4, 5, 6, 9 sq.in. (for IC manufacturing).
- Area: transparent (clear) and opaque formed by lithography and etching.

Mask: same pattern size as on the wafer (1x) → for mask aligner and 1x stepper.

Reticle: 4x-5x magnification pattern size → for Stepper & Scanner



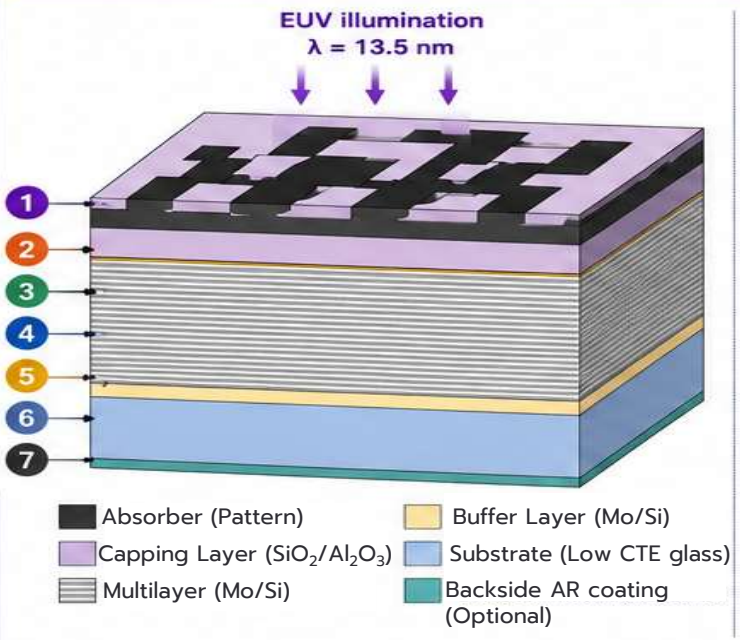
Clear Data = Non-Inverted (Quartz)

Dark Data = Inverted (Chrome)

Chrome up & Wrong Reading = Chrome down & Right Reading

EUV Reflective Mask Structure

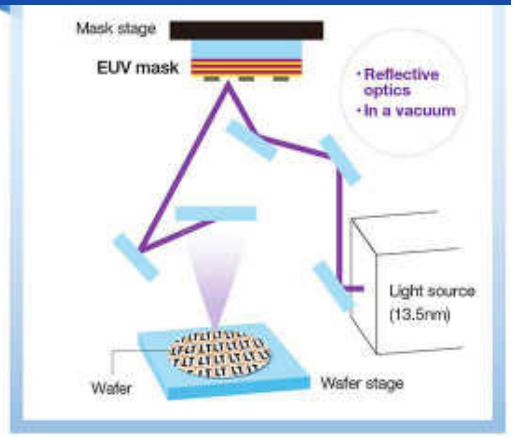
1. EUV Reflective Mask Layer Stack (Typical)



1. **Ru Cap (2-5 nm)**
   - Protects absorber from oxidation
   - Reduces defectivity
2. **Absorber (TaBN/Ta/Cr-based)**
   - Absorbs EUV light (60-70% absorption)
   - Defines the pattern (opaque areas)
3. **Capping Layer (SiO_2/Al_2O_3, 2-3 nm)**
   - Absorbs EUV light (60-70% absorption)
   - Defines the pattern (opaque areas)
4. **Multilayer Mirror (Mo/Si)**
   - 40-60 bilayers, period 6.9 nm
   - Reflects EUV (65-70%)
5. **Buffer Layer (Mo/Si)**
   - Smooths interface
   - Improves reflectivity
6. **Substrate (Low CTE glass)**
   - Ultra-flat, low defect
   - e.g. Quartz, Zerodur®
7. **Backside AR coating (optional)**
   - Reduces backside reflections
   - Improves illuminance uniformity

EUV masks are reflective, not transmissive
Patterns are formed by absorbing (dark field) EUV light instead of transmitting it.

EUV Lithography system





LTEM: Low Thermal Expansion materials (TiO_2 doped SiO_2 amorphous glass)

https://www.lasertec.co.jp/en/ir/individuals/euv.html
http://www.jommpublish.org/p/29/

Chemicals Used in Photomask Making Process

Mask Blank	Cr / CrO_x / CrON	Chromium-based absorber stack	Light-blocking absorber layer
	Quartz / Fused Silica	SiO_2	Transparent mask substrate
Pre-clean	SPM	$H_2SO_4 + H_2O_2$	Organic contamination removal
	SC-1	$NH_4OH + H_2O_2 + H_2O$	Particle / organic cleaning
	DIW / UPW	Deionized / Ultrapure Water	Rinse
Adhesion Promotion	HMDS	Hexamethyldisilazane	Improve resist adhesion
Resist Coating	PR / E-beam Resist	Chemically amplified resist, ZEP, PMMA, etc.	Patterning layer
Resist Solvent / EBR	PGMEA	Propylene Glycol Monomethyl Ether Acetate	Resist solvent / edge cleaning
E-beam Resist Developer	MIBK / IPA	Methyl Isobutyl Ketone + Isopropyl Alcohol	PMMA development
	n-Amyl Acetate	Amyl acetate	Developer for some e-beam resists such as ZEP-type processes
CAR Developer	TMAH	Tetramethylammonium Hydroxide, commonly ~2.38 wt%	Develop aqueous-developable resist
Cr Wet Etch	Ceric Ammonium Nitrate (CAN)	$(NH_4)_2Ce(NO_3)_6$-based solution	Chromium absorber etching
	Nitric Acid	HNO_3	Common component of formulated Cr etchant
Cr Dry Etch	$Cl_2 + O_2$	Chlorine / Oxygen plasma	Plasma etching of Cr-based absorber
Quartz Etch	CF_4 / CHF_3 / C_4F_8	Fluorocarbon plasma chemistry	Etch SiO_2 for phase-shift structures
PR Strip	NMP	N-Methyl-2-pyrrolidone	Resist removal
	DMSO-based stripper	Dimethyl Sulfoxide formulation	NMP-free resist stripping
	O_2 Plasma	Oxygen	Dry resist strip / ash
Final Clean	SPM	$H_2SO_4 + H_2O_2$	Organic/resist residue removal
	SC-1 / APM	$NH_4OH + H_2O_2 + H_2O$	Particle removal
	Ozonated DIW	$O_3 + H_2O$	Organic cleaning
	DIW / UPW	H_2O	Final rinse

Summary

Mask Blank	Cr / CrO_x / CrON	Chromium-based absorber stack	Light-blocking absorber layer
	Quartz / Fused Silica	SiO_2	Transparent mask substrate
Pre-clean	SPM	H_2SO_4 + H_2O_2	Organic contamination removal
	SC-1	NH_4OH + H_2O_2 + H_2O	Particle / organic cleaning
	DIW / UPW	Deionized / Ultrapure Water	Rinse
Adhesion Promotion	HMDS	Hexamethyldisilazane	Improve resist adhesion
Resist Coating	PR / E-beam Resist	Chemically amplified resist, ZEP, PMMA, etc.	Patterning layer
Resist Solvent / EBR	PGMEA	Propylene Glycol Monomethyl Ether Acetate	Resist solvent / edge cleaning
E-beam Resist Developer	MIBK / IPA	Methyl Isobutyl Ketone + Isopropyl Alcohol	PMMA development
	n-Amyl Acetate	Amyl acetate	Developer for some e-beam resists such as ZEP-type processes
CAR Developer	TMAH	Tetramethylammonium Hydroxide, commonly ~2.38 wt%	Develop aqueous-developable resist
Cr Wet Etch	Ceric Ammonium Nitrate (CAN)	$(NH_4)_2Ce(NO_3)_6$-based solution	Chromium absorber etching
	Nitric Acid	HNO_3	Common component of formulated Cr etchant
Cr Dry Etch	Cl_2 + O_2	Chlorine / Oxygen plasma	Plasma etching of Cr-based absorber
Quartz Etch	CF_4 / CHF_3 / C_4F_8	Fluorocarbon plasma chemistry	Etch SiO_2 for phase-shift structures
PR Strip	NMP	N-Methyl-2-pyrrolidone	Resist removal
	DMSO-based stripper	Dimethyl Sulfoxide formulation	NMP-free resist stripping
	O_2 Plasma	Oxygen	Dry resist strip / ash
Final Clean	SPM	H_2SO_4 + H_2O_2	Organic/resist residue removal
	SC-1 / APM	NH_4OH + H_2O_2 + H_2O	Particle removal
	Ozonated DIW	O_3 + H_2O	Organic cleaning
	DIW / UPW	H_2O	Final rinse

Lithographic Alignment & Exposure

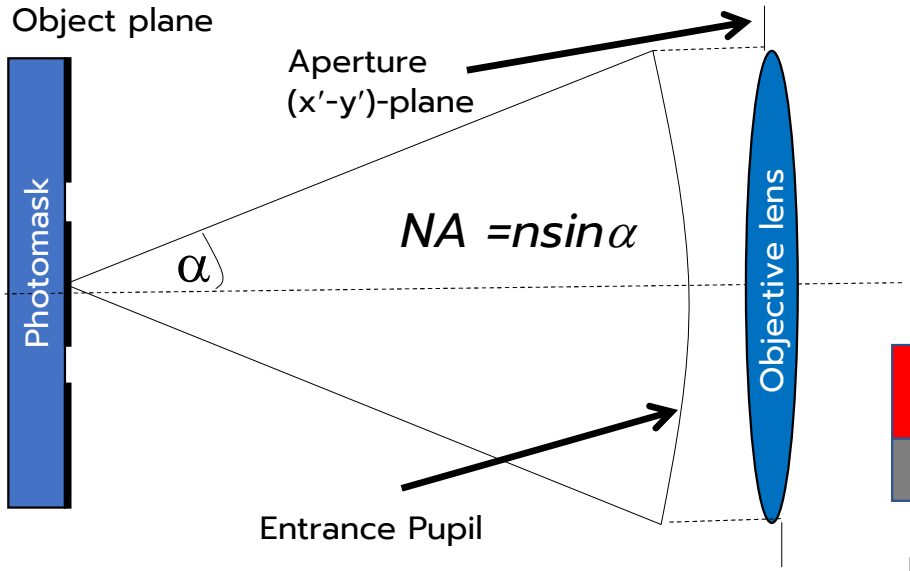
Numerical Aperture (NA) and DOF

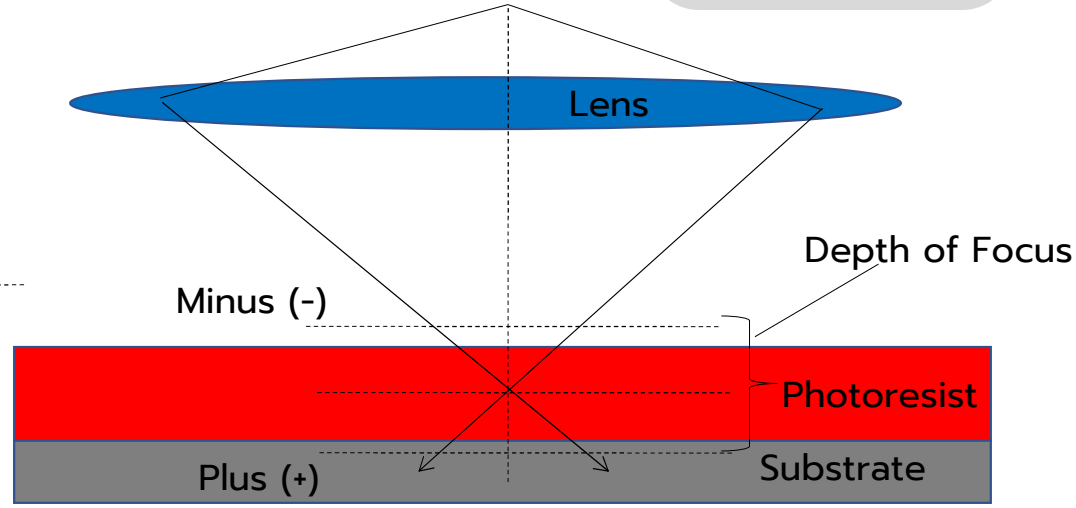
Numerical Aperture (NA): the sine of the maximum half-angle of diffracted light that can enter the lens times the index of refraction of the surrounding medium.

Depth of Focus (DOF): The total range of focus that can be tolerated and printed feature within a variety of specifications.

$$R = k_1 \frac{\lambda}{NA}$$

$$DOF = k_2 \frac{\lambda}{NA^2}$$





R = Resolution, DOF = depth of focus
K_1, K_2 = lithography constant, process parameters
λ= light wavelength

Chris A.Mack, Field guide to Optical Lithography. Washington, 2006.

Lithography Wavelength Selection

Why lithography wavelengths are 436, 365, 248, 193, 157, 13.5 nm

Longer λ (earlier) → Shorter λ (Later)

WAVELENGTH (nm)	LIGHT SOURCE	WHY THIS WAVELENGTH	TYPICAL USE/ NODE ERA	MAIN LIMITATION	NOTES
436 nm g-line	Mercury (Hg) arc lamp	• Strong emission line of Hg • Quartz optics work well • First widely used for ICs	> 1.0 μm (1980s and early)	Limited resolution	Older generation mercury lamp lithography
405 nm h-line	Mercury (Hg) arc lamp	• Shorter than 436 nm • Better Resolution	~ 1.0 μm – 0.5 μm (1980s – early 1990s)	Resolution limit	Interim improvement before i-line
365 nm i-line	Mercury (Hg) arc lamp	• Strong Hg line at 365 nm • Excellent transmission in fused silica (Qz) • Photoresist work well • Mature, reliable, low cost	~ 0.8 μm – 0.35 μm (1990s)	Could not keep up with shrinking dimensions	Workhouse of the 1990s
248 nm KrF	KrF excimer laser	• Naturally lasers at 248 nm • High pulse energy • Narrow linewidth • Scannable exposure systems	~ 0.8 μm – 0.35 μm (1990s)	More complex photoresist chemistry	Enabled significant scaling for many generations
193 nm ArF	ArF excimer laser	• Shorter wavelength • Fused silica optics still usable • PR can be engineered • Manageable transmission losses • Success with immersion, OPC, PSM, MPs	~ 90 nm – 7 nm and beyond (2000s - today) *** With multiple patterning**	Requires multiple patterning (MP) to continue scaling	Most successful DUV wavelength in industry
157 nm F_2 (VUV)	F_2 excimer laser	• Intended to replace 193 nm • Shorter wavelength	Largely abandoned	• Fused silica absorbs strongly (need CaF_2 optics – difficult & costly) • Air absorbs (need vacuum/N_2 purge) • PR absorb more & contamination	Technical and cost barriers too high
13.5 nm EUV	EUV plasma (source)	• Conventional optics stop working below deep UV, 13.5 nm is the sweet spot for mirror reflectivity and source efficiency • Uses reflective multilayer mirrors in vacuum	7 nm and beyond nodes **(EUV lithography)**	Extremely complex system with very high cost	Industry jumped to EUV after limits of 193 nm

WHY ONLY A FEW WAVELENGTHS ARE POSSIBLE

1. Light source availability — Needs high power, stable, narrow linewidth sources.
2. Optical Materials Transparency — Lenses, prisms and windows must transmit the light efficient.
3. Photoresist Compatibility — PR must be transparent and processable
4. Manufacturable Optics — Optics must be fabricable, reliable, and cost-effective.
5. Economic Feasibility — The whole system must be practical for HVM.

EVOLUTION OF LITHOGRAPHY WAVELENGTHS

436 nm g-line → 405 nm h-line → 365 nm i-line → 248 nm KrF → 193 nm ArF → 157 nm F_2 VUV → 13.5 nm EUV

Mercury Lamp Era | Excimer Laser Era | Deep UV Era | Abandoned | EUV Era

KEY TAKEAWAY

Lithography wavelengths are determined by the intersection of physics, materials, and engineering. The few wavelengths listed are the only ones where the entire ecosystem- light source, optics, photoresist, process, and cost – could actually work at industrial scale. As technology advanced, shorter wavelengths were used until practical limits pushed the industry to EUV.

HVM is High-volume manufacturing
VUV is Vacuum Ultraviolet

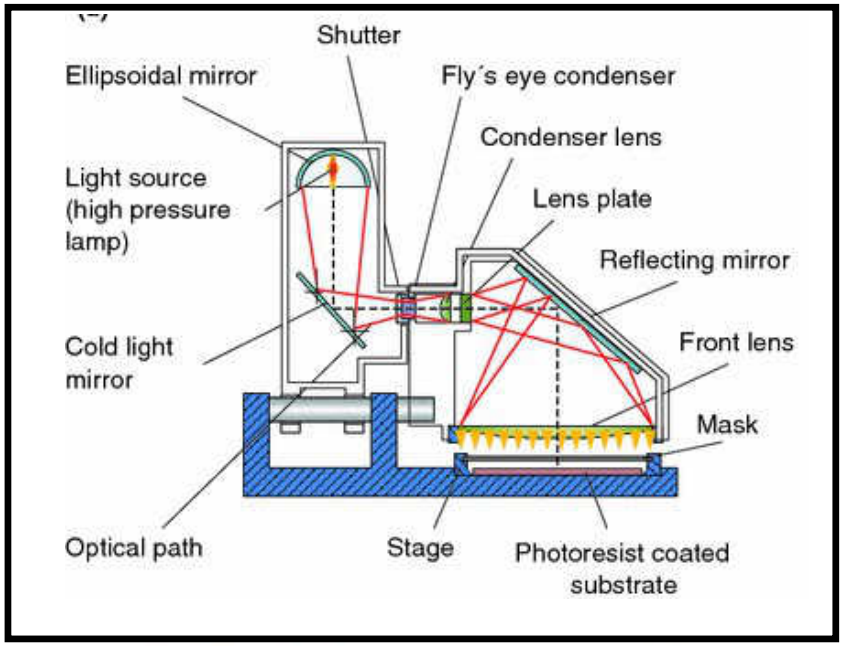
The letters **g, h, and i** are simply historical labels assigned to specific mercury emission lines.

VUV and EUV (Extreme-UV) are strongly absorbs O_2, H_2O vapor, organic materials. Optical materials Vacuum, or N_2-purged environments with extremely low oxygen and moisture levels

Contact Mask Aligner (M/A)

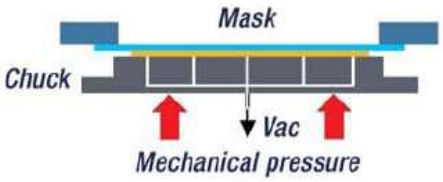
Resolution: $\frac{3}{2}\sqrt{\lambda\left(g+t/2\right)}$

(g=gap, t=resist thickness)

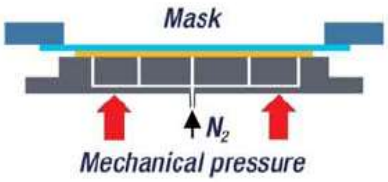


Parameter	Vacuum Contact	Hard contact	Proximity contact
Mask–Wafer Gap	~0 μm + vacuum	~0 μm	~1–300 μm
Contact Method	Vacuum pulls mask/wafer into intimate contact	Mask physically contacts wafer	No physical contact
Resolution / CD	Best: <1 μm	High: ~1 μm	Lower: ~3–10 μm
Diffraction	Minimum	Low	Higher
Mask Damage	Highest risk	Medium–high	Low
Particle Sensitivity	Very high	High	Lower
Mask Lifetime	Shorter	Medium	Longest

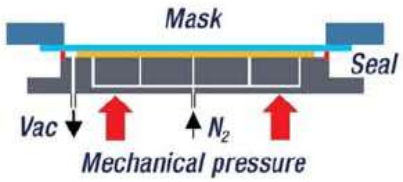
Soft Contact

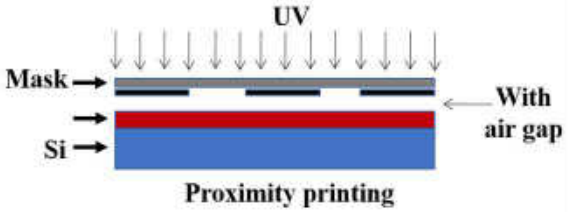


Hard Contact

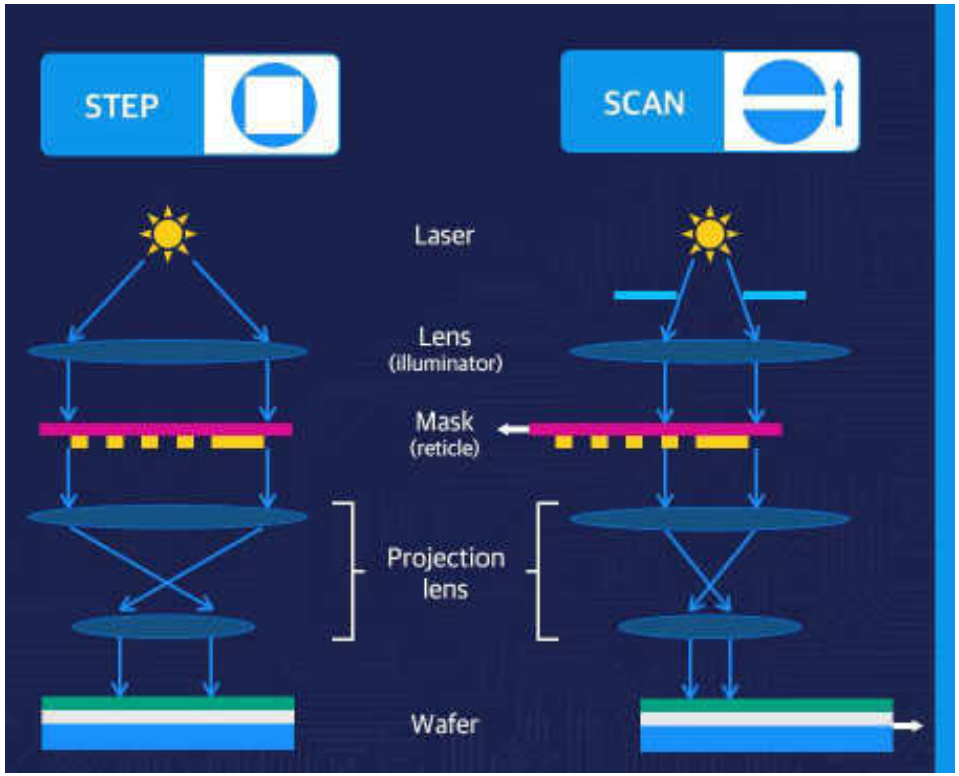


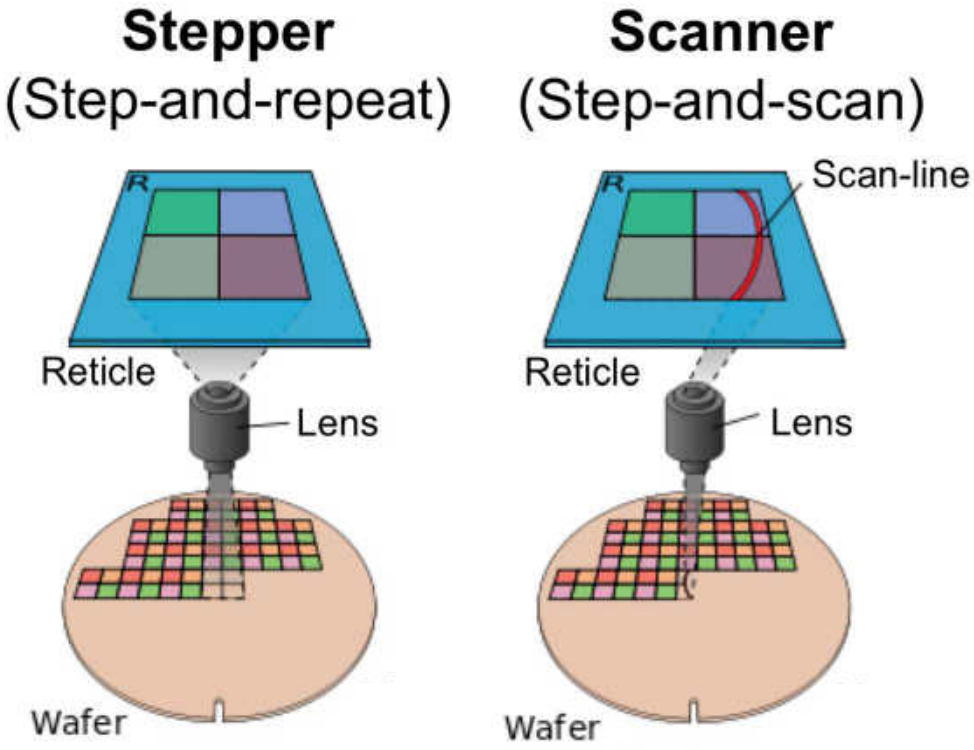
Vacuum Contact



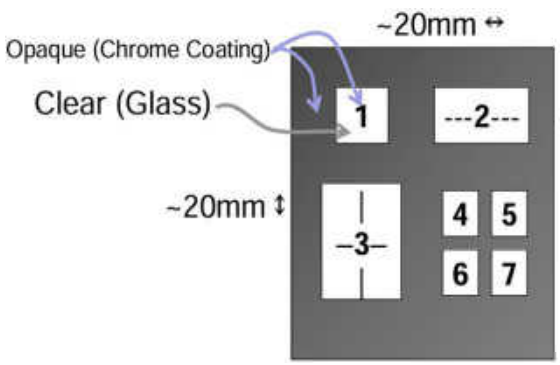


Stepper & Scanner Exposure tools

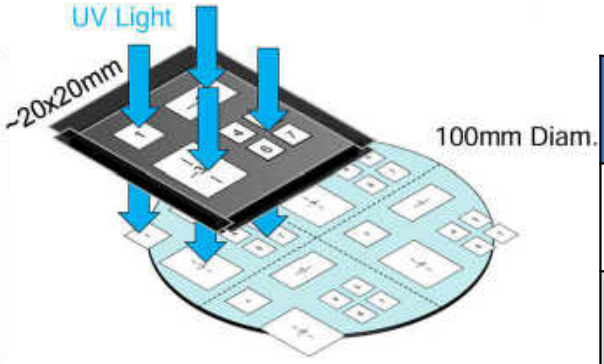




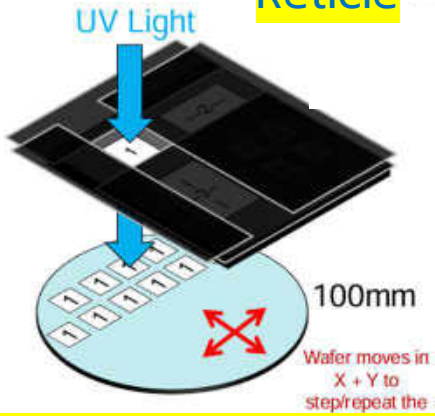
Stepper/scanner blade-off for exposure



Reticle



Fully-exposure



Exposing only pattern#1



Exposing only pattern#2

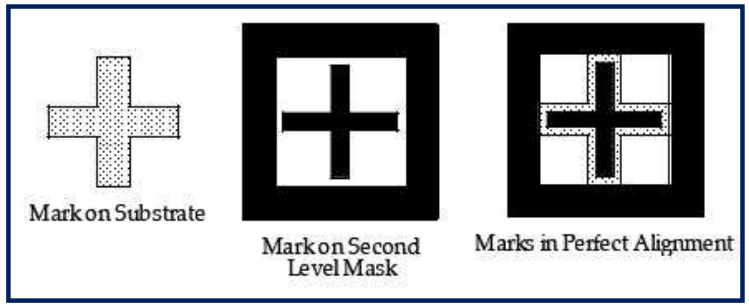
Tool Type	Wavelength	Max Field Size (mm)
i-line Stepper	365 nm	22x22 mm
KrF Stepper	248 nm	22x22 mm
ArF Stepper (DUV)	193 nm	22x22 mm
DUV Scanner (ArF dry)	193 nm	26x33 mm
DUV Immersion Scanner	193i	26x33 mm
EUV Scanner	13.5 nm	26x33 mm

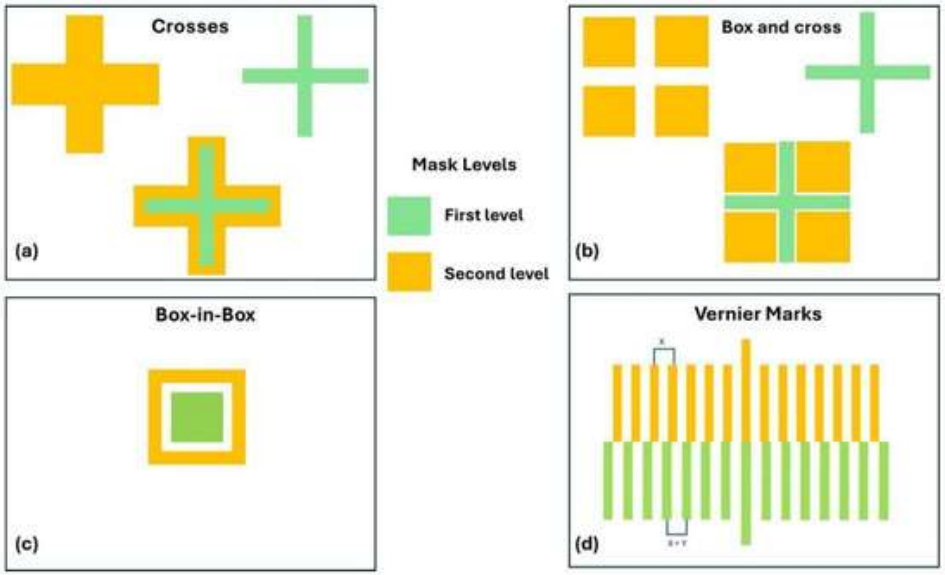
https://wiki.nanofab.ucsb.edu/w/images/c/cb/Demis_D_John_-_Stepper_Reticle_Layout_vs_Wafer_Layout.pdf

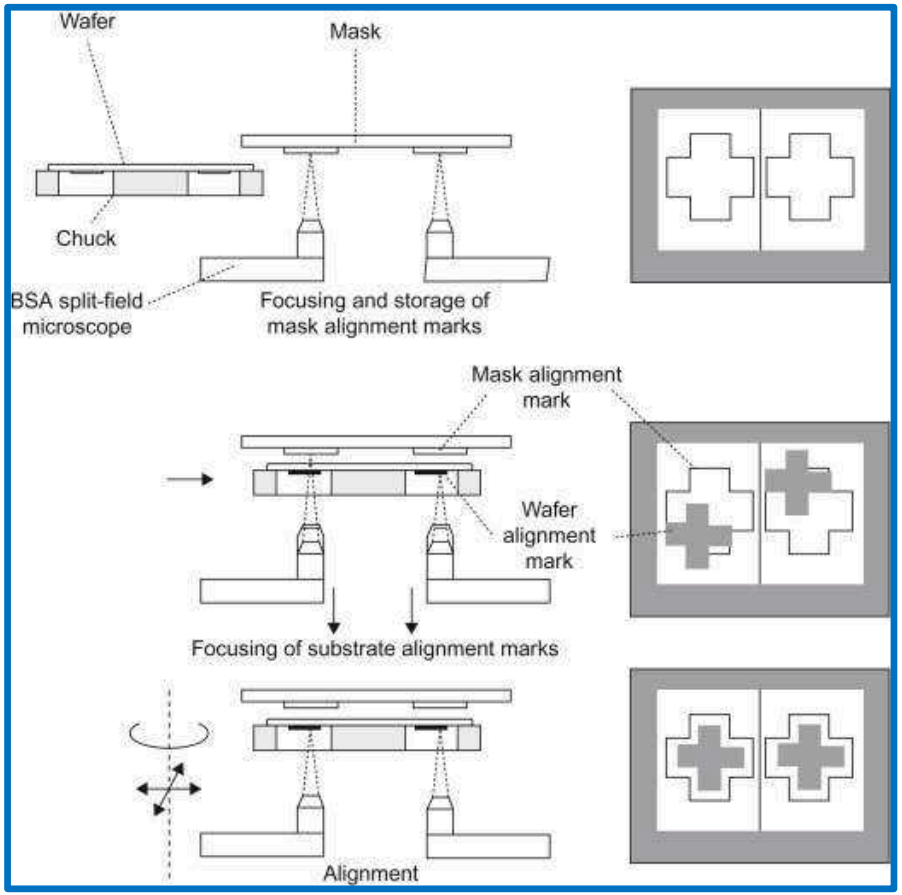
Lithography Exposure Tool Comparison

Parameter	Mask Aligner	Stepper	DUV Scanner	EUV Scanner
Exposure	Contact / proximity	Step & repeat	Step & scan	Step & scan
Imaging	1×	4× / 5×	4×	4×
Wavelength	436–365 nm	365–193 nm	248 / 193 nm	13.5 nm
Typical CD	~1–10 μm	~0.1–1 μm	~20–150 nm	~8–30 nm
Mask	Full-wafer mask	Reticle	Reticle	Reflective reticle
Complexity	Low	Medium	High	Extreme
Sweet Spot	MEMS / Packaging	Mature IC / MEMS	Advanced IC / SiPh	Leading-edge IC
Cost Level	$	$$	$$$$	$$$$$

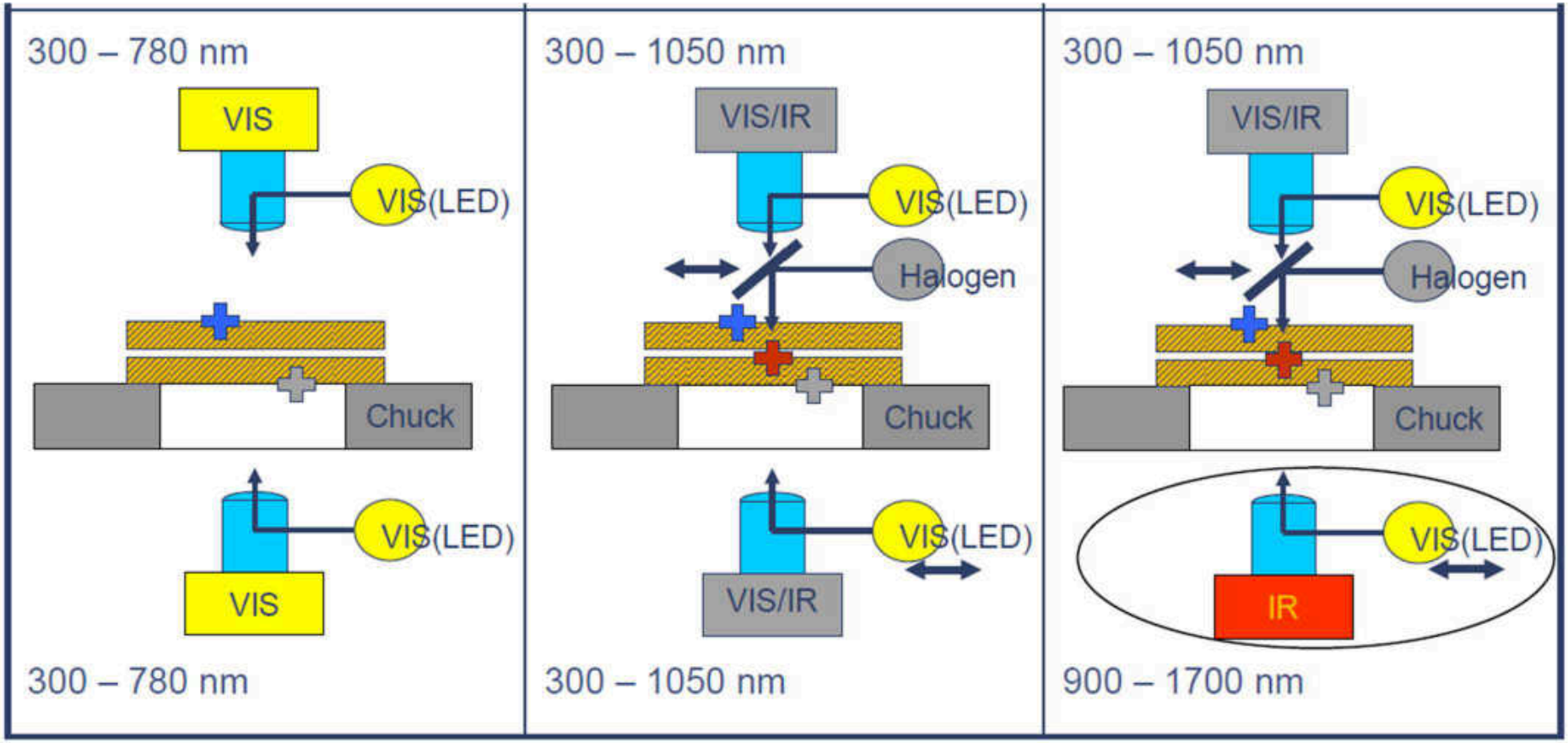
Pattern alignment and overlay accuracy







Front-side alignment (FSA) by VIS and Back-side alignment (BSA) by IR



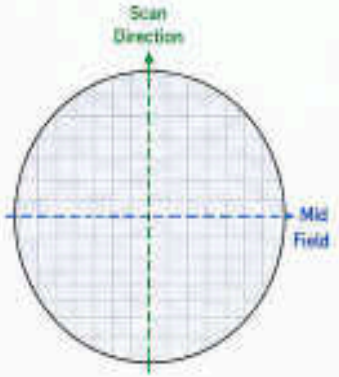
https://www.suss.com/en/products-solutions/mask-aligner

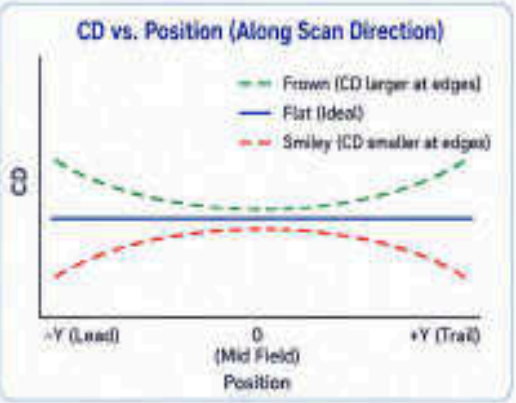
Smiley Curve & E-F Matrix

To understand and control wafer-level CD uniformity

SMILEY (RADIAL CD VARIATION)

Smiley is the radial (within-field) CD variation along the scan direction caused by optical and process effects. It appears as a "smile" or "frown" when CD is plotted across the field.





KEY POINTS

- Smiley affects CD uniformity within a die/field.
- Typically measured in nm across the scan direction.
- Control is essential for tight CD specifications and overlay accuracy.

MEASUREMENT

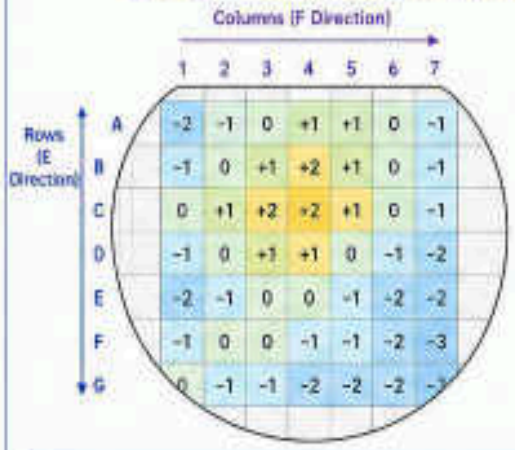
Measure CD at multiple points along the scan direction (e.g., -Y to +Y) at mid-field and calculate the variation (peak-to-peak or 3σ).

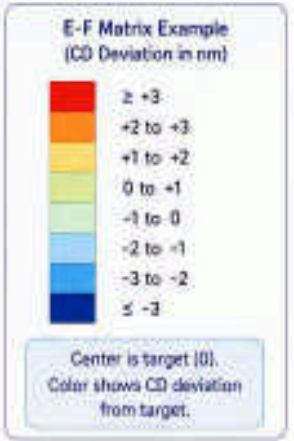
E-F MATRIX (WAFER-LEVEL CD UNIFORMITY)

The E-F matrix maps CD variation across the entire wafer.

E (Elliptical) = variation along Row (Scan direction)

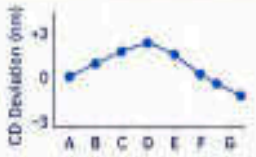
F (Focus) = variation along Column (Slit/Step direction)





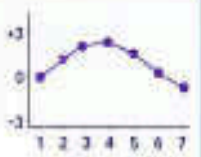
E (Row) Profile Example

Average CD of each row → shows variation along scan direction (Smiley/Frown trend)



F (Column) Profile Example

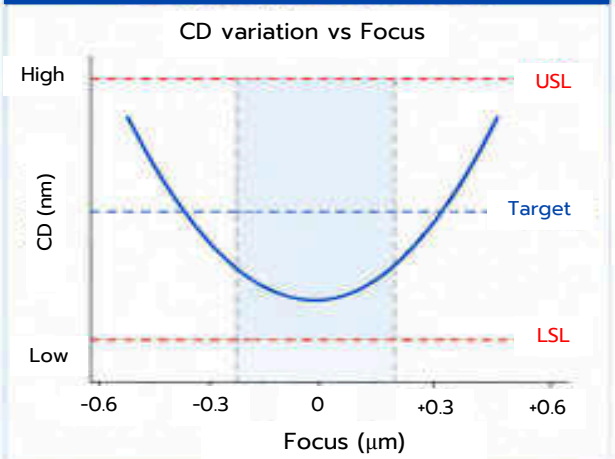
Average CD of each column → shows variation along slit/step direction



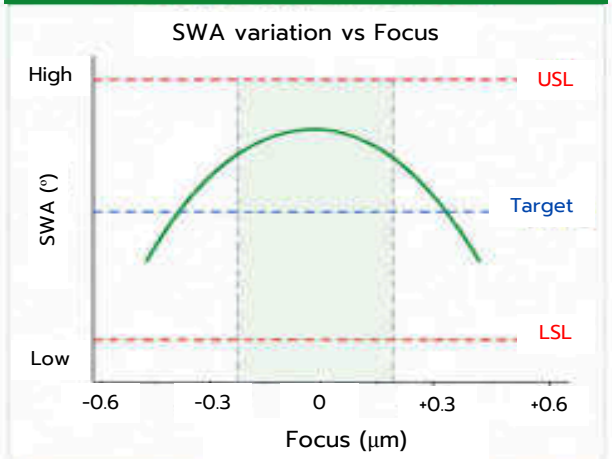
Lithography Process Window Curves

The range of focus (or dose) where key parameters meet specifications

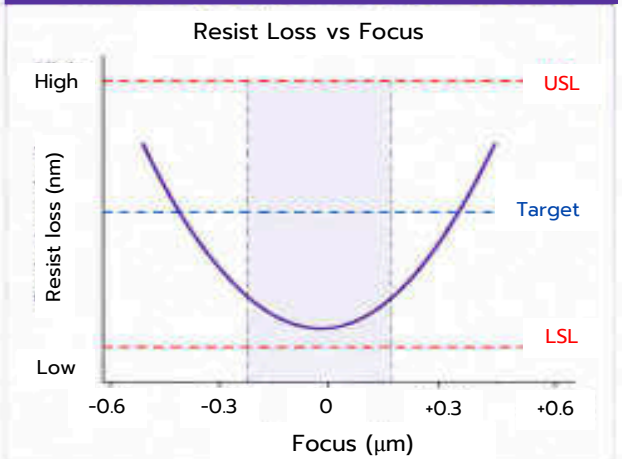
1. CD (Critical dimension)



2. SWA (Sidewall angle)

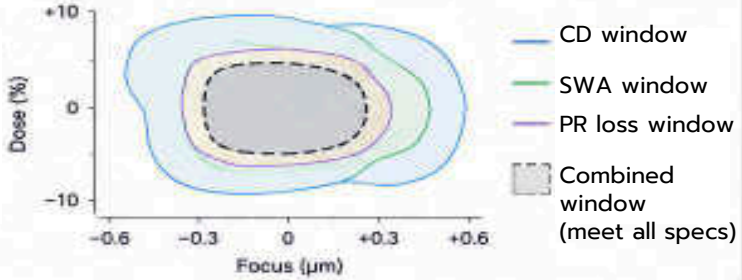


3. Resist loss (Footing/Undercut)



4. Combined process window (All specs)

The overlap of all individual windows gives the true process window where all specifications are met simultaneously



LEGEND

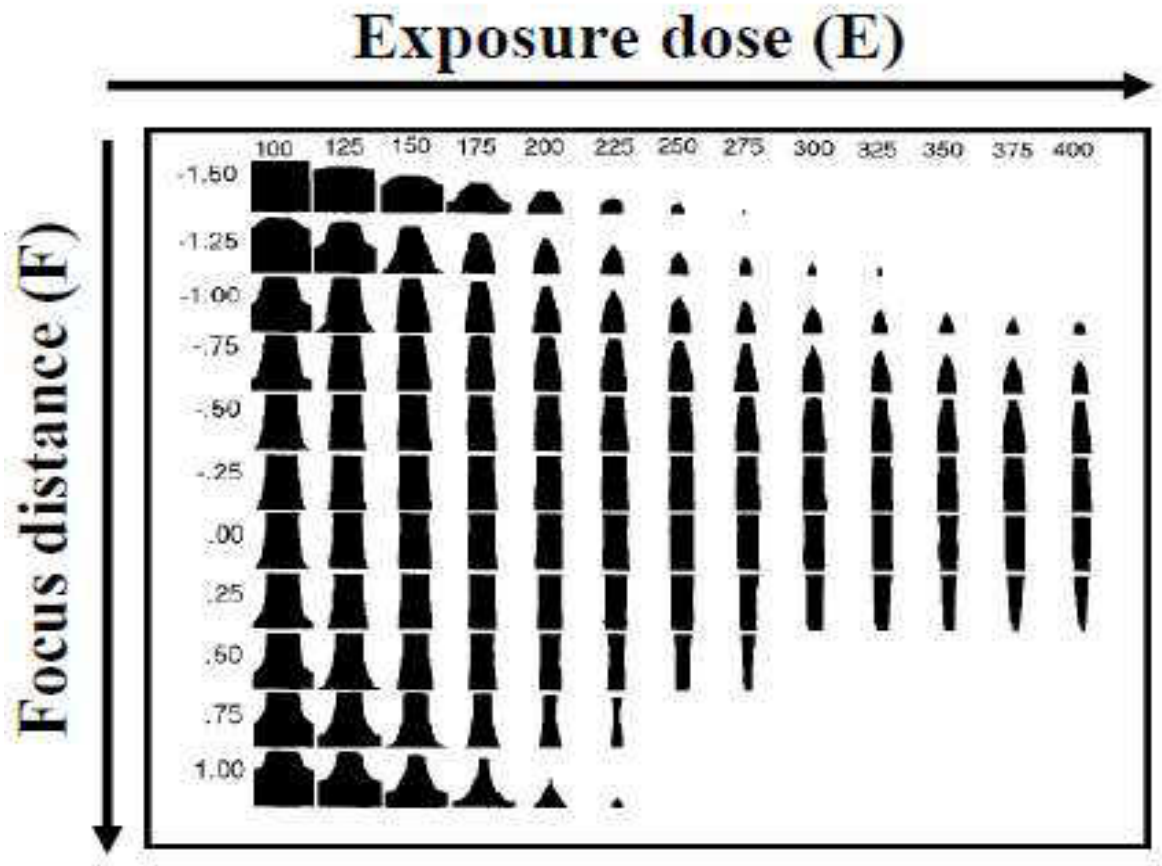
- Individual process window
- Target (nominal)
- USL/LSL (spec limits)
- Window boundaries

KEY TAKEAWAY

Each parameter has its own acceptable range.

The combined window (overlap) is the actual operating region where all specs are satisfied

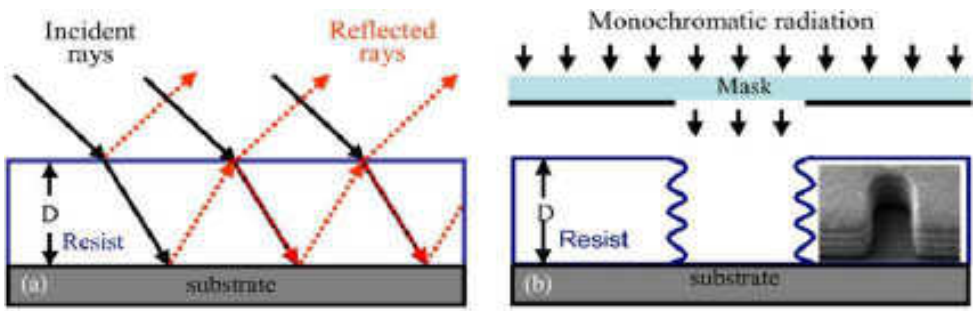
Energy-Focus (E-F) matrix (PR x-view)

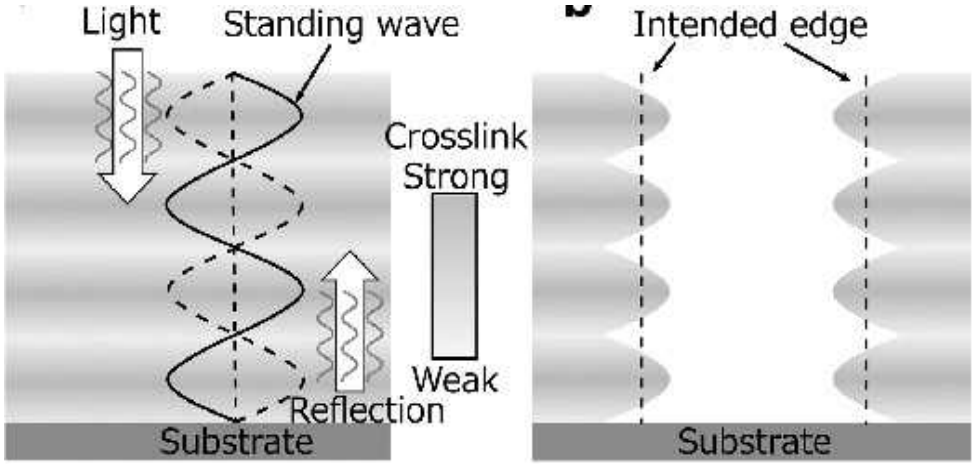


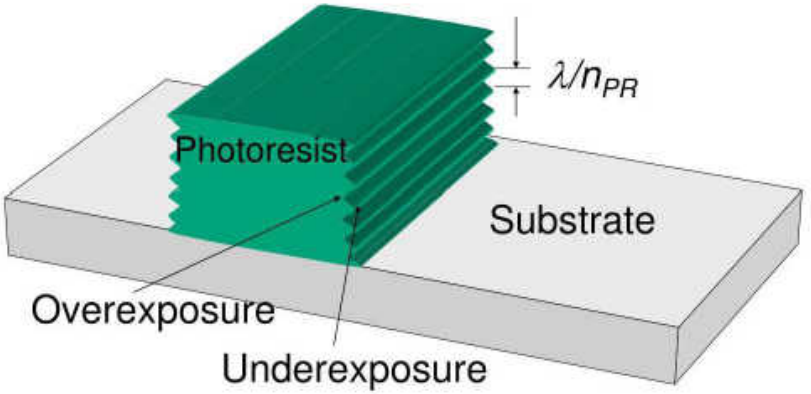
Post-Exposure Bake (PEB), Develop, Hardbake

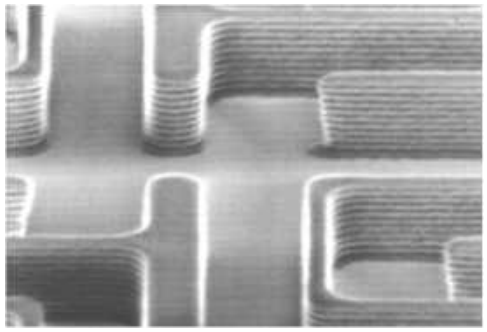
Standing wave effects

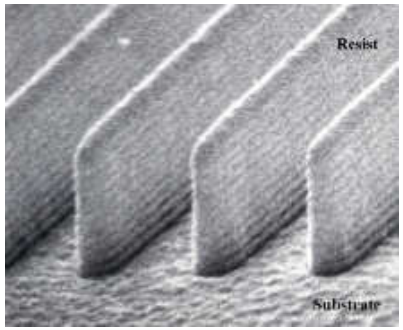
The standing wave effect is a photoresist profile defect caused by interference between incident light and light reflected from the wafer substrate.











Post-exposure bake (PEB)

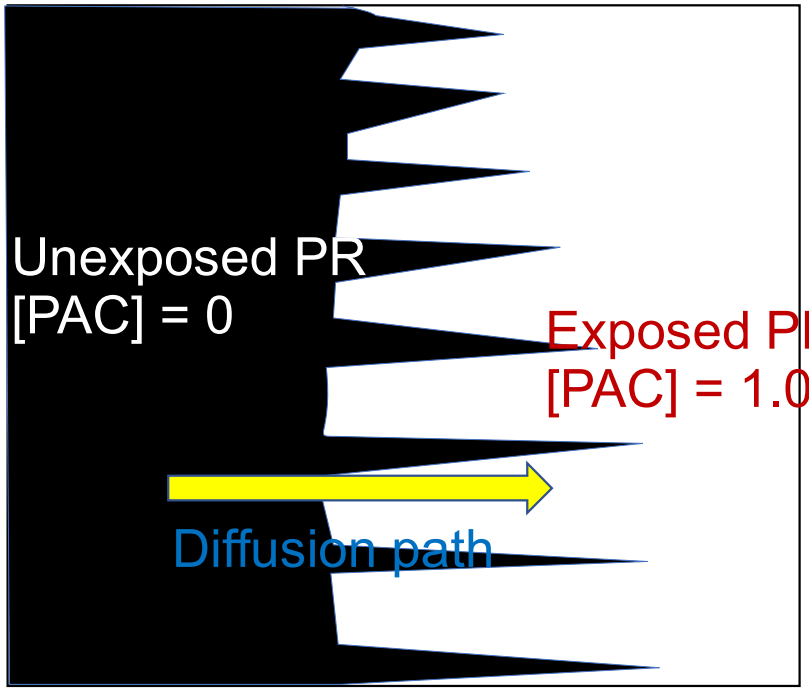
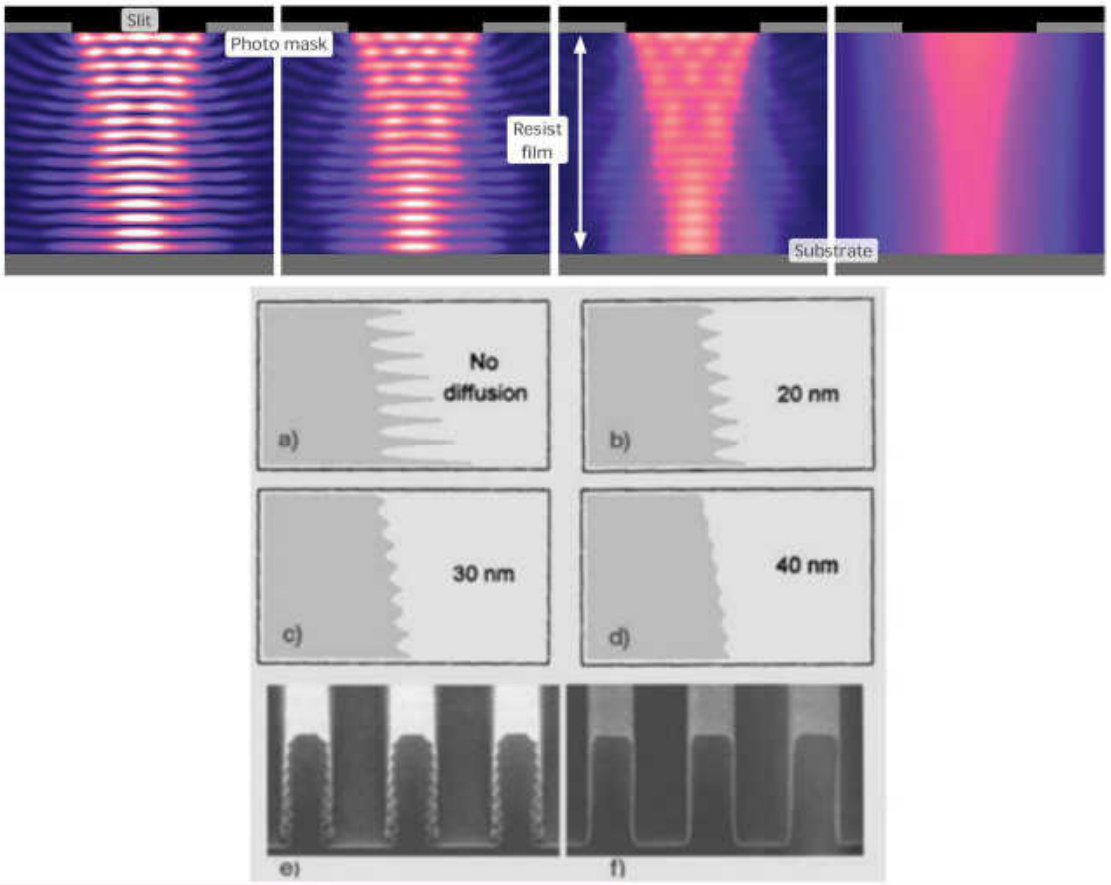
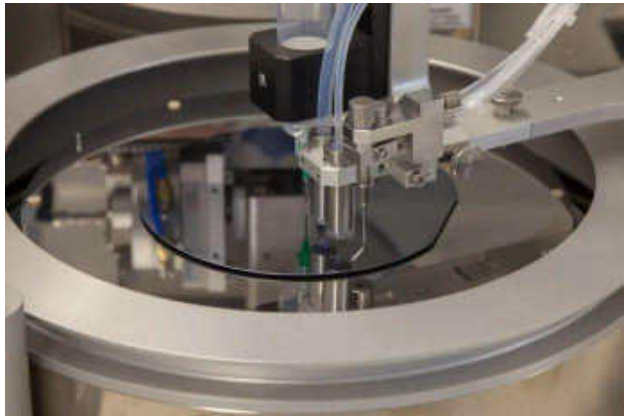
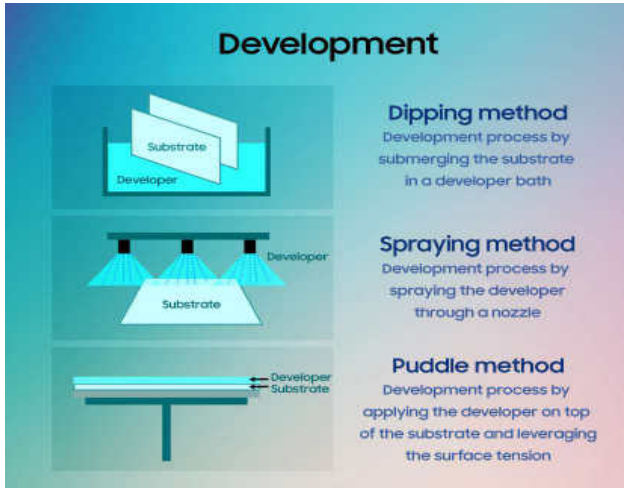


Photo active compound (PAC) can effectively diffuse through the polymer matrix to reduce effective "standing wave" in the developed resist sidewall



Develop





Dip / Immersion	Wafer immersed in bath	No nozzle	R&D, legacy
Puddle Develop	Developer dispensed to form puddle	Straight / single-point nozzle	IC, MEMS
Scan Puddle	Nozzle scans across wafer	Scanning nozzle	Large wafer, better uniformity
Spray Develop	Fine droplets sprayed on wafer	Spray / atomizing nozzle	Thick PR, MEMS
Fan Spray	Wide developer spray	Fan / slit nozzle	Large-area substrate
Spin Develop	Developer dispensed during rotation	Single-point / stream nozzle	Automated track
Linear / Slit Develop	Developer applied across substrate width	Slit / line nozzle	FPD, large-area mask
Dry Develop	Plasma removes resist	No liquid nozzle	EUV / advanced

https://global.samsungdisplay.com/29487/

Photoresist / PI / PBO Developer Types

PR → 2.38% TMAH
PS-PBO → TMAH-based developer
PSPI → TMAH or organic solvent, depending on formulation

TMAH	Tetramethylammonium Hydroxide	$(CH_3)_4NOH$	2.38 wt% (0.26 N) common	Positive PR / CAR	⭐⭐⭐ Mainstream i-line, KrF, ArF
TMAH	Tetramethylammonium Hydroxide	$(CH_3)_4NOH$	~2–2.5 wt%, formulation-dependent	Photosensitive PBO (PS-PBO)	⭐⭐⭐ Advanced packaging / passivation / RDL
TMAH	Tetramethylammonium Hydroxide	$(CH_3)_4NOH$	~2–2.5 wt%, formulation-dependent	Aqueous-developable PSPI	Advanced packaging / passivation
KOH	Potassium Hydroxide	KOH	~0.5–2 wt%*	Positive PR	Legacy lithography
NaOH	Sodium Hydroxide	NaOH	~0.5–2 wt%*	Positive PR	Legacy; less preferred in IC fabs
Sodium Metasilicate	Sodium Metasilicate	Na_2SiO_3	Formulation-dependent	Positive PR	Legacy / specialty PR
nBA	n-Butyl Acetate	$C_6H_{12}O_2$	Neat / high-purity solvent	Negative-tone PR	ArF NTD / specialty PR
PGMEA	Propylene Glycol Monomethyl Ether Acetate	$C_6H_{12}O_3$	Neat / formulated	Negative PR / some PSPI	Specialty / thick-film patterning
Cyclopentanone	Cyclopentanone	C_5H_8O	Neat / formulated	Some PSPI / polymer systems	Solvent-developed polymer patterning
NMP	N-Methyl-2-pyrrolidone	C_5H_9NO	Neat / formulated	Some PI/polymer systems	Solvent development / processing; formulation-specific
MIBK / IPA	Methyl Isobutyl Ketone / Isopropyl Alcohol	—	Typically 1:3	PMMA	E-beam lithography
Proprietary Developer	Supplier-specific aqueous/organic formulation	—	Supplier-specific	PI / PBO / thick PR	Advanced packaging / MEMS

Developer Kinetics

Developer Kinetics = Dissolution Rate of Resist in Developer → controlled by exposure dose, developer chemistry/concentration, temperature, time, and mass transport.

Under-development → PR residue / incomplete opening
Optimum development → correct CD + clean profile
Over-development → CD enlargement, resist loss, profile degradation

$$\text{Contrast} \propto \frac{\text{dissolution rate of region to be removed}}{\text{dissolution rate of region to remain}}$$

Dissolution Rate (DR)	Rate at which PR dissolves in developer, typically nm/s or μm/min
Exposure Dose	Changes resist solubility; controls dissolution rate after exposure
Developer Concentration	Higher activity generally → faster dissolution, but may reduce process margin
Development Time	Must be sufficient for complete pattern opening; excessive time can cause CD loss
Temperature	Higher temperature generally → faster dissolution
Agitation / Flow	Improves developer transport and removal of dissolved resist
PR Chemistry	Positive, negative, CAR, thick PR, PI/PBO have different dissolution behavior
Developer Type	TMAH concentration and organic-solvent chemistry strongly affect dissolution rate
Resist Thickness	Thick PR generally requires longer development and good mass transport

Why 2.38 % TMAH is Used for PR Development

2.38 wt% TMAH (≈0.26 N) = Metal-Ion-Free aqueous developer providing a good balance of dissolution rate, contrast, CD control, and process repeatability.

High PR selectivity	Efficiently dissolves the developer-soluble region while minimizing attack on the region intended to remain
Good development rate	2.38 wt% provides practical dissolution kinetics for production lithography
High lithographic contrast	Produces a sharp difference between soluble and insoluble resist → better pattern definition
Good CD control	Controlled dissolution helps maintain critical dimension (CD) and resist profile
Metal-ion free (MIF)	TMAH contains no Na^+ or K^+, reducing mobile-ion contamination risk in semiconductor devices
Aqueous developer	Water-based process; compatible with DI-water rinse and automated track systems
Process repeatability	Mature, standardized chemistry with stable and reproducible development behavior
Broad PR compatibility	Widely used for i-line, KrF and ArF positive-tone resist systems
Industry-standard strength	2.38 wt% ≈ 0.26 N, a commonly standardized developer concentration

TMAH is a serious acute toxicity hazard, including through skin exposure. Fab handling therefore requires appropriate chemical PPE, engineering controls, and site-specific emergency procedures—not just treatment as a conventional dilute base.

Why 2.38 % TMAH is Used for PR Development

The concentration is essentially a **process optimization point**.
Too low TMAH concentration
→ slower dissolution → longer development → possible incomplete development / residue.

~2.38 wt% TMAH
→ suitable dissolution rate + high contrast + good CD control + practical process time.

Too high TMAH concentration
→ more aggressive development → greater risk of resist loss, CD change, profile degradation, or reduced process latitude depending on the resist.

So **2.38% is not a universal chemical optimum for every resist**; rather, it became a widely adopted standard concentration matched to many semiconductor resist systems.

Why TMAH instead of KOH or NaOH?

- NaOH / KOH → Na^+ / K^+ mobile-ion contamination
- TMAH → Metal-Ion-Free (MIF) developer

Alkali-metal contamination such as Na^+ and K^+ is undesirable in semiconductor manufacturing because mobile ions can affect dielectric/device electrical characteristics.

Hardbake (Post bake)

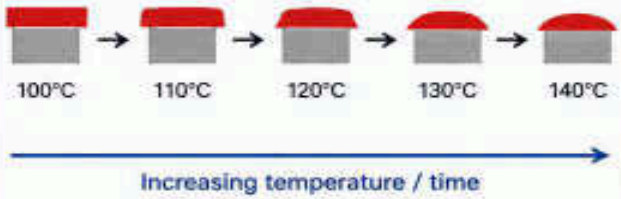
PURPOSE

- Stabilize and harden developed photoresist.
- Main parameter is the plastic flow (glass transition) temperature.
- Remove remaining traces of coating solvent or developer.
- Eliminate solvent burst effects in vacuum processing.
- Introduce some stress; some shrinkage of the photoresist may occur.
- Longer or hotter postbake makes resist removal much more difficult.

HOW IT WORKS

With sufficient time and/or temperature, the photoresist undergoes plastic flow.

– Resist reflow can be used for tailoring sidewall angles.



WHEN TO USE

USE (Firm postbake needed)
For acid etching, e.g. BOE.

NOT NEEDED
For processes where a soft resist is desired, e.g. metal liftoff patterning.

TYPICAL CONDITIONS (EXAMPLES)

Temperature	Time
~100 – 140 °C	Typically 30 sec – 10 min (depends on resist & thickness)

KEY POINTS / CAUTIONS

- Postbake introduces stress and may cause shrinkage.
- Solvent removal prevents solvent burst in vacuum tools.
- Longer or hotter postbake makes resist removal much more difficult.
- Control temperature and time to balance resist stability and stripability.

WHERE IT FITS IN

Coat Resist → Expose → Develop (Pattern) → **Hard Bake (Postbake)** → Etch / Implant / Plating / etc.

Major Photoresist & Lithography Material Suppliers

Lithography Material	TOK JP	JSR JP	Shin-Etsu JP	Fujifilm JP	Sumitomo JP	Dongjin KR	Merck/AZ DE	Specialty Suppliers / Brands
HMDS / Adhesion	✓						✓	MicroChemicals
Photoresist								
g-/h-line PR (436/405 nm)	✓	✓		✓			✓	Shipley
i-line PR (365 nm)	✓	✓	✓	✓	✓	✓	✓	Shipley
KrF PR (248 nm)	✓	✓	✓	✓	✓	✓	✓	DuPont
ArF PR (193 nm)	✓	✓	✓	✓	✓	✓		DuPont
EUV PR (13.5 nm)	✓	✓	✓	✓	✓	△		Inpria
MEMS / Thick PR	✓			✓	✓		✓	Kayaku: SU-8, KMPR; DuPont
Dry Film PR (DFR)								Riston, SUNFORT, PHOTEC
PSPI / PBO		✓		✓				HD MicroSystems, Toray
TARC / BARC	✓	✓	✓			✓	✓	Brewer Science
EBR / Thinner	✓			✓		✓	✓	Kayaku
Developer	✓			✓			✓	Kayaku
Photoresist Stripper	✓			✓			✓	EKC®, ALEG™ / J.T.Baker

✓ = Supplier / product available △ = Emerging / limited portfolio

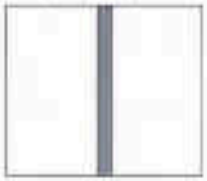
Lithography Pattern Quality Inspection & Characterization

Lithography Pattern Types

1. ISOLATED LINE (Lowest Density)

Single lines separated by large spaces; no neighboring lines.

Top View (Layout)



Cross-Section (After Develop)



Key Characteristics

- 1D feature: single line
- CD = line width
- Minimal pattern density effects
- Good process latitude

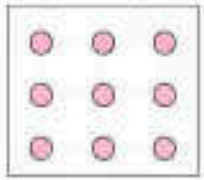
Typical Applications



2. ISLAND PATTERN (PILLAR)

Resist left standing as isolated features (pillars) after development.

Top View (Layout)



Cross-Section (After Develop)



Key Characteristics

- 2D features: isolated posts/pillars
- Size defined by pillar CD
- Height (resist thickness) critical
- Affected by collapse, aspect ratio, and adhesion

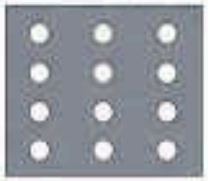
Typical Applications



3. HOLE PATTERN (VIA)

Holes etched/opened in the resist or transferred into the underlying layer.

Top View (Layout)



Cross-Section (After Develop)



Key Characteristics

- 2D features: openings (no resist remaining in opening)
- Size defined by hole diameter
- CD critical in X and Y
- Sensitive to footing, corner rounding, and CD variation

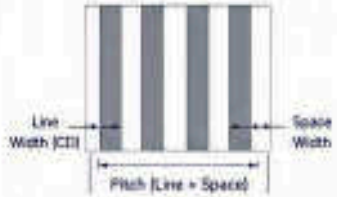
Typical Applications



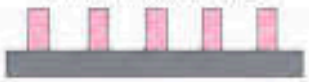
4. LINE-AND-SPACE (L/S)

Alternating lines and spaces with well-defined pitch.

Top View (Layout)



Cross-Section (After Develop)



Key Characteristics

- 1D features: lines and spaces
- Pitch = line CD + space CD
- Critical for resolution and overlay
- Most common and challenging pattern type

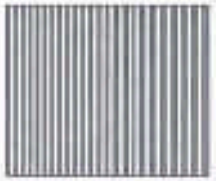
Typical Applications



5. DENSE LINE (LINE ARRAY) (Highest Density)

Many parallel lines with small pitch (line/space = low or no space).

Top View (Layout)



Cross-Section (After Develop)



Key Characteristics

- 1D feature: repeating lines with minimum pitch
- Limited or no space between lines
- High aspect ratio and pattern density
- Higher risk of line-width variation, collapse, and resist scumming

Typical Applications

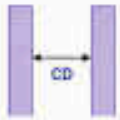


Lithography Process Metrology & Inspection

KEY METROLOGY & INSPECTION ITEMS

1 CD MEASUREMENT (CRITICAL DIMENSION)

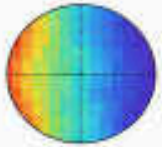
Measures feature width and space.



- Optical / SEM CD tools
- Monitors critical dimensions at target locations

2 CDU (WITHIN-WAFER UNIFORMITY)

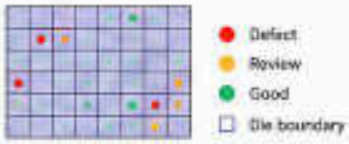
Evaluates uniformity of CD across the wafer.



- Reported as 3σ or Range
- Critical for yield and overlay control

3 DIE-TO-DIE DEFECTS INSPECTION

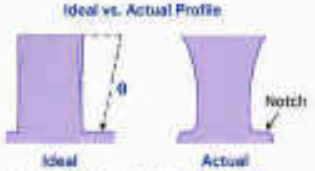
Detects pattern defects on each die.



- Bridges, opens, particles, line breaks, pinholes, etc.
- Full wafer die map

4 SIDEWALL PROFILE (RESIST / ETCH PROFILE)

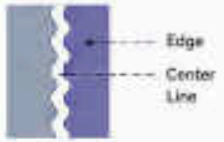
Measures the vertical profile of resist or etched features.



- Angle (θ), Foot / Notch, Taper, Bow
- SEM cross-section or AFM profilers

6 LER (LINE EDGE ROUGHNESS)

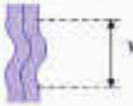
High-frequency variation of the edge position.



- 3σ of edge deviation
- Typical range: 1 – 5 nm (tech dependent)

6 LWR (LINE WIDTH ROUGHNESS)

Variation of line width along the line length.



- 3σ of width variation
- Typical range: 2 – 10 nm (tech dependent)

7 SWR (SIDEWALL ROUGHNESS)

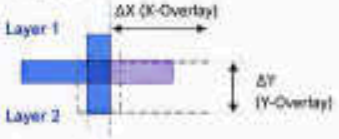
Variation of sidewall position along the vertical direction.



- 3σ of sidewall deviation
- Typical range: 1 – 10 nm (tech dependent)

8 OVERLAY (REFERENCE)

Alignment accuracy between layers.



- Vector overlay (ΔX, ΔY)
- Typical range: < 2 – 5 nm (advanced nodes)

9 FILM / RESIST THICKNESS

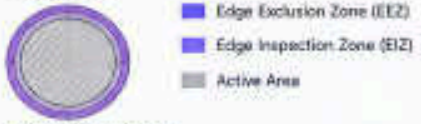
Ensures proper process window and etch control.



- Measured by ellipsometry, reflectometry, XRR
- Typical spec: 10 nm – several µm

10 WAFER EDGE INSPECTION

Inspects defectivity and pattern quality at wafer edge and exclusion zone.



- Edge chips / notches
- Resist scumming / peel-off
- Pattern defects near edge
- Particle contamination

WHY IT MATTERS

- Process Control
- Yield Improvement
- Defect Reduction
- Root Cause Analysis

NOTES

- Measurement results depend on tool, technology node, and process.
- Specs vary by design, layer, and customer requirements.
- Regular monitoring and trending are essential for continuous improvement.

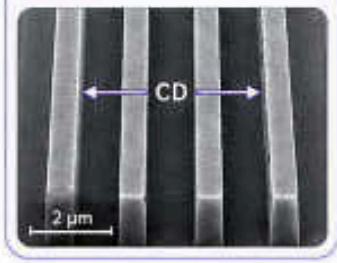
ABBREVIATIONS

CD: Critical Dimension	SWR: Sidewall Roughness
CDU: CD Uniformity	EEZ: Edge Exclusion Zone
LER: Line Edge Roughness	EIZ: Edge Inspection Zone
LWR: Line Width Roughness	

Lithography Photoresist Pattern Quality

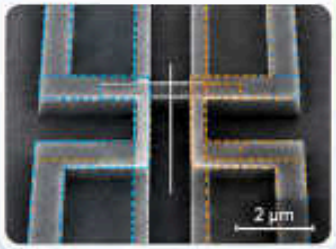
CD CONTROL

- Linewidth accuracy
- Uniformity



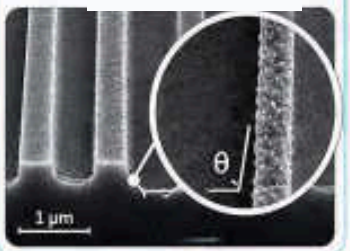
Overlay Accuracy

- Alignment error
- Layer-to-layer shift



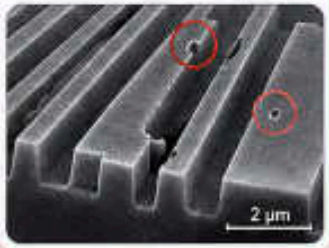
Sidewall Quality

- LER/ LWR
- Profile angle



Defects

- Particles
- Pattern collapse



PATTERING ARTIFACTS

T-Topping



Resist top wider than bottom
→ Causes CD variation and poor transfer

Footing



Resist spreads at the bottom
→ Reduces CD control and CD

Bridging/Scumming



Residue or resist bridges between line
→ Leads to shorts and pattern defect

PATTERN QUALITY METRICS

CD & CDU
Critical dimension & Uniformity

Overlay
Alignment Accuracy

LER / LWR
Line Edge Roughness
Line Width Roughness

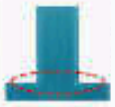
Defect Density
Particles, Voids, Pattern Collapse

Better pattern quality → Higher yield → Better device performance → Reliable MEMS

NOTE
Patterning artifacts can degrade pattern quality, affect yield, and compromise device performance



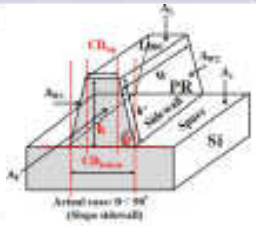
T-Topping
Resist top wider than bottom
→ Causes CD variation and poor transfer



Footing
Resist spreads at the bottom
→ Reduces CD control and CD



Bridging/Scumming
Residue or resist bridges between line
→ Leads to shorts and ,pattern defect



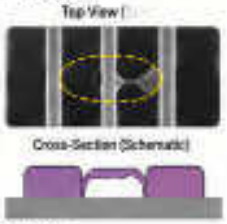
Pattern quality metrology & inspection

Category	Acronym, parameter	Description	What it measures	Metrology Tool	Inspection Tool	Patterned Inspection Method	Defects Review & FA Tool	Impact on Device
CD Control	CD	Critical Dimension	Target linewidth or feature size	CD-SEM, OCD	BF/DF	D2DB (design reference)	FE-SEM, DR-SEM	Performance variation
	CDU	CD uniformity	Variation of CD across wafer/lot	CD-SEM, OCD	–	–	–	Yield variation
	CD Bias	PR vs etched CD	Difference between PR CD and etched CD	CD-SEM	–	–	FE-SEM, TEM	Device mismatch
Edge Placement	EPE	Edge placement error	Edge deviation from intended design location by PR blur	CD-SEM, OCD	–	Design-based , D2DB	FE-SEM, DR-SEM	Pattern placement accuracy
Profile Control	SWA	Sidewall angle	Sidewall angle of pattern	CD-AFM, OCD, 3D-OP	–	–	FE-SEM, FIB, TEM	Etch transfer, electrical behavior
	LES	Line edge slope	Local edge slope (edge sharpness)	CD-SEM, AFM	–	–	FE-SEM	Pattern fidelity
	Bowing	Curved sidewall	Curved sidewall	CD-SEM, 3D-OP	–	–	FE-SEM	Profile distortion
Roughness	LER/LWR	Line edge/width roughness	Edge/width deviation along line by process noise	CD-SEM, AFM	–	–	FE-SEM	Variability, leakage, Local CD variation
	SWR	Sidewall roughness	Roughness along sidewall by plasma damage	AFM, OCD, 3D-OP	–	–	FE-SEM, TEM	Resistance, reliability, Optical loss
Overlay	Overlay Error	Layer misalignment	Misalignment between layers due to Stage drift, distortion	Overlay tool	–	D2DB, design-based	FE-SEM	Interconnect failure
Advanced Patterning	Multi-pattern error	Overlay mismatch	Misalignment in multi-patterning	CD-SEM	e-beam	D2DB + design-based	DR-SEM, TEM	Catastrophic
Design-Sensitive Defects	Hotspot defect	Layout-sensitive failure	OPC / pattern density	CD-SEM	e-beam	Design-based inspection (DBI)	FE-SEM, TEM	Yield limiter

Pattern Defects in Lithography Process

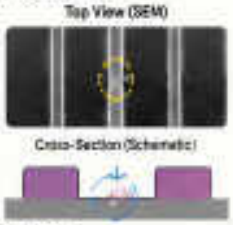
COMMON PATTERN DEFECTS

1 BRIDGING
Resist remains between features that should be separated.



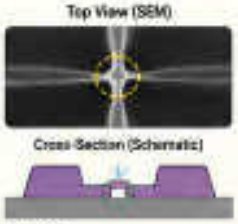
Causes:
- Excess resist thickness
- Insufficient exposure
- Development issues

2 SCUM / RESIDUE
Thin resist film remains on the bottom or sidewall after development.



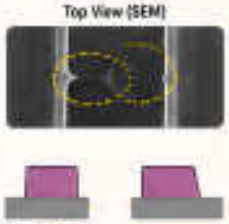
Causes:
- Insufficient development
- Resist chemistry
- High aspect ratio patterns

3 MICROBRIDGE
Very small bridge between adjacent lines or spaces.



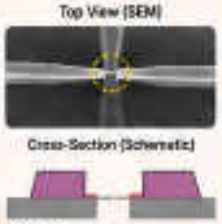
Causes:
- CD too small
- Process variation
- Resist footing

4 OPEN
Line or contact is missing or not printed.



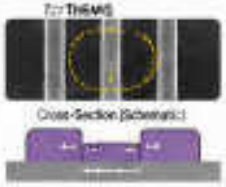
Causes:
- Insufficient exposure dose
- Mask defect
- Poor resist adhesion

5 BREAK
Line is partially missing or has a discontinuity.



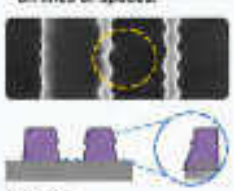
Causes:
- Particle contamination
- Resist defect
- Process non-uniformity

6 CD VARIATION
Line/space width variation from target (too wide or too narrow).



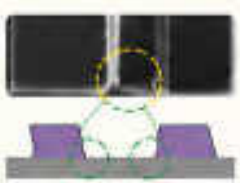
Causes:
- Focus / dose variation
- Resist process variation
- Tool instability

7 EDGE ROUGHNESS (LER)
Irregular, rough edges on lines or spaces.



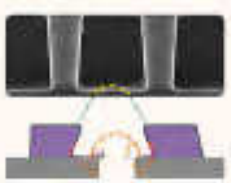
Causes:
- Resist noise
- Photon shot noise
- Mask roughness

8 FOOTING
Undesired resist remains at the bottom of the sidewall.



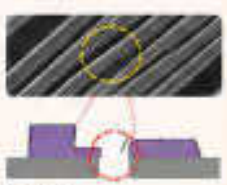
Causes:
- Over-development
- Resist-thickness interaction
- Etch / develop bias

9 T-TOPPING
Resist profile narrower at the bottom than the top.



Causes:
- Insufficient exposure
- Standing waves
- Thick resist processing

10 COLLAPSE
High aspect ratio resist structures fall over or collapse.



Causes:
- High aspect ratio
- Capillary forces during dry
- Poor resist mechanical strength

ORIGIN OF PATTERN DEFECTS

MASK
- Mask defect (particle, pinhole)
- Pattern error / OPC limits
- Phase defect (for PSM)

TOOL / EQUIPMENT
- Focus / alignment error
- Stage vibration
- Illumination non-uniformity

PROCESS
- Resist coating variation
- Exposure dose variation
- PEB / Develop conditions

MATERIALS
- Resist age / contamination
- Resist-outgassing
- Developer contamination

ENVIRONMENT
- Particle contamination
- Temperature / humidity
- Air flow

Pattern defects metrology & inspection

Category	Acronym, parameter	Description	What it measures	Metrology Tool	Inspection Tool	Patterned Inspection Method	Defects Review & FA Tool	Impact on Device
Stochastic Defects	Missing hole	Random missing via/contact in array	EUV photon noise	CD-SEM	e-beam inspection	D2DB + e-beam inspection	DR-SEM, TEM	Catastrophic
	Bridging	Random short between lines vs design	Dose fluctuation	CD-SEM	e-beam	C2C + D2DB	DR-SEM	Short
	Local CDU	Random local CD variation	Stochastic	CD-SEM	—	—	FE-SEM	Variability
Photoresist Defects	Line collapse	Pattern collapse due to capillary force	Capillary force	CD-SEM	BF/DF	D2D / C2C	DR-SEM	Yield loss
	Bridging	PR lines connected	Overexposure	CD-SEM	BF/DF, e-beam	D2D / C2C	DR-SEM	Short
	Missing pattern	Pattern loss	Underexposure	CD-SEM	BF/DF	D2D / D2DB	FE-SEM	Open
	Residue (scum)	Residual PR after develop	Developer issue	CD-SEM	BF/DF	D2D	FE-SEM, TEM	Etch failure
	PR thickness non-uniformity	PR film thickness variation	Thickness mapping	Ellipsometer, 3D-OP	—	—	—	CDU impact
Particle Defects	Particles	Contamination	Tool/env	—	BF/DF, e-beam	D2D / D2DB	FE-SEM, TEM	Killer defect
	Mask print defect	Mask-origin defect transfer	Mask issue	—	BF/DF, e-beam	D2DB (mask vs wafer)	FE-SEM, TEM	Yield loss
Etch Defects	Microloading	CD variation due to pattern density	Pattern density	CD-SEM, 3D-OP	—	—	FE-SEM	CDU loss
	RIE Lag	Slower etch in narrow features (Depth/CD vs feature size)	Transport limit	CD-SEM, 3D-OP	—	—	FE-SEM, FIB	Profile distortion
	Notching	Undercut at interface	Charging	CD-SEM	—	—	FIB + TEM	Failure
	Grass	Needle-like residues due to micro-masking	Particle masking	—	BF/DF, e-beam	D2D / anomaly detection	FE-SEM, TEM	Leakage
	Polymer residue	Etch byproduct residue	Chemistry	—	BF/DF	D2D	FE-SEM, TEM	Contact failure

Lithography & etch defects inspection tools matrix

CD-SEM	✅	✅	✅	⚠	✅	⚠	❌	⚠		⚠	–	⚠
FE-SEM	✅	⚠	✅	✅	✅	✅	⚠	⚠		✅	–	✅
OCD	✅	✅	✅	✅	❌	⚠	❌	❌		❌	–	❌
AFM	⚠	❌	⚠	✅	✅	✅	❌	❌		❌	–	⚠
Overlay Tool	❌	❌	⚠	❌	❌	❌	✅	❌		❌	–	❌
BF Inspection	❌	❌	❌	❌	❌	❌	❌	⚠		✅	D2D / C2C	❌
DF Inspection	❌	❌	❌	❌	❌	❌	❌	⚠	✅ (high sensitivity)		D2D / C2C / anomaly	❌
e-beam Inspection	⚠	❌	⚠	❌	⚠	❌	❌	✅		✅	D2DB / DBI / hotspot detection	⚠
3D Optical Profiler	❌	⚠	❌	⚠ (macro)	❌	⚠	✅ (excellent)	✅ (Ra, Rq)		❌	⚠	❌
DR-SEM	⚠	❌	⚠	✅	✅	⚠	❌	✅		✅	–	✅
FIB	❌	❌	❌	✅	❌	✅	❌	❌		❌	–	✅
TEM	❌	❌	❌	✅	⚠	✅	❌	⚠		❌	–	✅
Computational Metrology & AI	⚠	✅	✅	✅	✅	⚠	⚠	✅		⚠	Design-based / ML anomaly detection	⚠

CD-SEM: Critical Dimension Scanning Electron Microscope
FE-SEM: Field-Emission Scanning Electron Microscope
OCD: Optical Critical Dimension
AFM: Atomic Force Microscope
BF: Bright-field inspection
DF: Dark-Field inspection
DR-SEM: Defect Review Scanning Electron Microscope
FIB: Focused Ion Beam
TEM: Tunneling Electron Microscope

3D-OP: 3-dimensional Optical Profiler
CD: Critical Dimension
CDU: Critical Dimension Uniformity
EPE: Edge placement error
LER: Line-edge Roughness
LWR: Line-width Roughness
LES: Line-edge slope
SWR: Sidewall roughness

D2D: Die-to-Die
(Compare one die with another die on the same wafer)
C2C: Cell-to-Cell
(Compare repeating unit patterns within a die)
D2DB: Die-to-Database
(Compare wafer pattern to design layout)
DBI: Design-based Inspection
(Inspect only critical layout regions (hotspots))

PR Stripping & Rework Process

Chemicals Used in Lithography Rework Process

Lithography Rework = PR Removal → Surface Cleaning → Re-prime → PR Recoat
The mildest chemistry compatible with complete PR removal and the underlying film should be selected, because rework must avoid changing oxide thickness, metals, low-k films, CD, or wafer surface condition.

PR Stripper	NMP — N-Methyl-2-pyrrolidone	Dissolve/remove baked or difficult PR	★★★ PR removal
DMSO Stripper	Dimethyl Sulfoxide-based stripper	NMP-free PR stripping	★★★ PR removal
Acetone	Propanone	Rapid dissolution of many PRs	★★ R&D / less critical processes
PGMEA	Propylene Glycol Monomethyl Ether Acetate	Dissolve fresh / lightly baked PR	★★★ Early-stage rework
PGME	Propylene Glycol Monomethyl Ether	PR solvent / formulated stripper component	★★ Rework / cleaning
ALEG™ / EKC™	Proprietary formulated strippers	PR + polymer/residue removal	★★★ Difficult PR / residue
IPA	Isopropyl Alcohol (2-Propanol)	Solvent rinse / displacement rinse	★★★ Final solvent rinse
DIW / UPW	Deionized / Ultrapure Water	Chemical removal and final rinse	★★★ Aqueous-process rinse
TMAH	Tetramethylammonium Hydroxide	Developer; sometimes used during develop-related rework, not normally the primary PR stripper	★ Process-dependent
SPM	Sulfuric Acid + Hydrogen Peroxide (H_2SO_4/H_2O_2)	Strong organic/resist residue removal	★★ Front-end, substrate-dependent
O_3 / DIW	Ozonated DI Water	Oxidative organic/resist cleaning	★★ Advanced cleaning option
Dilute HF (DHF)	Hydrofluoric Acid + DIW	Surface oxide/residue conditioning	★ Only when process specifically requires it

Chemicals Used in Lithography Rework Process

Typical rework flow: For a wafer rejected after PR coating / bake / develop but before irreversible pattern transfer, the simplified flow is:

Failed Lithography → PR Strip → Rinse/Clean → Dry → Dehydration Bake → HMDS → Recoat PR → Re-exposure

After PR coat / before bake	PGMEA / PR-compatible solvent
After soft bake	PGMEA or dedicated PR stripper
After exposure / PEB	Dedicated NMP/DMSO/formulated stripper
After development	PR stripper → rinse → surface clean
After hard bake / plasma exposure	Strong formulated stripper such as ALEG/EKC-type chemistry, sometimes combined with plasma/oxidative clean
After etch / implant	Much more difficult; may require post-etch residue cleaner and may not be fully reworkable

Photoresist stripping (PGMEA vs. NMP vs. DMSO)

PGMEA → best suited to fresh PR / coating rework
NMP → traditional solvent for baked and thick PR stripping
DMSO-based stripper → important NMP-free approach,

Full Name	Propylene Glycol Monomethyl Ether Acetate	N-Methyl-2-pyrrolidone	Dimethyl Sulfoxide
Formula	$C_6H_{12}O_3$	C_5H_9NO	C_2H_6OS
Main Lithography Role	PR solvent / EBR / BSR / early rework	PR stripper	PR stripper / stripper base solvent
Fresh / Soft-baked PR	★★★	★★★	★★★
Baked PR	★	★★★	★★★*
Thick PR	★	★★★	★★★*
Crosslinked / Difficult PR	Poor	Moderate–Good*	Good when formulated*
Typical Strip Temperature	Often room temp / process-specific	Often heated	Often heated when formulated
Water Miscibility	Moderate	Complete	Complete
Volatility	Higher	Low	Low
Key Advantage	Fast dissolution of fresh PR; already common in lithography track	Strong solvency for many baked/thick PRs	Strong polar solvent; useful basis for NMP-free stripper formulations
Key Limitation	Not ideal for hard-baked or crosslinked PR	Significant health/reproductive-hazard concern	Pure DMSO is not universally an effective stripper; performance depends strongly on formulation
Typical Semiconductor Use	EBR, BSR, PR thinning/rework	Front-end / MEMS / advanced packaging PR strip	Advanced packaging / modern formulated PR strip
Overall Role for PR Strip	★ Limited	★★★ Traditional choice	★★★ Important NMP-free platform*

Flood Exposure

Flood Exposure = Maskless, blanket UV exposure of the entire PR layer.
For positive-PR rework, the easy-to-remember concept is:
Flood Exposure → Remaining PR becomes soluble → Developer (TMAH) → PR Removal

Definition	Blanket UV exposure of the entire wafer without a patterning mask
Light Source	UV; wavelength depends on resist/process, e.g. i-line (365 nm)
Mask Required?	❌ No patterned mask
Area Exposed	Entire wafer / resist surface
Main Purpose	Change PR chemistry uniformly across the wafer
Typical Uses	Lithography rework, image reversal, resist conditioning, specific lift-off/process schemes
Positive PR Effect	Makes exposed PR more soluble in developer
Negative PR Effect	Generally causes exposed regions to become less soluble / crosslinked, depending on chemistry
Key Parameters	Exposure wavelength, intensity, dose (mJ/cm²), resist type and thickness

Flood Exposure

This approach is **resist- and process-dependent**; it is not a universal replacement for wet PR stripping, especially after hard bake, plasma, ion implantation, or severe resist crosslinking

	Normal Exposure	Flood Exposure
Patterned mask	✅ Yes*	❌ No
Purpose	Transfer circuit pattern	Uniformly expose PR
Exposure area	Selected pattern	Entire wafer
Creates pattern	🟩 Yes	❌ No
Common use	Normal lithography	Rework, image reversal, special resist processes

Summary

Successful Lithography = Right Material + Right Chemical + Right Process + Right Pattern + Right Control

Clean → Prime → Coat → Bake → Align/Expose → PEB → Develop → Inspect → Etch/Implant → Strip

- **Lithography** = Key Patterning Process
  Transfers circuit patterns onto the wafer and determines resolution, CD, overlay, pattern fidelity and yield.
- **Surface Preparation & Adhesion**
  Dehydration Bake + HMDS remove surface moisture and improve photoresist adhesion.
- **Photoresist & ARC Materials**
  Positive/Negative PR, CAR, BARC/TARC, PI/PBO are selected according to wavelength, resolution, film thickness and application.
- **Coating & Exposure Control**
  PR thickness/uniformity and exposure dose, wavelength, focus & alignment are critical to achieving the target pattern.
- **Development & Pattern Formation**
  TMAH (typically 2.38 wt%) is a key developer for positive PR; development conditions directly affect CD and pattern profile.
- **Strip, Rework & Process Control**
  PGMEA, NMP/DMSO-based and formulated strippers support PR removal/rework, while metrology verifies CD/CDU, overlay and defects before subsequent processing.

Related media for semiconductor industry



https://waa.inter.nstda.or.th/stks/pub/ebook/sarawit-pdf/Sarawit-Issue145.pdf

https://www.nstda.or.th/sci2pub/wp-content/uploads/2025/04/ST_TMEC_Semiconductors_Ecosystem_NA_250502.pdf



https://www.youtube.com/watch?v=ZMXWttDPGHA



https://www.mmthailand.com/tmec-semiconductor-future-of-thailand/



https://www.mmthailand.com/tmec-semiconductor-future-of-thailand/



https://www.youtube.com/watch?v=l7dlWs3n1Xg



https://www.youtube.com/watch?v=dX9CGRZwD-w&t=11s



https://www.youtube.com/watch?v=sjyKWUBgIco&t=2960s



https://www.youtube.com/watch?v=FugKCV7IUu4



https://www.youtube.com/watch?v=o16wy8R_k1c



https://www.youtube.com/watch?v=c804RyLsWBo&t=41s

TMEC
Thai Microelectronics Center

Contact us

51/4 Moo 1, Wangtakien District,
Amphur Muang, Chachoengsao,
Thailand 24000

+66 38 857 100

info-tmec@nectec.or.th

www.tmec.or.th

Thai Microelectronics Center-TMEC

